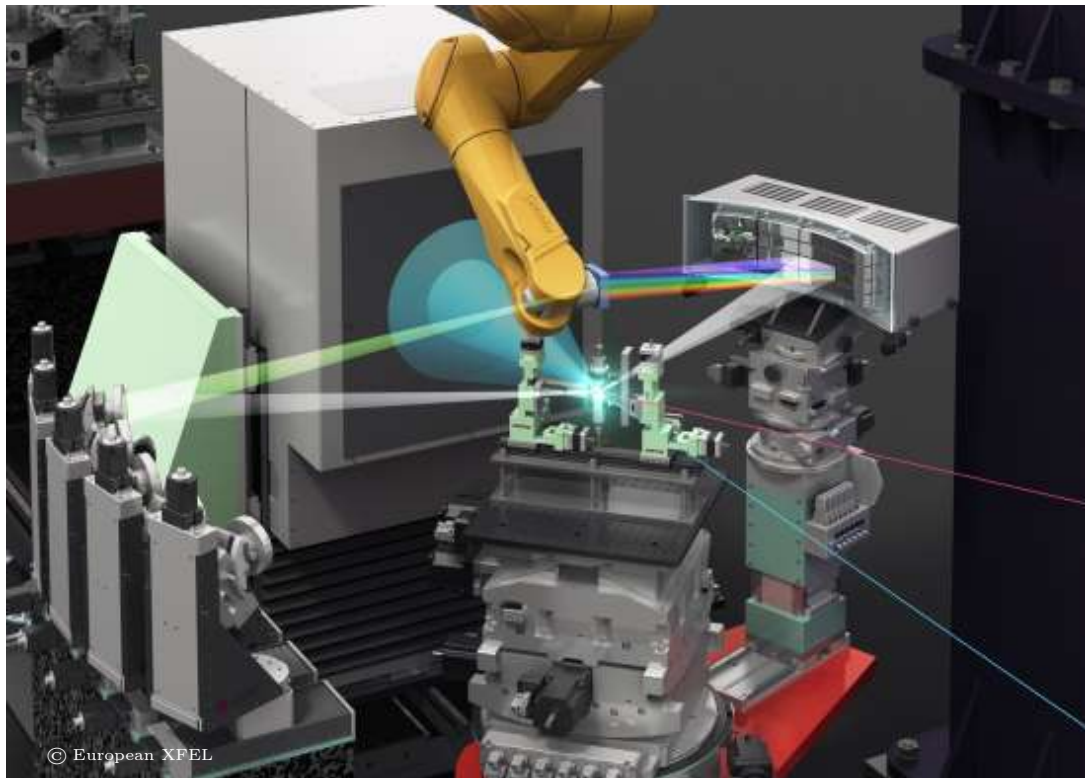


ADVANCED LAB PRACTICAL IN PHYSICS

Ultrafast X-Ray Experiments

Virtual Experiments at the X-Ray Laser European XFEL

Version 2.0 (July 2024)
Script in Progress



Credits

Conceived by C. Bressler, the software environment including the scientific and technical details of this virtual XFEL experiment were originally developed in close collaboration with the Southern Federal University Rostov/Don and its Spin-Off Company Univirlab GmbH until 02/2022 (Content Managers: A. Guda, M. Soldatov, D. Vlasenko; Programming: V. Popuzin, V. Myalkina; 3D-Visualization: G. Okoneshnikov; Scientific Advisor: A. V. Soldatov) and funded by European XFEL.



The script iterations including the first steps towards the present document were supported by European XFEL, the Hamburg Centre of Ultrafast Imaging, and funded by the Joachim Herz Foundation (2020/2021). Dr. Katharina Kubicek, Master student Kay Ebbesen and Dr. Alexander Britz were key players to this first version.



Further software improvements including a complete overhaul of the original program structure executed by Benjamin Stahl Softwareentwicklung, and the necessary overhaul of the present document by Kay Ebbesen, Dr. Alexander Britz and Christian Bressler, better match the increasing capabilities of the software environment, meanwhile called "VLab", and funded by Universität Hamburg under the Excellence Strategy of the Federal Government and the Länder.



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How to use this Script

The experiment can be fully executed in about five 6-hour days - a short version for only 3 lab practical days is also possible (see Chap. 5). This script should be used in the following way: Chapters 1 to 4 should be worked through and largely understood **prior** to the introduction discussion session at the beginning of this lab practical experiment (Sec. 1.3 delivers a very brief summary of the entire contents, this may be comprehensible only after working through the entire experiment). This concerns especially the several figures provided therein¹. Chapter 5 contains the detailed work program. The remaining chapter 6 and the annex provide additional information about this complex experiment and the basic functions of VLab. These can and should be used as a dictionary to look up certain details whenever needed.

After having worked through this experiment, the participants should have acquired the following abilities:

- A deeper understanding about the purpose, construction and required handling of an X-ray end station at a large scale facility (here: the FXE instrument at European XFEL).
- Basic hands-on knowledge about ultrafast pump-probe spectroscopy.
- Basic understanding of X-ray emission spectroscopy and related X-ray techniques.
- Self-organized skills about data acquisition strategies, especially in view of the required collection of saved data. Generating a suitable data file system including meaningfully filenames beforehand is extremely helpful.
- Structured data download, analysis and graphic display of extracted results (together with suitable fit functions) and meaningful interpretation.
- A deeper understanding about the physical chemistry of reacting molecules on the relevant femtosecond timescale.
- Properties of the investigated species after photoexcitation, especially their involved electronic states and relaxation channels, as is done in photophysical (and chemical) dynamics studies.

¹This especially applies to Figs. 2.3, 3.6, 4.2 and 4.3

1 An Overview

1.1 Synchrotron Radiation and Free Electron Lasers

Since the discovery of X-radiation by Wilhelm Conrad Röntgen in 1895, it has developed into an invaluable tool for the structural elucidation of matter. In addition to the “classical” laboratory-based X-ray sources, namely the X-ray tube with the rotating anode, *synchrotron radiation* (SR) was added from around the 1950s (Fig. 1.1). It is emitted tangentially by charged relativistic electrons or positrons when they are deflected from their straight flight path. SR was first observed in the late 1940s in a powerful class of ring accelerators, the *synchrotrons*,² and understood as being due to energy loss of accelerated particles. Many useful properties of this new type of light source quickly became apparent:

- A continuous spectrum (Fig. 1.2) from the far infrared to the range of hard X-radiation. This allows freely selecting an appropriate frequency window for various types of experiments.
- A very high intensity compared to all conventional X-radiation sources.
- A very good focus; this leads to a correspondingly high *brilliance*.
- SR is pulsed, with a pulse duration of about 100 ps.
- SR is theoretically[1] and experimentally very well understood.

Since the 1970s, most second-generation synchrotron facilities were no longer built for their original purpose, particle physics, but specifically as highly specialized light sources for (photon) research. Another milestone was the development of wigglers and especially *undulators* (Sec. 6.2.1). These assemblies branch off tangentially at short straight sections of a storage ring. They consist of alternately arranged dipole magnets and force the charged particles onto a transversally sinusoidal flight path. Each change of direction in the electron path produces radiation. This radiation is not only extremely intense, or highly brilliant³, but also somewhat coherent. There are now several dozen third-generation synchrotron facilities worldwide, and each one equipped with many points of use, the *beamlines*.

²The term synchrotron is composed of *synchronous* and *cyclotron*. These terms describe the interplay between acceleration of electrons to higher relativistic energies and the higher magnetic field strength required for constant curvature of the trajectory.

³The characteristic term *brilliance* (App. B) includes the focusability of the X-ray beam

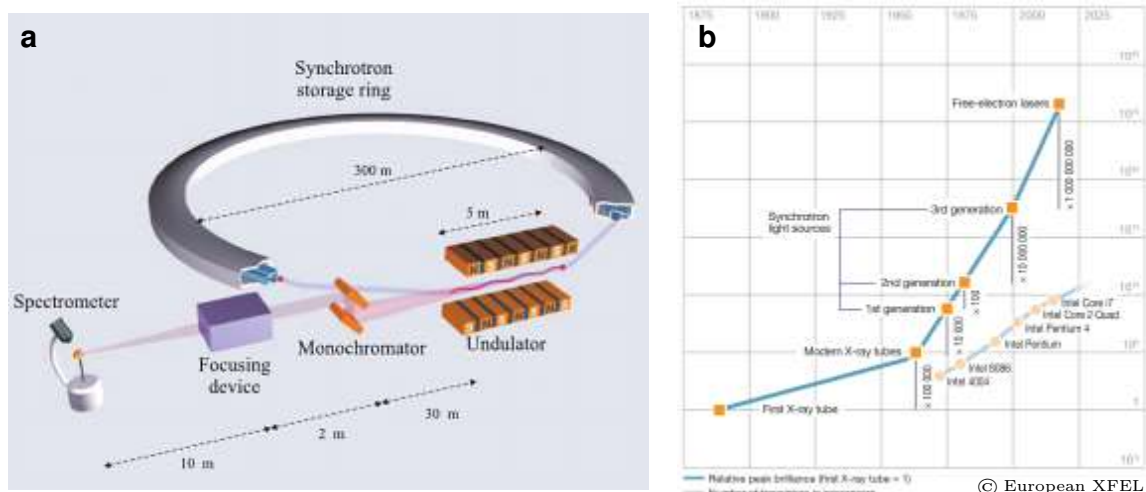


Figure 1.1: (a) Schematic structure of a third generation synchrotron with one exemplary X-ray beamline. (From [2]) (b) The development of artificial X-radiation sources since the beginning of the 20th century - from the first X-ray tubes to the FELs. The brilliance (on the y -axis in logarithmic units) marks their most important characteristic. The strong increase in brilliance since the mid-1960s resulted from the systematic development of better synchrotron radiation sources. For comparison: the progress in the performance of computer processors is significantly smaller.

Another giant leap in the intensity of X-radiation sources - by about four orders of magnitude in single pulse intensity and up to six orders of magnitude in brilliance (Fig. 1.1b) - with full lateral coherence is achieved with linear constructed *Free Electron Lasers* (FEL, Chap. 6). The duration of an X-ray pulse can also be reduced to a few femtoseconds (or even less). Powerful linear accelerators and much longer undulators are used for this purpose. These machines benefit from crucial progress compared to previous synchrotron sources: the *SASE* effect (*Self-amplified spontaneous emission*, Sec. 6.2.1) amplifies the emerging radiation much more than would be expected from the total undulator length. Over the long undulator path, there is an intensive interaction between the electron packets and the already generated accompanying radiation.

The present virtual experiment - *VLab* - realistically replicates the actual *FXE*-experiment (Sec. 6.1) of *European XFEL* in Hamburg/Schenefeld, and can thus be regarded as a *digital twin*⁴. To extract information about the system under investigation, essentially three X-ray methods are available: *X-ray scattering* (in forward direction) provides information about the geometric structure of the sample, including all molecules present inside. *X-ray absorption spectroscopy* (XAS, App. A) provides information about the local unoccupied electronic states and structural environment of a selected element. *X-ray emission spectroscopy* (XES, Chap. 4) shows the type of bound electronic and spin states. We will deal with the latter in this version of the *VLab*.⁵

⁴Original CAD design data were used for the 3D programming.

⁵Further virtual experiments are planned, including an already implemented ultra-fast scattering experiment.

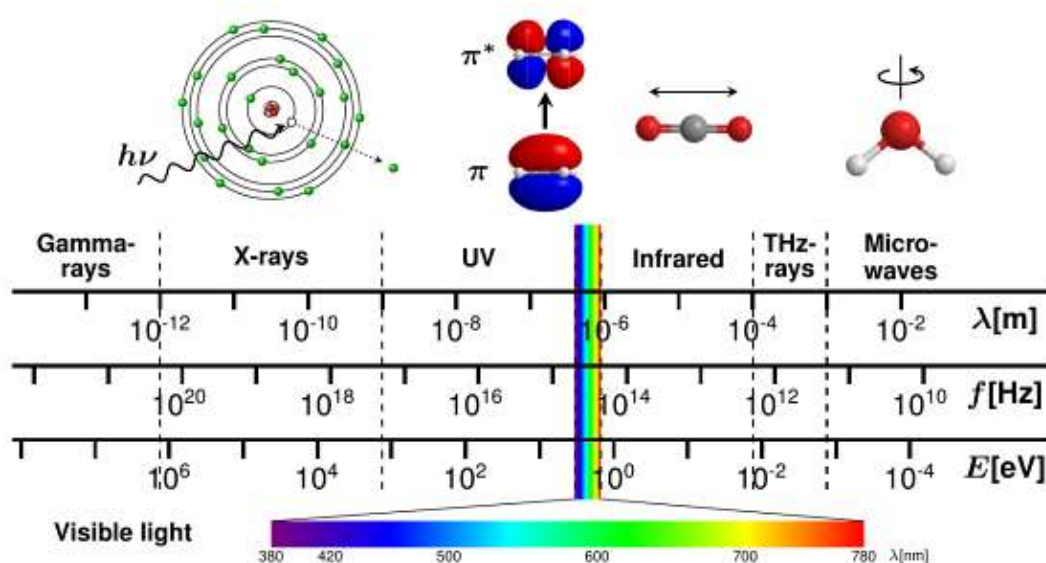


Figure 1.2: Electromagnetic spectrum with typical processes triggered by light absorption in the respective wavelength range: (core shell) electron excitations in the keV X-ray range, electronic transitions in the UV-VIS or eV range, molecular vibrations in the infrared and molecular rotations in the THz/microwave range.

At every beamline there is an *optical hutch* or *optical bench* (Sec. 6.4) upstream from the experiment to manipulate the X-ray beam, visualize it and diagnose its properties, depending on the requirements of the experiment. Important components are a *primary monochromator*, *attenuator units*, *focusing lenses* and/or *focusing mirrors* and *beam imaging cameras*. Experimental data from the sample can be spectrally decomposed with *secondary spectrometers* and recorded with detectors. We will deal intensively with these components during the course of this virtual experiment. You will learn to understand their function and to adjust your beam parameters very realistically.

1.2 Principle of Pump-Probe Experiments

Chemical reactions - bond breaking, formation and structural reorganization - vibrational motion, energy and charge transfer processes in molecules and solids typically occur on the timescale of a few femtoseconds (10^{-15} s) to a few hundred picoseconds (10^{-10} s). To spectroscopically capture the temporal evolution of such ongoing reactions including possible intermediate states, we need light pulses with a similar pulse duration. Optical lasers do this already since several decades; ultrafast X-ray sources are just these years available with the advent of FEL sources. This opens up the method of *Ultrafast X-Ray Spectroscopy* on a timescale down to about 10 fs to 50 fs.

This VLab experiment uses *pump-probe spectroscopy* (Fig. 1.3) in the X-ray probe domain. The system is first excited specifically, typically by an ultrashort optical laser pulse in the visible range or even in the ultraviolet or near infrared (Fig. 1.2). A time-delayed probe pulse - here an X-ray pulse - measures the system's response

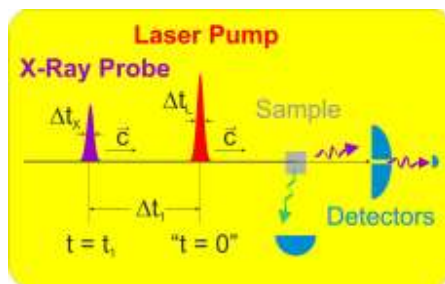


Figure 1.3: Principle of a pump-probe experiment. The time difference Δt_1 determines the excited state one will find. Various detectors can then measure different properties of this state. The measurement is then repeated at different time delays Δt between pump and probe pulses to create a “molecular movie” over the relevant time frame.

due to its ongoing reaction. The so-called *transient* signal results from the difference between the measured X-ray signal after laser excitation to the X-ray signal of the (unexcited) ground state. Specifically, the *transmission* and *emission* are recorded in X-ray spectroscopy. Variation of the pump-probe delay times then allows monitoring the dynamics of this light-triggered reaction and intermediate states. In this way, chemical reactions can be virtually “filmed”. By varying the pump excitation wavelengths, possible different processes can be investigated as well.

In this virtual experiment, X-ray pulses from the European XFEL laser are used to measure the reaction. We will often refer to “laser pulses” and “X-ray pulses”. By this, we mean light pulses from an optical laser - the pump pulse - as well as light pulses from the X-ray laser - the probe pulse - even though both light sources equally provide laser radiation.

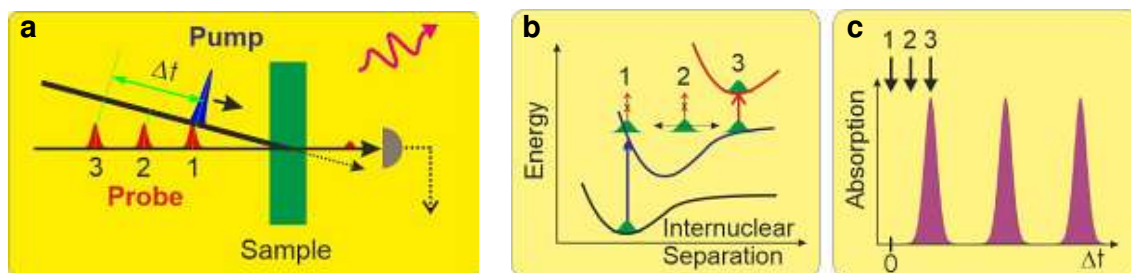


Figure 1.4: Example of a pump-probe experiment. **(a)** The pump pulse is accompanied by three probe pulses at different time delays, and the transmission of the probe pulses is measured. **(b)** In the energy level scheme, the pump pulse excites the molecule. This takes place at the left potential edge of the electronically excited state - according to the Franck-Condon principle. The subsequent movement - here vibration - of the molecule only allows absorption into the higher electronic state for time difference 3. **(c)** This is detected by the time-dependent absorption. The interpretation allows monitoring an ultrafast vibration and determine its precise vibrational frequency.

A Typical Example of a Pump-Probe Experiment

A molecule is placed into an electronically excited state by a pump pulse (Fig. 1.4a). This usually occurs at the boundary points of the potential curves (Franck-Condon

principle). Right after the excitation, the molecule begins to vibrate. Hereby, an atom swings within the molecule from the left turning point to the right (and then back again), with typical frequencies of about 10^{14} Hz. The probe pulse serves to excite that vibrating molecule into an even higher electronic state (Fig. 1.4b). In this example, this only succeeds at the right turning point, since the probe photon energy is not sufficient to reach the upper excited state for earlier times. Thus, the optical absorption requires a well-defined time delay (probe pulse No. 3 in Fig. 1.4). If the time difference between the pump and probe pulse is moved in small steps over a larger time range, this molecular vibration can be visualized by the measurement (Fig. 1.4c).

1.3 The XES Experiment in a Nutshell

We aim to measure a photophysical reaction of an iron complex in aqueous solution ($[\text{Fe}(\text{bipy})_3]^{2+}(\text{aq})$, Fig. 1.5) in a pump-probe experiment with femtosecond time resolution. With ultrashort pulses, molecular movements can be captured as almost “frozen” in time. The chosen method of X-ray emission spectroscopy (XES, Chap. 4) can additionally capture the momentary electronic structure and the spin state of the species - here, we measure locally the properties of the central Fe atom.

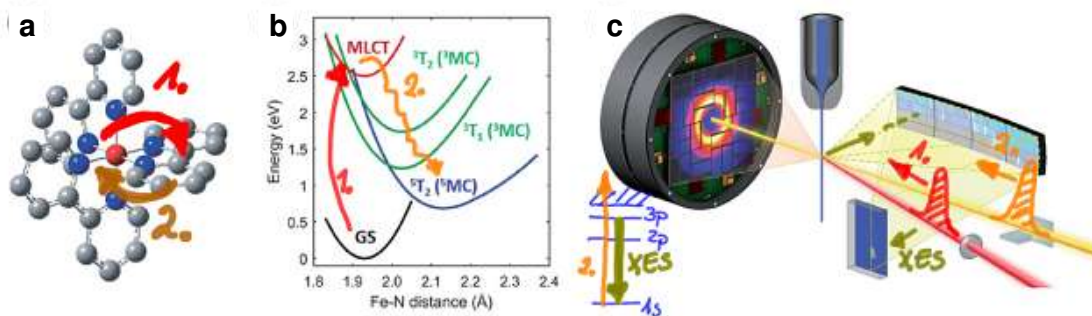


Figure 1.5: The virtual experiment, “in a nutshell”. (a) Structure of the investigated iron complex $[\text{Fe}(\text{bipy})_3]^{2+}$. An optical laser pulse stimulates an MLCT transition (1.) in $[\text{Fe}(\text{bipy})_3]^{2+}$. (b) The subsequent ultrafast relaxation up to a short-lived HS state is to be probed with an X-ray pulse (2.). (c) The two pulses have a different and variable time offset Δt . The current state in the term scheme (bottom left) is detected by means of K β XES (green arrows). (Adapted from [3])

An optical laser pulse starts the reaction, and a time-delayed X-ray pulse examines the current status at the time $\Delta t = t_{\text{X-ray}} - t_{\text{laser}}$ (Fig. 1.5c). This pump-probe sequence is repeated for different delay times Δt . From this, a sort of *molecular movie* of the ongoing reaction can be generated. Two experimental conditions are prerequisite for success:

- Both laser and X-ray pulses must have ultra-short (femtosecond) pulse durations.
- The time difference Δt must be known to within femtoseconds.

These conditions exist at the real European XFEL facility and are equally matched in VLab. The time delay can be measured and possibly corrected with femtosecond precision for each laser-X-ray pulse pair using the *arrival time detector* (Sec. 6.4.6).

The emission is dispersed in an X-ray spectrometer placed perpendicular to the X-ray beam, and its spectrum recorded on a 2D detector. At the same time, wide-angle scattering in the forward direction can be measured with another detector. The actual state in the term scheme (Fig. 1.5c and 3.6) can thus be monitored for each time delay Δt , and the transient emission spectra can be displayed for each time delay (Fig. 1.6a). Plotting of selected transient emission intensities at certain discrete energies provides *kinetic traces*. These represent the rise and decay behavior of certain states (Fig. 1.6b).

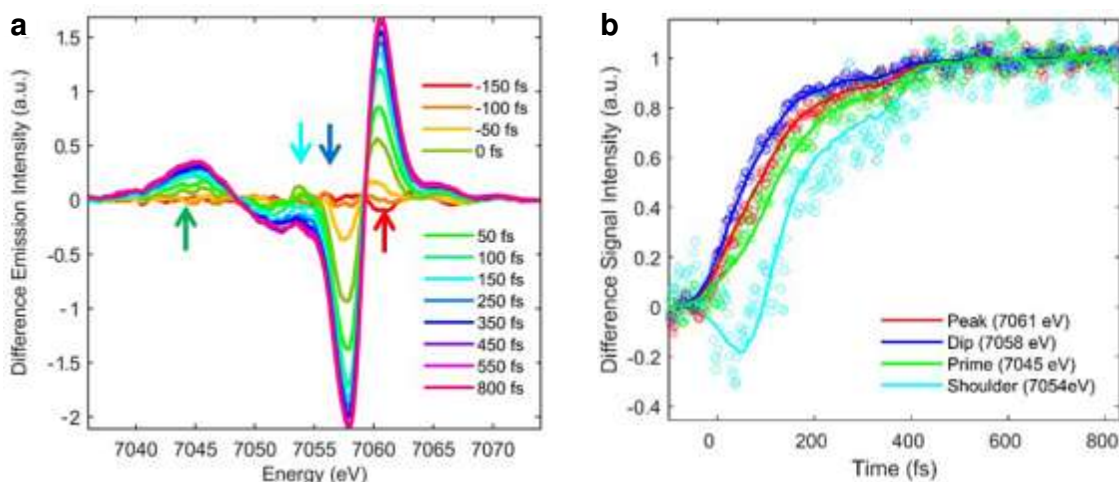


Figure 1.6: The expected measurement results. (a) Transient spectra of the $K\beta$ X-ray emission line at different pump-probe time delays. The 4 arrows show the energies of the kinetic traces plotted in b). (b) Kinetic traces at selected emission energies (vertical arrows in a). The $K\beta'$ line at 7045 eV (green trace) is particularly spin sensitive, and shows the transition from the LS ground over the MLCT manifold to the HS excited state. (Adapted from [3])

Now that we know how the experiment is conducted and how it can be interpreted, the underlying physical and chemical principles will be explained in more detail in the following three chapters.

2 The Investigated Molecule

2.1 Transition Metal Complexes

All chemical elements of groups III to XII in the periodic table, the transition or d block elements⁶, are metals. Hereby only a small energy difference separates the outermost s orbitals from the second outermost d orbitals (4s and 3d for the fourth period containing e.g., Fe, Co, Ni). The physical and chemical properties of transition elements are primarily determined by the d valence electrons⁷.

Due to the small energy gap between different valence orbitals, the ions of transition metals exist in several oxidation states. In the case of iron, Fe^{2+} and Fe^{3+} are particularly stable, but other oxidation states are also known. Transition metal ions are also classified based on the number of valence d electrons. For example, Fe^{2+} has a d^6 electron configuration with the orbital filling $[\text{Ar}] 3d^6 4s^0$, and Fe^{3+} has a d^5 configuration.

Transition metals have a strong tendency to form chemical *coordination compounds*, also known as *complex compounds*. These are molecules or ions with a neutral or charged metallic central atom (M), also referred to as the *coordination center*. Attached to it are neutral or charged *ligands* (L), consisting of mono- or polyatomic groups. Coordination compounds are denoted in square brackets for nomenclature, with the total charge written outside the brackets, for example: $[\text{Fe}(\text{bipy})_3]^{2+}$. The physical and chemical properties of complex compounds differ significantly from those of the isolated bonding partners. For example, the concentration of free cyanide ions in an aqueous solution of $\text{K}_4[\text{Fe}(\text{CN})_6]$ is extremely low, making this compound an approved food additive, despite the fact that free $(\text{CN})^-$ is highly toxic. The $[\text{Fe}(\text{CN})_6]^{4-}$ complex practically does not dissociate in water.

In complex compounds, the central atom usually acts as an electron acceptor, while the ligand acts as an electron donor. According to the *valence bond (vb) theory*, the bonding occurs through covalent overlap between an occupied ligand orbital and an unoccupied orbital of the central atom. Typically, only the ligand provides a lone pair of electrons, as opposed to a purely covalent bond in which each partner contributes one electron, as in the H_2 molecule, for example. Therefore, ligands

⁶According to IUPAC[4], transition elements are defined by partially filled d orbitals. Not all elements in the d-block belong to this category. However, both terms are often used interchangeably.

⁷Standard textbooks on inorganic or physical chemistry, such as [5, 6, 7], are used as references for this and the following chapters.

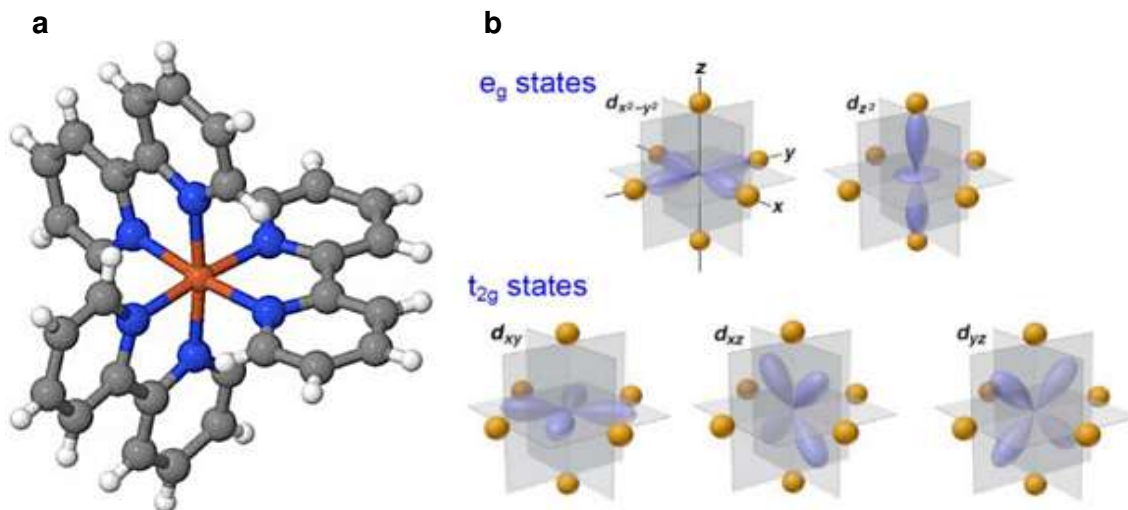


Figure 2.1: (a) Structure of the $[\text{Fe}(\text{bipy})_3]^{2+}$ complex (Fe: red, N: blue, C: gray, H: white). The six nitrogen atoms of the three bidentate bipyridine ligands are arranged at the vertices of an octahedron centered around the iron atom. Nitrogen contributes its lone pair of electrons for the coordinative bond. (b) The involved Fe d orbitals for ligand binding (b-f): The d_{z^2} and $d_{x^2-y^2}$ orbitals (b,c) point along the octahedral Cartesian axes containing the six N atoms (blue dots) with the lone pairs. This leads to an increased repulsive electrostatic interaction and consequently both d_{z^2} and $d_{x^2-y^2}$ orbital energies increase. The energy levels of the electrons in the other three d orbitals (d-f), reaching out between the axes, are lowered, preserving a mean energy for all five d orbitals (Fig. 2.3)

are characterized (mostly) by at least one lone pair of electrons, such as halide ions, oxygen atoms in water, carbon atoms in cyanide ligands, or nitrogen atoms in hetero aromatic pyridine rings. In addition to such cases with *monodentate ligands*, there are also numerous *multidentate* or *polydentate ligands* that contain more than one donor atom. For example, two bridged pyridine rings form the uncharged bidentate ligand bipyridine (specifically, 2,2'-bipyridine, abbreviated as bipy, Fig. 2.1a), and both nitrogen atoms can bind coordinative to a given central metal atom.

The *coordination number* denotes the total number of donor atoms in all binding ligands of the complex. Common coordination numbers are 2, 4, and especially 6. The spatial arrangement of the ligands, also known as the *coordination sphere*, is described by *coordination polyhedra*. These are mostly linear, tetrahedral, square planar, or octahedral, each with different *crystal field splitting patterns* (Sec. 2.2). Here, only octahedral - thus 6-fold coordinated - complexes will be discussed. The *oxidation state* is the (formal) charge of the metal center if all ligands, including shared electron pairs, were removed. For example, formal removal of 6 $(\text{CN})^-$ ions from $[\text{Fe}(\text{CN})_6]^{4-}$ or removal of 3 $(\text{bipy})^0$ ligands from $[\text{Fe}(\text{bipy})_3]^{2+}$ deliver a remaining iron(II) (Fe^{2+}).

Transition metal complexes play a significant role in nature, especially as active centers in enzymes. A well-known example is the oxygen-binding heme group in hemoglobin in red blood cells. An iron(II) ion is octahedrally coordinated, to a planar tetradentate porphyrin ligand via four nitrogen atoms, and proximal to the nitrogen of histamin. An oxygen molecule can coordinate on the remaining vacant

distal site (Fig. 2.2).

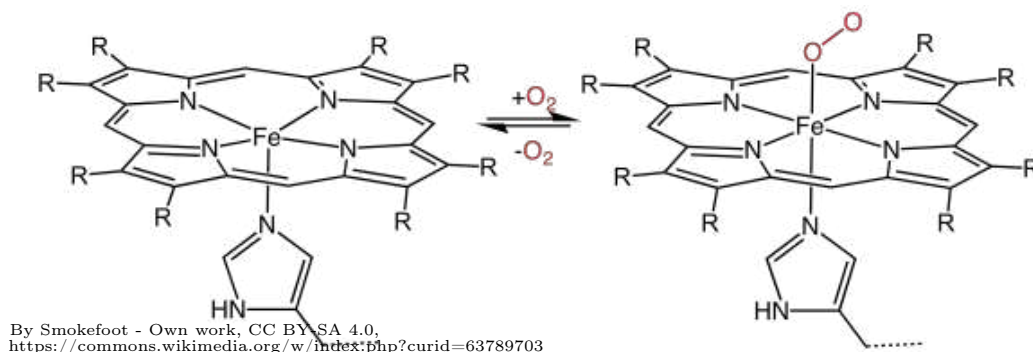


Figure 2.2: Reaction center for oxygen uptake and release inside hemoglobin. The Fe^{2+} ion is slightly too large to be precisely centered in the porphyrin ring and is pushed to approximately 0.04 nm below the plane (left). A bonded O_2 molecule draws electron density away from the iron, resulting in a formal loss of one electron. Subsequently, the iron changes his oxidation state from Fe(II) to Fe(III) , with a slightly smaller ionic radius, now fits into the center of the porphyrin (right). This leads to a decreasing Fe-N bond length. [8].

Coordination compounds are also important in inorganic, organic, and especially technical chemistry. One example is the ruthenium-based Grubbs catalyst (R.H. Grubbs, Nobel Prize in Chemistry, 2005) used in alkene metathesis in organic synthesis and petrochemistry. Complex compounds also have great potential for various technical applications, such as (enzymatic) catalysts, photochemical switches, color-sensitive photovoltaic cells, medicine (chelation therapy), and they are also used as color pigments.

2.2 Crystal Field Theory

However, the valence bond theory fails to provide a satisfactory explanation for the actual electronic state, particularly the excited state(s), as well as the intense coloration observed in most complexes. For example, $[\text{Fe}(\text{bipy})_3]^{2+}$ appears deep dark red, while $[\text{Ru}(\text{bipy})_3]^{2+}$ exhibits a bright fluorescent red color. Valence band theory also fails to explain the absorption spectra (Chap. 4 and App. A) and other properties, such as magnetic behavior, next to the preferred oxidation states and coordinations. To address these issues, the *crystal field theory* (along with its similar counterpart, the *ligand field theory*) was developed. In this theory, the ligands are considered as negative point charges. Their electrostatic interactions with the d electrons of the central atom explain the properties of the complex. No covalent overlap between metal and ligand orbitals is taken into account.

In the free metal ion in the gas phase, the five d orbitals are energetically equivalent (degenerate) and are occupied by electrons according to Hund's rule, maximizing spin multiplicity (Fig. 2.3). The influence of a (hypothetical) spherical ligand field increases the energy of these orbitals due to electrostatic repulsion. When the binding ligand atoms are positioned at specific locations, here at the vertices of an octahedron (in Fig. 2.1 thus along the Cartesian axes), the metallic d orbitals pointing

2. The Investigated Molecule

to the ligands are further energetically raised due to repulsive interactions. Those d orbitals not pointing towards the ligand atoms are energetically lowered against the spherical ligand energy. The resulting difference of energy levels is called the *strength of the crystal field*.⁸

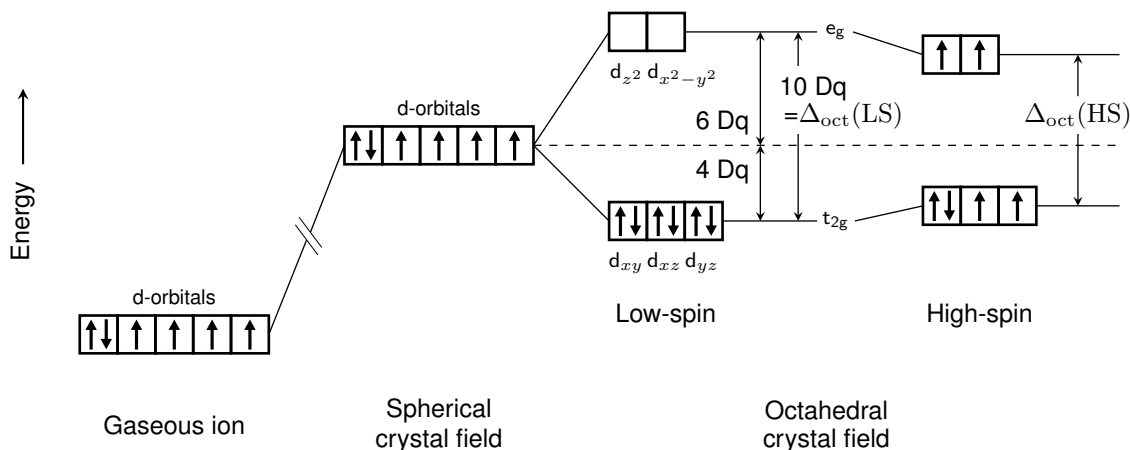


Figure 2.3: Energy level diagram of the d orbitals of a d^6 transition metal ion in an octahedral crystal field. An assumed spherical ligand field raises the energy of the d orbitals uniformly by approximately 20 eV to 40 eV. The actual octahedral field of the point-charged ligands removes the degeneracy of the d orbitals: the energy of those d orbitals that point directly at the ligands (e_g) is raised, while the energy of those that point between the ligands are lowered (t_{2g} , compare Fig. 2.1). The energy splitting Δ_{oct} is smaller for the HS configuration, since the electrons in the upper e_g level (located on the octahedral axes) push the ligand atoms further away than in the LS configuration, leading to a reduced octahedral field acting on the d orbitals.

This energy separation in the octahedral field is denoted by Δ_{oct} or $10 Dq$ and is typically in the range of 1 eV to 4 eV, while the energetic barycenter of the d orbitals remains unchanged. The energy of both energetically raised e_g orbitals, as denoted by symmetry considerations, is raised by $6 Dq$, while the energy of the three favorable t_{2g} orbitals is lowered by $4 Dq$. The absolute value of this energy differences can be experimentally determined through spectroscopy and depends on the charge and atomic number of the metal ion, as well as on the nature of the ligands. It increases with the charge of the central particle and its principal quantum number: $10 Dq(3d) < 10 Dq(4d) < 10 Dq(5d)$. Octahedral complexes formed by transition metals of the fifth and sixth period stabilize quasi exclusively in the low-spin form, even in a weak ligand field (see the next section).

⁸Other coordination geometries, such as tetrahedral coordination, result in different crystal field splitting patterns: in Fig. 2.3, the e_g and t_{2g} orbitals interchange their energy positions because the d_{z^2} and $d_{x^2-y^2}$ orbitals - creating the e_g orbitals - now pointing between the 4 ligand atoms, resulting in an energy-lowering effect, while the d_{xy} , d_{yz} , and d_{xz} orbitals interact repulsively and are thus energetically raised.

2.3 High-Spin, Low-Spin Complexes

When distributing electrons among the d orbitals in the ground state, the lower-energy t_{2g} orbitals are filled first one by one with electrons of the same spin, following Hund's rule. For octahedrally coordinated complexes with 1-3 and 8-10 d electrons at the metal center, there is only one possible electron configuration. For complexes with 4-7 electrons, there are two possibilities: In the *low-spin* (LS) state, the central ion has the minimum number of unpaired d electrons, while in the *high-spin* (HS) state, it has the maximum number. In the case of a d^6 metal ion, the LS state has no unpaired electron (Fig. 2.3). Thus, the total spin quantum number is $S = 0$ corresponding to a singlet state with the term symbol 1A_1 .⁹ In HS configuration the ion has four unpaired electrons, with $S = 2$, representing a quintet state with the term symbol 5T_2 .

The stability of these two states is determined by the magnitude of Δ_{oct} (strength of crystal field) and the spin pairing energy. If the spin pairing energy is greater than Δ_{oct} , the HS state is favored; otherwise, the LS state is favored. The *spectrochemical series* orders various ligands via their action on the central metal ion, and thus Δ_{oct} , from weak to strong. For some frequently encountered ligands, it is as follows:¹⁰



For example, $[Fe(H_2O)_6]^{2+}$ is a HS complex, while $[Fe(CN)_6]^{4-}$ is a LS complex.

A *Tanabe-Sugano diagram* is used to plot the energy differences of the excited electronic states of the central ion compared to the ground state as a function of the strength of the crystal field for transition metal complexes with various ligation. Thereby, energies of the ground state are taken to be zero for all field strength. For a d^6 metal ion, the initially higher-energy singlet state (LS) approaches the quintet state (HS) as Δ_{oct} increases. At a certain field strength, the relationship reverses, and the LS region begins, with the 1A_1 state becoming more stable and thus the ground state (Fig. 2.4, hint: imagine the spectrochemical series at the x-axis).

In some LS transition metal complexes (particularly those with Fe^{2+} , Fe^{3+} , and Co^{3+}), certain ligands result in a small value of Δ_{oct} , which could lead to excitation to the HS state even with low thermal energy input. This phenomenon, known as *spin crossover* (SCO, or thermal spin transition), can even lead to *spin equilibria*. A 1:1 HS-LS equilibrium occurs at a temperature $T_{1/2}$, depending on the difference in zero-point energies: $\Delta E_{\text{HL}}^0 = E_{\text{HS}}^0 - E_{\text{LS}}^0$ (Fig. 3.6). In the HS state, more electrons occupy antibonding orbitals, and less bonding orbitals, resulting in an elongation of the metal-ligand bond length $\Delta r_{\text{HL}} = r_{\text{HS}} - r_{\text{LS}}$. Therefore, both external pressure and crystal lattice structures of solid-state materials can influence the HS-LS ratio. Additionally, excitation to the HS state can occur through photoexcitation (as in this experimental setup), for example, via higher-lying MLCT (metal-to-ligand charge transfer, Sec. 3.2) bands followed by relaxation [9].

⁹Hint: Quantum numbers are variables and are therefore marked in italics (S), while term symbols are labeled roman (S , singlet state).^[4]

¹⁰Bidentate ligands: en = ethylenediamine, bipy = 2,2'-bipyridine, phen = phenanthroline

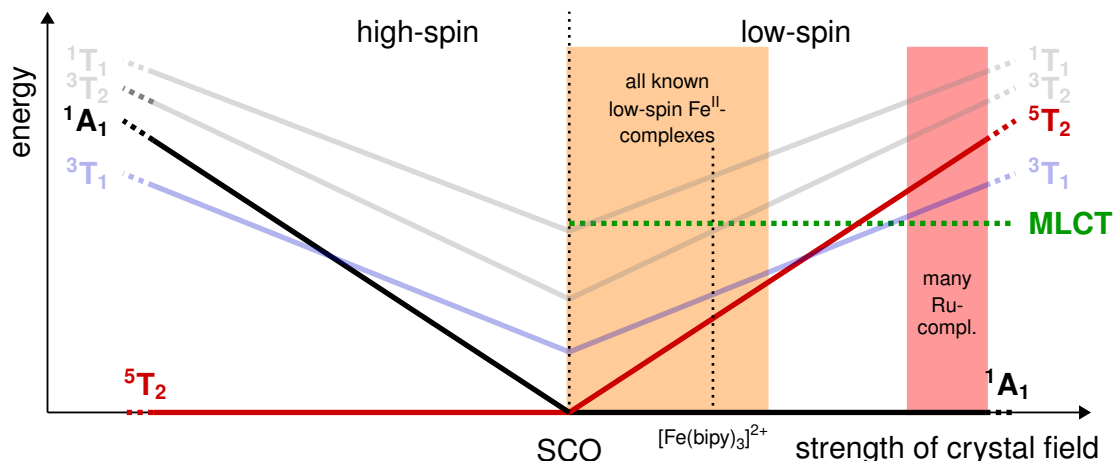


Figure 2.4: Excerpt of the Tanabe-Sugano diagram for octahedral d^6 complexes. In a weak ligand field, e.g., in $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, the HS state (red line) is the energetic lowest (ground) state, and in a strong ligand field, e.g., in $[\text{Fe}(\text{CN})_6]^{4-}$, the LS state (black line) is energetically lowest. Only the most relevant states are displayed. The approximate energy of the MLCT state (Sec. 3.2) of d^6 tris(bipyridine) complex compounds and the approximate strength of the crystal field of $[\text{Fe}(\text{bipy})_3]^{2+}$ are indicated by dashed lines. In the case of LS-Fe complexes, the energy of the HS state is lower than that of the MLCT, unlike for most Ru complexes. This provides a fast relaxation pathway into the HS excited state after photoexcitation into the MLCT state. (Adapted from [10, 11])

In this laboratory experiment, we aim to investigate the dynamics of electron transfer after photoexcitation of transition metal complexes in solution using experimental pump-probe techniques. The primary system of interest is $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ ¹¹ (Fig. 2.1a). The substance is highly water-soluble, stable in air, and relatively resistant to X-rays. It is well-suited as a calibration and test substance at synchrotron and free-electron laser facilities, and is currently used as a reference compound for measurements on other (iron) complexes.

Due to the fixed distance between the two nitrogen atoms (donors) of the bidentate ligands, the symmetry of the $[\text{Fe}(\text{bipy})_3]^{2+}$ complex is not exactly octahedral (O_h) but exhibits D_3 symmetry [10]. However, this deviation is small enough to be neglected hereafter.

At room temperature, in the electronic ground state, $[\text{Fe}(\text{bipy})_3]^{2+}$ is a LS complex. Photoexcitation to the MLCT state (Chap. 3.2) leads to the formation of the HS state. For a long time, the relaxation pathway from the MLCT state to the HS state remained unknown. But this ultrafast deactivation to the HS state has prevented the use of this class of complexes in photovoltaic applications. The relatively inexpensive-to-produce iron complexes have great potential to replace more expensive compounds based on rare metals, such as Ru. As a result, numerous theoretical and X-ray spectroscopic studies have been conducted to investigate the LS-HS con-

¹¹The correct IUPAC formula is $[\text{Fe}(\text{2,2'}\text{-bipyridine})_3]\text{Cl}_2$, named Tris(2,2'-bipyridine)iron(II)chloride. However, the commonly used abbreviated notation will be used here. In laboratory jargon, this compound is often referred to as irontrisbipyridin or simply ironbipy.

2. The Investigated Molecule

version mechanism after photoexcitation of iron complexes [9, 12, 13], with the goal to eventually stabilize newly synthesized Fe compounds in their MLCT states, as needed for energy applications. However, this compound is excellently suited for initial pump-probe experiments at synchrotron radiation sources and X-ray free-electron lasers, as provided in this virtual laboratory experiment. The results obtained in the experiment can be directly compared with published results.

3 The Pump Pulse: Interaction Between the Optical Laser and the Molecule

3.1 Fundamentals of Spectrophotometry

When a molecule or an atom absorbs a photon, its energy increases by promoting an electron into an excited energy level. Similarly, when a molecule emits a photon, its energy decreases accordingly. The type of excitation is dependent on photon energy E , and thus wavelength λ (Fig. 1.2). Absorption of microwave radiation causes molecules to rotate, while absorption of infrared radiation leads to molecular vibrations. Both of these happen within the same electronic state. Visible and ultraviolet light (UV/VIS) additionally cause excitation to another electronic state. These excitations in the single-digit eV range only affect the outermost valence electrons. Short-wave UV radiation and X-rays can excite more strongly bound electrons. Hard X-rays excite the electrons up to the innermost K-shell electrons (from the 1s orbital) (Sec. 4.2).

According to the *Lambert-Beer law*, the intensity I of a light beam decreases exponentially on its way through a sample, compared to its incident intensity I_0 (Fig. 3.1):

$$I(\lambda) = I_0(\lambda) \cdot e^{-N\sigma(\lambda)d} \quad (3-1a)$$

The wavelength dependent *absorption cross-section* $\sigma(\lambda)$ (in cm^2) is a measure of the probability of absorption by one molecule, N represents the number of sample molecules per unit volume, and d is the thickness of the illuminated sample. An equivalent expression is particularly preferred in chemistry and laser spectroscopy,

$$I(\lambda) = I_0(\lambda) \cdot 10^{-\epsilon(\lambda)cd} \quad (3-1b)$$

with the molar(!), decadic *attenuation coefficient* $\epsilon(\lambda)$ (in $\text{l mol}^{-1} \text{cm}^{-1}$) or *absorptivity*¹², and the *amount (of substance) concentration* c (in mol/l). The latter, which

¹²Strictly speaking, both terms are not identical. Attenuation takes scattering and luminescence processes into account beside pure absorption.

3. The Pump Pulse

is also known as *molarity* (obsolete)¹³, is defined as the *amount of sample species* n (in mol) per volume V :

$$c = \frac{n}{V} \quad (3-2)$$

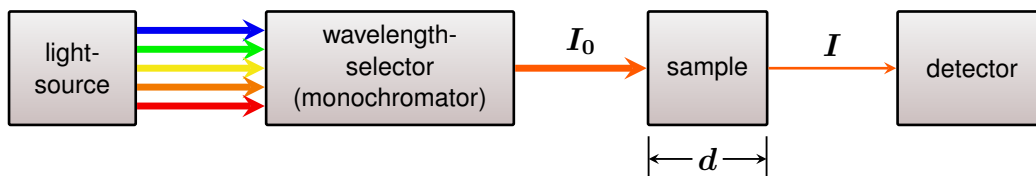


Figure 3.1: Schematic construction of a spectrophotometer. The broad-band incident light is dispersed by a monochromator, passing only a small bandwidth around the chosen wavelength to the sample. This absorbs a certain fraction of the incident light, and the transmitted intensity $I(\lambda)$ is recorded with a suitable detector. To obtain a full spectrum, the monochromator scans through the desired wavelength range. $I_0(\lambda)$ is either previously determined with an empty sample holder (cuvette), or recorded in parallel using a double beam spectrophotometer setup (not shown). (Adapted from [15]).

The *transmittance* T is the fraction of the originally incident light that passes through the sample:

$$T = \frac{I}{I_0} \quad (3-3)$$

The dimensionless *absorbance* A is defined as the logarithm of the ratio between irradiated and transmitted radiation. Its numerical value depends on the selected base, and is given by equation eq. (3-1a) via the natural logarithm,

$$A = -\ln T = \ln \left(\frac{I_0}{I} \right) = N\sigma d \quad (3-4)$$

while the *optical density*¹⁴ (OD) utilizes the decadic logarithm from eq. (3-1b):

$$A_{\text{OD}} = -\lg T = \lg \left(\frac{I_0}{I} \right) = \epsilon cd \quad (3-5)$$

Note that albeit their similarities their numerical values are not identical, so be careful when using either equation.

Spectroscopic Selection Rules

The selection rules (formally) constrains the possible electronic transitions of a system from one quantum state to another:

The *Laporte rule* applies to centrosymmetric wave functions, in both atoms and

¹³According to NIST[14], *molarity* and the unit M should not be used anymore, but only its correct unit mol l^{-1} . However, one can still find these obsolete terms frequently, e.g., ϵ in $\text{M}^{-1} \text{cm}^{-1}$.

¹⁴This term is obsolete as well[4], but is frequently used in daily laboratory jargon too.

molecules, i.e., species with an inversion center. It states that electronic transitions that conserve parity are (*dipole*) *forbidden*: $g \leftrightarrow g$ or $u \leftrightarrow u$. This leads to the selection rule: $\Delta l = \pm 1$. Allowed transitions are thus: $s \rightarrow p$, $p \rightarrow d$, $d \rightarrow f$; and forbidden are: e.g., $s \rightarrow s$, $p \rightarrow p$, $d \rightarrow d$, $s \rightarrow d$. However, molecular vibrations in the reactant molecule result in a temporary loss of centrosymmetry; this allows mixing of d with p orbitals, which can then deliver a small dipole-allowed contribution to the otherwise forbidden transition. Since the vibrational period ($\approx 10^{-13}$ s) is much longer than the electronic transition itself ($\approx 10^{-18}$ s), this mechanism, called *vibronic coupling*, introduces a weak attenuation coefficient: $\epsilon \approx 1 \text{ L mol}^{-1} \text{ cm}^{-1}$ to $10 \text{ L mol}^{-1} \text{ cm}^{-1}$ (if the transition is spin allowed) for centrosymmetric complexes, and up to $10^3 \text{ L mol}^{-1} \text{ cm}^{-1}$ for others.

The *spin selection rule*: $\Delta S = 0$. A change in spin multiplicity is forbidden, so transitions can occur e.g., from singlet to singlet, or from triplet to triplet, but not from singlet to triplet state (Fig. 3.2 shows two complexes, where these rules apply differently). However, such states can be mixed by *spin-orbit coupling* and spin-forbidden transitions become “allowed”. For first row transition elements (e.g. Mn, Fe, Co) the degree of mixing is small, and the absorption bands are therefore typically very weak ($\epsilon < 1 \text{ L mol}^{-1} \text{ cm}^{-1}$).



Figure 3.2: Examples for applying the spin selection rule: **a)** is a d^5 Mn(II) complex. A t_{2g} to e_g transition would change its spin multiplicity, and is therefore forbidden. **b)** is a d^7 Co(II) complex. Here, one-electron transitions are possible, which cause no change in spin multiplicity, and are therefore allowed. Note: H_2O is a weak ligand, thus both species are HS complexes (Sec. 2.3). They are octahedrally coordinated, and all possible transitions are Laporte-forbidden. (Adapted from [16]).

3.2 Optical Behavior of Transition Metal Complexes

Transition metal compounds are generally characterized by a distinctive and intense coloration.¹⁵ In order to understand this, as well as the UV/VIS absorption spectrum of $[\text{Fe}(\text{bipy})_3]^{2+}$ (Fig. 3.3), we compile the known excitations from low to high photon energies: First, there are the $d \rightarrow d$ transitions of electrons within the coordination center. They are dipole forbidden, but due to the vibronic coupling (Sec. 3.1), they are observed as very weak to moderately weak *metal-centered* (MC) features in the longer-wavelength infrared spectral range ($\lambda = 1000 \text{ nm}$ to 2500 nm , for this experiment we can ignore this contribution). Second, there are the important *charge-transfer* (CT) excitations, which causes electron transfer between

¹⁵In contrast, compounds of the main group elements typically absorb in the ultraviolet region and are thus often colorless in the visible region.

metallic center(s) and ligands. They are subject to other selection rules than those for $d \rightarrow d$ transitions. Therefore, the *CT bands* are usually moderately to highly intense and often occur in the visible to shorter-wavelength region ($\lambda < 350$ nm). Due to these intense absorptions, transition metal compounds with CT transitions are often used as color pigments, such as *Berliner Blau* ($[\text{Fe}^{\text{II}}\text{Fe}^{\text{III}}(\text{CN})_6]^-$) and the ancient *red lead* ($[\text{Pb}_2^{\text{II}}\text{Pb}^{\text{IV}}\text{O}_4]$) (both contain Metal-to-metal transitions, see further below).

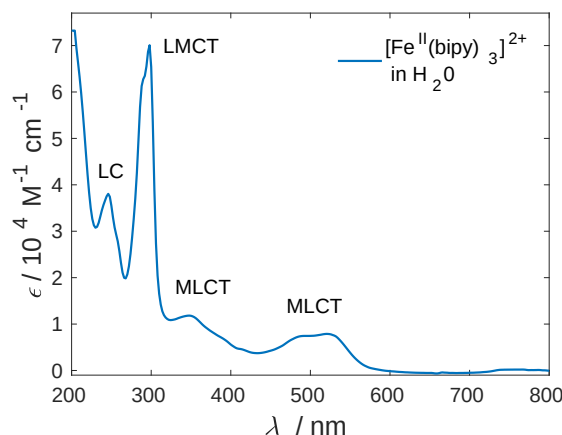


Figure 3.3: UV/VIS absorption spectrum of an aqueous $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ solution. The different CT transitions are indicated. The measured molar absorptivity ϵ values lie in the expected range of $10^{-4} \text{ l mol}^{-1} \text{ cm}^{-1}$ to $10^{-5} \text{ l mol}^{-1} \text{ cm}^{-1}$ for dipole-allowed transitions [11, 17].

A CT transition is an intramolecular charge transfer involving an electron donor and acceptor site and can therefore be seen equivalent to a redox reaction. These transitions can be described by the more complicated *molecular orbital theory*. Here, we just provide a brief overview (more details can be found in Appendix ??) to understand the UV/VIS absorption spectrum (and the deep red color) of our iron complex, its electronic excitation by the optical laser in the experiment, and the subsequent electronic and structural changes.

In molecular orbital theory, the linear combination of atomic (AO) and ligand orbitals with the same symmetry and similar energy leads to the formation of *molecular orbitals* (MOs) that can be delocalized over the entire molecule. These MOs can be bonding (b), non-bonding (nb), or antibonding (*) and predominantly possess for our iron complex metal (M) or ligand (L) character. The spatial part of the MOs is not characterized by the quantum numbers known from AOs, but rather by the point group symmetry in the molecule. Here, the rotationally symmetric (σ) and antisymmetric (π) MOs with respect to the bonding axis are relevant. The number of MOs is always equal to the sum of all AOs included in the treatment. The highest occupied MO is called the *HOMO* (Highest Occupied Molecular Orbital), and the lowest unoccupied MO is called the *LUMO* (Lowest Unoccupied Molecular Orbital). Additionally, by linear combination, *ligand group orbitals* (LGOs) can be formed. In the case of our octahedral complex, there are six nearly degenerate LGOs with three different symmetries ($1 \cdot a_{1g}$, $3 \cdot t_{1u}$, $2 \cdot e_g$).

3. The Pump Pulse

Generally, for CT transitions in complex compounds, four types of electronic excitations can be distinguished depending on the donor and acceptor. In an absorption spectrum, these electronic excitations appear as discrete and partly overlapping bands:

- *Metal-to-Ligand CT transition* (MLCT, Fig. 3.4): The metal center acts as electron donor, and the ligand as the acceptor. Photoabsorption promotes an electron from an MO with metal character, in our case a t_{2g} MO, which corresponds to the HOMO, to an unoccupied MO with ligand character, here an antibonding π^* MO of the aromatic system of the bipyridine ligand(!). Although this electron weakens the aromatic system, it does not lead to bond breaking. MLCT transitions are preferred in organometallic complexes (i.e., complexes with organic ligands) with aromatic systems and easily oxidizable metal centers (i.e., transition metals in their lower oxidation states).
- *Ligand-to-Metal CT transition* (LMCT): The electron is excited from the ligand (donor) to the metal (acceptor).
- *Metal-to-Metal CT transitions* (MMCT): These only occur in multi-metallic complexes.
- *Ligand-centered* (LC) or *Ligand-Ligand* (LL) transitions: Transition of a ligand electron to a higher energy ligand state.

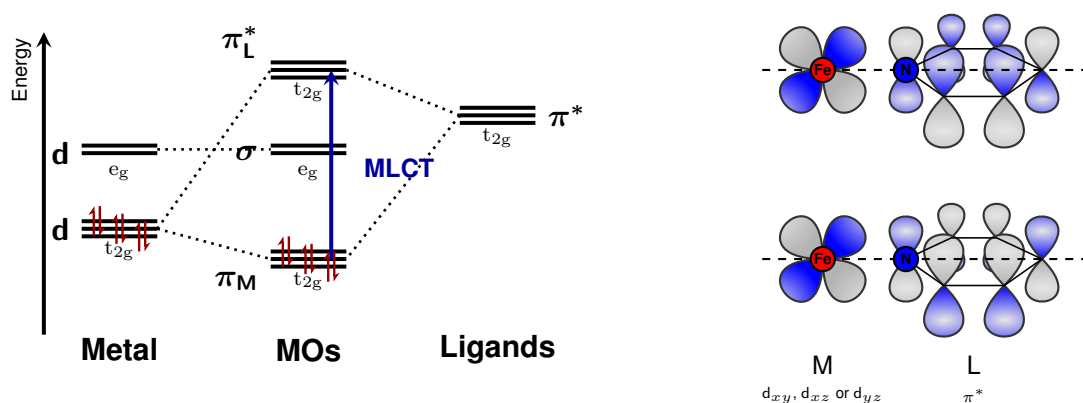


Figure 3.4: Simplified energy level diagram of an octahedral organometallic complex with a d^6 metal center and π -acceptor ligands (such as bipyridine) showing an MLCT transition. Here, only π interactions perpendicular to the Fe-N axis are considered (see app. ??). Excitation occurs from a binding t_{2g} MO (bottom right) with predominantly metal character to an antibonding MO (top right) with predominantly ligand character. Direct excitation to the e_g MO (high-spin) is forbidden according to the selection rules, but can be populated within a relaxation cascade (quasi “from above”).

3.3 Relaxation Processes

An excited molecule can deactivate its excited state energy in three different processes: (i) by emission of radiation, (ii) via non-radiative energy dissipation through rotation, translation, or vibration, or (iii) by inter- or intramolecular chemical reactions.

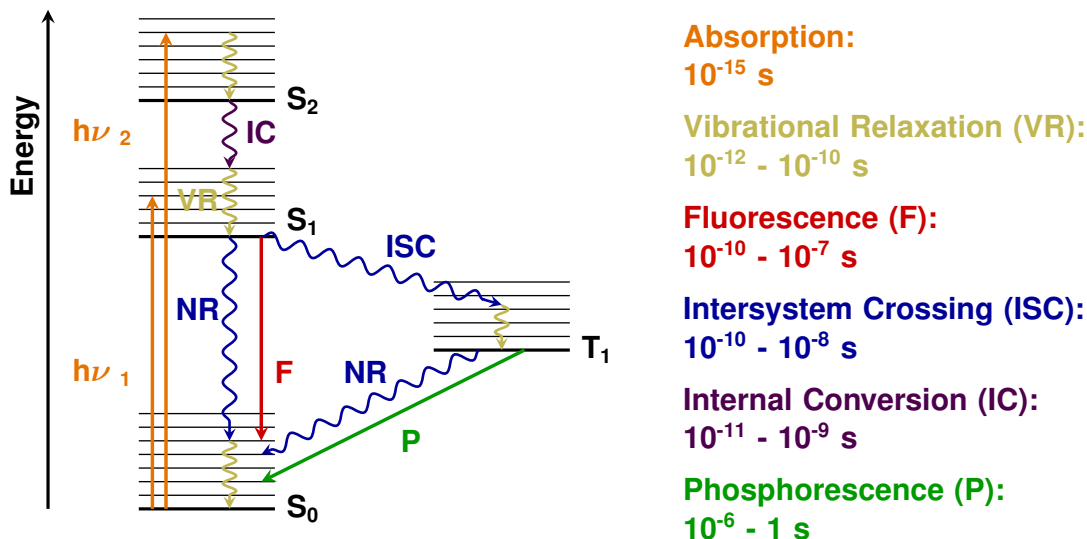


Figure 3.5: Jablonski diagram for ground and excited states in a molecular system. Each electronic state is indicated by a thick horizontal line (and labeled), its (ro)vibrational levels are depicted by the equidistant horizontal lines above. Also indicated are the relaxation processes following optical excitation (upward vertical arrows): by emission (downward straight arrows) and intramolecular non-radiative processes (wiggly arrows). The typical timescale for each relaxation process is given on the right. (Adapted from [10, 18])

The *Jablonski diagram* (Fig. 3.5) illustrates these intramolecular relaxation processes: Absorption of a photon with energy of either $h\nu_1$ or $h\nu_2$ leads to an electronic excitation from the singlet ground state, S_0 , to the excited singlet state, S_1 or S_2 , respectively. For excitation to a higher vibrational level of the electronically excited state, which is the typical case, relaxation usually occurs first through *vibrational relaxation* (VR) on a picosecond timescale. This involves energy transfer to surrounded atoms, which now vibrate instead of the originally excited vibration.

Subsequent deactivation processes can occur: *internal conversion* (IC) involves a quasi isoenergetic transition from the vibrational ground state of an electronically excited state (e.g., S_2) to an excited vibrational state of the next lower-lying electronic state (e.g., S_1) without changing its multiplicity, i.e., without spin state changes. In contrast, *intersystem crossing* (ISC) denotes the non-radiative transition to another electronically excited state with a different multiplicity, often achieved by a spin flip of one electron (here: singlet to triplet $S_1 \rightarrow T_1$). In states with identical nuclear configurations, spin-orbit coupling enables the ISC process.

The triplet state can act as a radiative reservoir: Since the optical transition to

3. The Pump Pulse

the ground state is spin-forbidden it is long-lived - up to microseconds and longer - which can lead to *phosphorescence* (P), the emission of radiation with a change in multiplicity. This distinguishes the process from *fluorescence* (F), the emission of radiation while maintaining the same multiplicity. These dipole-allowed transitions in the visible range always have lifetimes in the range of 10 ns to 100 ns. Various non-radiative deactivation processes, not further described here, are summarized as *non-radiative* (NR) processes in Figure 3.5.

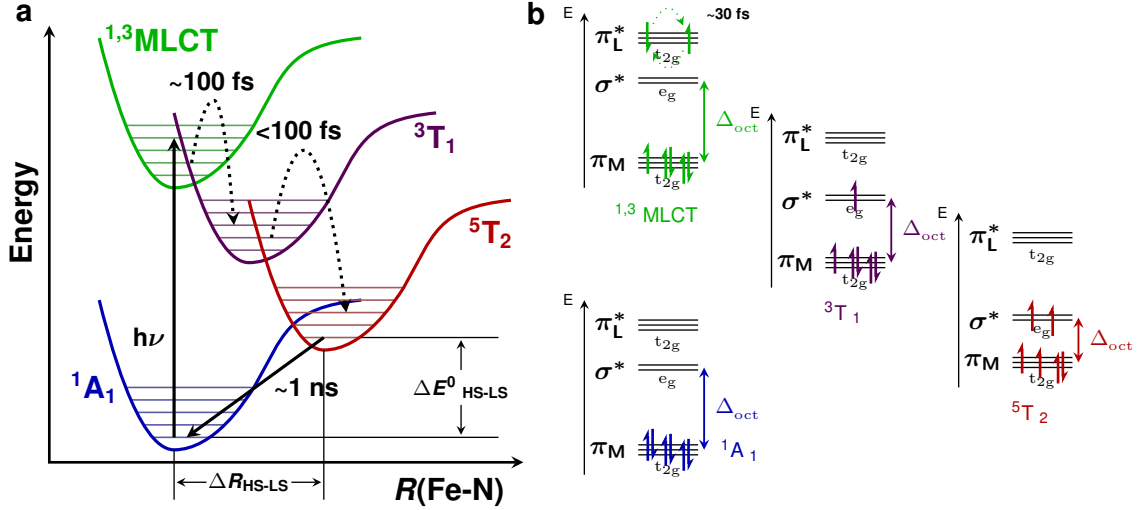


Figure 3.6: (a) Potential energy curves describing the excited state dynamics of $[\text{Fe}(\text{bipy})_3]^{2+}$ in H_2O following photoexcitation into its MLCT band. Relaxation channels and average lifetimes are given. Theoretically, there are two possible pathways for the relaxation from the $^1\text{MLCT}$ state: either via direct transition into the HS state ($^5\text{T}_{2g}$) or indirectly by passing over intermediate $^3\text{T}_{1g}$ transient states. (b) The corresponding energy level diagram of the states in (a) showing the electron occupancy on the central metal site. The strength of the crystal field Δ_{oct} decreases with the number of electrons in the antibinding σ^* -MO by a factor of about 1.75 [19] (i.e. from about 2.5 eV to 1.4 eV) leading to the increase of the Fe-N distance.

In the case of $[\text{Fe}(\text{bipy})_3]^{2+}$, direct excitation from the low-spin (LS) ground state to the high-spin (HS) state is spin-forbidden ($\Delta S = 2 \neq 0$). Therefore, in the experimental setup, the HS state is accessed via optical excitation by fs laser pulse at an appropriate wavelength to populate the $^1\text{MLCT}$ state. This is followed by a very fast (≤ 20 fs in $\text{H}_2\text{O}(\text{l})$ at room temperature) ISC to the $^3\text{MLCT}$ state. For the subsequent deactivation, two possible pathways exist (Fig. 3.6): direct deactivation to the HS state ($^5\text{T}_{2g}$) or indirect deactivation via the $^3\text{T}_{1g}$ transient state. The calculated potential energy surfaces (PES) for either pathway shows a minimal reaction barrier (Fig. 3.7), making both processes equally possible. Direct deactivation to the HS state requires simultaneous promotion of a second electron from the π_{M} (t_{2g}) orbital to the σ_{M}^* state (e_g). However, spin-orbit coupling primarily involves single-electron operators. In the case of indirect deactivation via the $^3\text{T}_{1g}$ transient state, each deactivation step involves only a single-electron transition, suggesting a two-step deactivation process via this transient state in this simplified picture.[12, 13, 20]

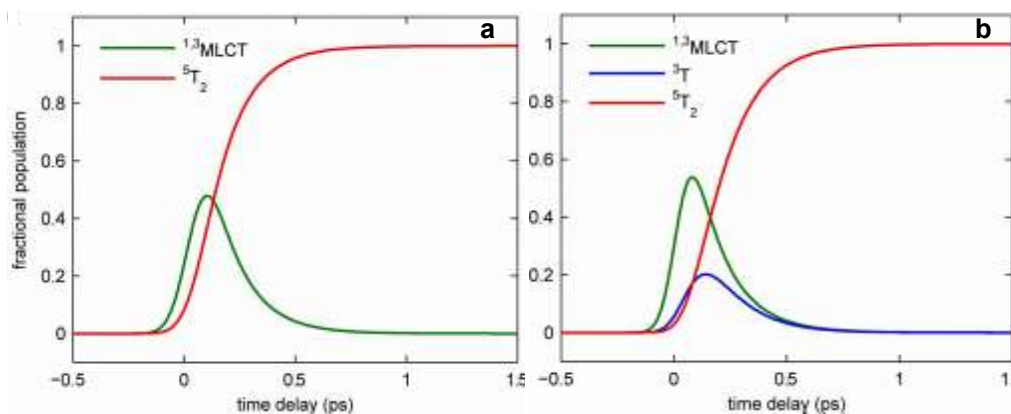


Figure 3.7: Plot of the temporal relaxation dynamics of $[\text{Fe}(\text{bipy})_3]^{2+}$ in H_2O after optical excitation (see Fig. 3.6), calculated with the rate constants using a time resolution of 120 fs. (a) For the direct $^{1,3}\text{MLCT} \rightarrow ^5\text{T}_2$ transition and (b) assuming passing through an intermediate triplet state within the relaxation cascade. (From [12])

One method to determine the actual state during an ongoing reaction is femtosecond time-resolved X-ray emission spectroscopy (Chap. 4), which is indeed a spin-sensitive method. This is the main focus of the experiment: use this method to investigate the ultrafast dynamics in $[\text{Fe}(\text{bipy})_3]^{2+}$ after its photoexcitation and elucidate the possible relaxation channels, including the accessed intermediate state(s): the short-lived $^{1,3}\text{MLCT}$, ^3T , and $^5\text{T}_2$.

4 The Probe Pulse: Interaction Between the X-ray and the Molecule

4.1 Why do we need X-Ray Pulses? Typical Energies

Short laser pulses in the UV/VIS range are much easier to generate than X-ray pulses, and they are available in many laboratories, including universities. UV/VIS laser pulses alone are readily used to study reaction dynamics of systems, including $[\text{Fe}(\text{bipy})_3]^{2+}$. So why should one consider a pump-probe experiment using X-rays from a large-scale research facility? The answer: X-rays offer additional observables that are inaccessible with optical radiation. These include the following properties: i) X-radiation can be used element-specific by choosing the precise binding energy of one of its core electrons, e.g., the Fe-1s electron in $[\text{Fe}(\text{bipy})_3]^{2+}$. The metal center and its local environment will be examined without being hampered by other elements in the sample, even in a disordered liquid. ii) X-ray emission spectroscopy is also spin-sensitive. The different molecular states can be reliably distinguished by the total spin state of the metal center. These two observables are usually not available with UV/VIS optical investigations, as these photons interact with valence orbitals stretching over several atoms, and overlapping bands make clear assignments challenging. A third unique ability of X-rays concerns their wavelength, which allows measuring atomic distances, e.g., via X-ray diffraction or EXAFS (App. A).¹⁶

As we show in the following, it is useful to selectively probe the iron 1s electron of $[\text{Fe}(\text{bipy})_3]^{2+}(\text{aq})$. This K shell electron has a binding energy of 7112 eV [21]. This is unique inside the solution sample, because all core shell binding energies of the elements in the second period are on the order of 10^2 eV to 10^3 eV (1s-electrons of C: 284 eV, N: 410 eV, O: 543 eV[21]). Beyond these *absorption edges* the absorption cross-section drops sufficiently with energy (Eq. (4-1) and Fig. 4.1), which then - in the hard X-ray region (ca. 5 keV to 20 keV) - allows probing selectively the iron center through bulk matter. If we had other elements besides iron with electrons of similar binding energies (such as chromium, copper, etc.), we can still address them individually by choosing the appropriate X-ray energy. This also applies to X-ray emission spectroscopy (XES, Sec 4.3), as the emission energies are also element-specific.

¹⁶Text books used for this chapter include, for example, [2, 22].

4.2 Element-specific X-Ray Absorption

For most elements between carbon and lead, in the broad range of about 0.3 keV to 100 keV, the photoelectric effect is the dominant process of X-ray-matter interactions (at higher energies other interactions, such as scattering processes and pair formations, become relevant as well). The atom absorbs the X-ray quantum $h\nu$ and is ionized by removing an electron from a deeply bound shell. The absorption cross-section σ increases strongly with the wavelength λ of the X-radiation and the nuclear charge number Z of the absorbing material (corresponding to more electrons being able to absorb X-rays). Combined with an element-specific constant C , or C' , the variation of σ can be described via

$$\sigma = C \cdot Z^4 \cdot \lambda^3 = C' \cdot Z^4 \cdot \frac{1}{E^3} \quad (4-1)$$

where E corresponds to the X-ray photon energy.

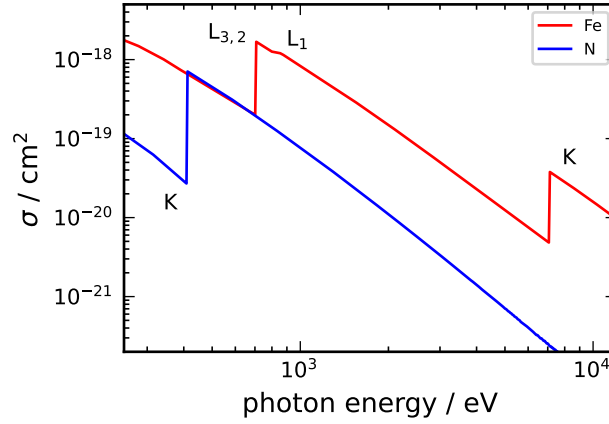


Figure 4.1: X-ray absorption cross-section of iron and nitrogen. It decreases with increasing photon energy, and shows discrete steps at the absorption edges. The Fe K edge absorption cross-section σ of about 10^{-20} cm^2 corresponds roughly to a radius of $0.01 \text{ \AA}$, which is about the size of a 1s orbitals in Fe. Data from [23].

If one would measure the absorption cross-section over a large wavelength range, one finds this monotonic decrease with energy, given by equation (4-1). However, at certain wavelengths, clear jumps occur, which are referred to as *absorption edges* at each critical energy E_{edge} (Fig. 4.1). Below this energy, only less tightly bound electrons from higher shells are excited, while at these absorption edges a new absorption channel opens up due to a now accessible deeper bound electron shell. This leads to an abrupt increase of the absorption cross-section. Above the E_{edge} , the absorption coefficient again falls according to equation (4-1).¹⁷

For the different inner shell electrons of all elements, their corresponding absorption edges appear at specific energies in the spectrum, and this also applies to

¹⁷This energy-dependent decrease in absorption cross-section with increasing energy is nicely covered by Fermi's Golden Rule, and yields the otherwise mystic appearing behavior given in eq. (4-1). Fermi's Golden Rule is derived in several atomic quantum physics text books.

their corresponding X-ray emission lines. This makes *X-ray absorption spectroscopy* (XAS) and *X-ray emission spectroscopy* (XES) a truly element-specific technique. The nomenclature for each absorption edge follows the historical shell designation: K-edge for 1s-electron removal, L_3 , L_2 , L_1 edge for removal of $2_{p3/2}$, $2_{p1/2}$ and 2_s electrons, respectively, and accordingly with M, N, O, etc., edges.

Looking into the details around a given absorption edge using *X-ray absorption fine structure spectroscopy* (XAFS) can provide information about the local electronic structure, as well as about the local geometric structure around a specific element up to approximal 6 Å to 10 Å. The present experiment does not use XAS, but to understand the subsequent emission process in XES, we have to consider the absorption process as well. More details on X-ray absorption spectroscopy are given in Appendix A.

4.3 X-ray Emission Spectroscopy

As we have seen, the absorption of an X-ray photon leads to excitation of a (core) electron, generating therefore a (inner shell) vacancy. This highly excited state decays by filling the core hole by a weaker bound electron from a higher shell, according to Fermi's Golden Rule. The excess energy is then removed by emission of one X-ray photon (possible XES lines are indicated in Fig. 4.2b). The energetic (or spectral) analysis of this emitted photon contains information about the higher shell energy level and also about the spin state of the atom. The emission of a photon must be accompanied by a change in the orbital angular momentum l of the atom by one. Thus, e.g., dipole transitions from the 3s or 2s to 1s, or from the 3d to 1s level are dipole forbidden (Sec. 3.1).

Besides this fluorescence process, there is a second dominant decay mechanism: nonradiative *Auger electron emission*, in which the excess energy is carried away by the ejection of another weaker bound electron from the same atom (not shown in Fig. 4.2). Both Auger and fluorescence processes occur in all atoms, however, in lighter elements the Auger process dominates, while in heavier elements X-ray emission dominates. Iron with its $Z = 26$ is entering the middle range: its total XES fluorescence yield Φ_f is about 30 %, and the Auger emission amounts to $\Phi_{\text{Aug}} \approx 70$ %. The latter process does not affect our experiment, but we have to keep it in mind, in regard to the overall radiative emission efficiency.

With respect to the excitation energy, we distinguish between two types of XES. If a core electron is excited with an energy near the absorption edge, it can be excited into empty levels (unoccupied orbitals) of the absorbing atom (Fig. 4.2a). This process influences the X-ray emission energy of *another* electron, e.g., the 3p electrons responsible for the $K\beta$ emission spectrum. This photon-in photon-out process is termed *resonant* XES. But if one probes the same occupied energy levels (e.g., the aforementioned 3p orbitals) of an atom without these additional effects, the core electron must be removed completely from the atom with higher X-ray energies into the continuum. Thus, the selected X-ray absorption energies for *non-resonant* XES have to be at least a few tens of electron volts above the selected absorption

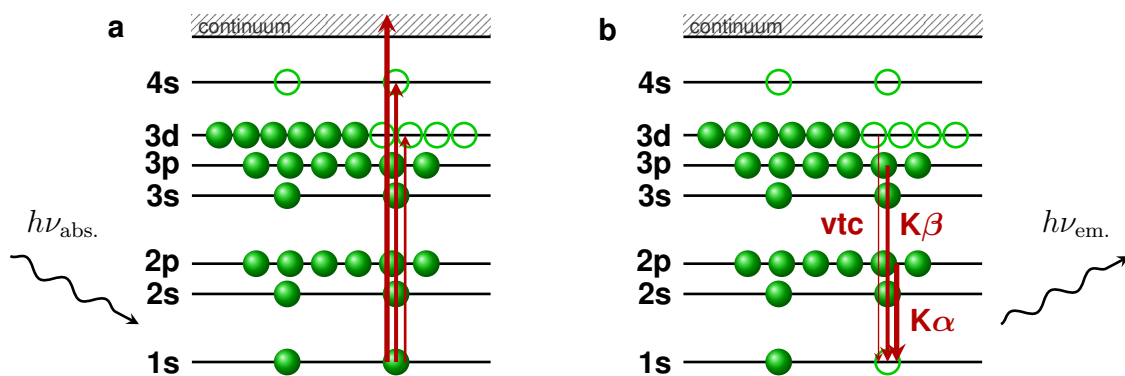


Figure 4.2: (a) Simplified energy diagram for the K-edge X-ray absorption and (b) the associated X-ray emission of a metal atom with $3d^6$ electron configuration. Emission extracted by exciting the core electron (into continuum) far above the absorption edge is called non-resonant XES, while emission extracted after exciting into bound levels of the absorbing atom is called resonant XES. The arrow thicknesses approximate the relative oscillator strength for each transition. (After [10])

edge, or even higher to avoid contributions to the absorption process by EXAFS (*extended X-ray absorption fine structure*, App. A.2).

The XES nomenclature is based on XAS lines (Sec. 4.2) - representing the vacancies - but denote all possible transitions by a subsequent Greek letter, respectively. The longest wavelength (smallest energy-difference) transition within a so-called *Siegbahn series* is named with an α , successive transitions are labelled β , γ , etc. The transition energy depends subtly on the orientation of the spin momentum vector with respect to the orbital angular momentum - “spin-orbit” interactions. These XES lines are further split and are labelled by the subscripts 1, 2, etc.¹⁸

Let’s consider the emission of an iron transition metal complex at X-ray energies above the K-edge of iron. The $2p \rightarrow 1s$ transition generates the strong $K\alpha_1$ and $K\alpha_2$ bands (Fig. 4.3) with $\Phi_{\text{H}} \approx 0.3$, comprising the spin-orbit split transitions $2p_{3/2} \rightarrow 1s$ and $2p_{1/2} \rightarrow 1s$ transitions. Since the $2p_{3/2}$ state is four-fold degenerate, and the $2p_{1/2}$ state two-fold degenerate, the $K\alpha_1$ line is twice as strong as the $K\alpha_2$ line.

The $K\beta$ emission spectrum originates from the different $3p \rightarrow 1s$ transitions and is about an order of magnitude weaker. Next to the stronger $K\beta_{1,3}$ peak is a weak shoulder in the spectrum, a so-called *satellite peak*. This $K\beta'$ line is made possible by exchange interactions between 3d electrons and the 3p vacancy, and is therefore highly spin-sensitive (see next section).

The weakest $K\beta_{2,5}$ and $K\beta''$ emission lines originate from $3d \rightarrow 1s$ transitions and are therefore also referred to as *valence-to-core* (vtc). They are the Fe X-ray emission lines with the highest energies, contain information about the bonding conditions

¹⁸The IUPAC notation is more logical and simple: the filled state is followed by the source state. Therefore, the $K\alpha_1$ line is called the $K-L_3$ and the $K\alpha_2$ is $K-L_2$.^[22] However, this notation is not widely used.

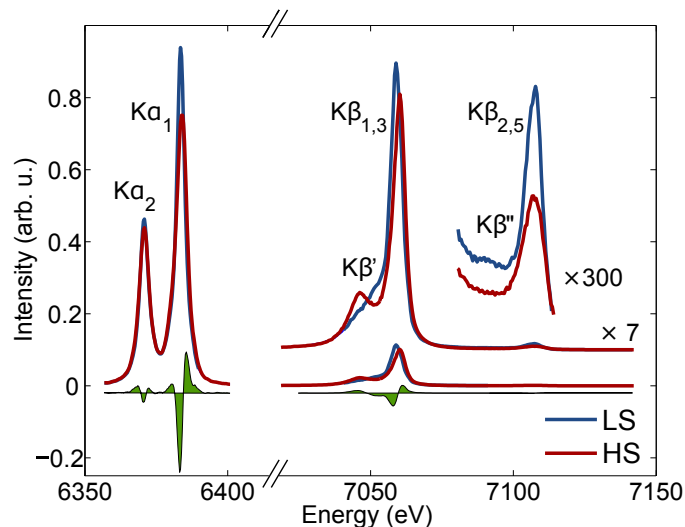


Figure 4.3: X-ray emission spectrum of a Fe^{2+} complex in its LS (blue) and HS (red) states. Shown are the $\text{K}\alpha$, $\text{K}\beta$ and valence-to-core spectra with their corresponding HS-LS difference spectra (green). The relative intensities correspond to the overlap integrals (from Fermi's Golden Rule) from each of these orbitals into the $1s$ state: $\text{K}\alpha$ is about 7-10 times stronger than $\text{K}\beta$, and about 500 times stronger than the $\text{K}\beta_{2,5}$ lines. (From [24, 25])

in the complex, and can supplement the structural information from the X-ray absorption spectroscopy.

4.4 Spin Sensitivity of $\text{K}\alpha$ and $\text{K}\beta$ Lines

As mentioned earlier, the $\text{K}\beta'$ line near 7046 eV is spin-sensitive. It is not visible for Fe^{2+} in its singlet spin state (Fig. 3.4), but it grows in with increasing spin multiplicity of iron (Fig. 4.4). Therefore, this line is well suited for distinguishing different electronic spin states of the metal center. As can be seen in Figure 4.3, the $\text{K}\alpha$ line also shows strong differences between LS and HS. However, these consist mainly of a broadening and shift of the lines, which may also be caused by other reasons (e.g., oxidation state changes also shift XES lines) than just the spin state change. Therefore, we prefer $\text{K}\beta$ -XES for determining the spin state.

The formation of a new $\text{K}\beta'$ feature near 7046 eV (Fig. 4.3) in $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ with increasing spin multiplicity will be used, also in a time-dependent manner. We will observe how the system, after photoexcitation from the LS ground state into the $^1\text{MLCT}$ state, releases its excess energy through relaxation and finally arrives into the metastable HS state (Fig. 3.6). With our pump probe setup (Sec. 1.2 and 1.3), we can also determine rise and decay times of the involved states.

4.5 The von Hamos Spectrometer

The X-ray emission is radiated isotropic from the sample. To generate an energy-dispersive emission spectrum on a detector, a spectral decomposition of the emitted

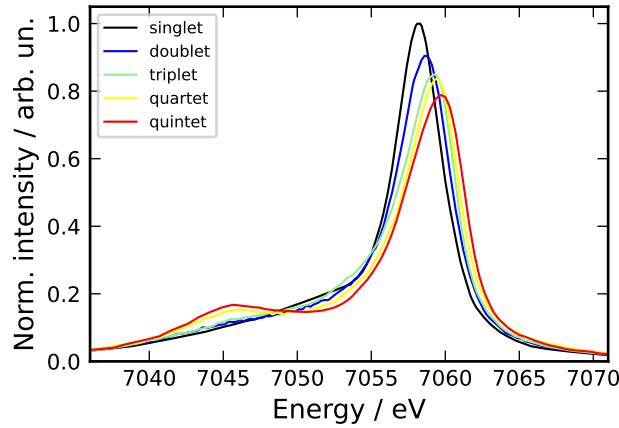


Figure 4.4: $K\beta$ fluorescence spectra of different Fe complexes, each exhibiting a different multiplicity in its ground state. Singlet: $\text{Fe}^{\text{II}}(\text{bipy})_3$, Doublet: $\text{Fe}^{\text{III}}(\text{bipy})_3$, Triplet: $\text{Fe}^{\text{II}}\text{Pc}$, Quartet: $\text{Fe}^{\text{III}}\text{Pc}$, Quintet: $\text{Fe}(\text{phen})_2(\text{NCS})_2$. Pc = Phthalocyanine, phen = Phenanthroline, NCS = Thiocyanate. The spin dependence of the satellite peak at about 7046 eV is clearly visible. This allows, in particular, good differentiation of the HS from the LS state. (Data from [12])

light is required, usually via an X-ray emission spectrometer. A spectrometer in the hard X-ray range is based on Bragg-diffraction by crystals, in the soft (< 2 keV) by diffraction from a grating. The diffraction of light on crystals is described by the Bragg equation:

$$n\lambda = 2d \sin(\theta) \quad (4-2)$$

Here, λ is the wavelength of incident light, θ is the diffraction angle (measured from the crystal plane to the X-ray beam), d is the chosen lattice plane spacing of the diffracting crystal, and n represents the order of diffraction.

In addition to the (isotropic) X-ray emission, there are also elastic scattering effects. However, the intensity of the scattered light depends on its direction from the atom and is smallest horizontally perpendicular to the X-ray beam.¹⁹ Therefore, the arrangement of an X-ray emission spectrometer in this spatial direction makes sense.

In the von Hamos geometry, cylindrically curved crystals are arranged so that they form with the source and the detector a plane which is perpendicular to the X-ray beam. The dispersion takes place along the crystal (Fig. 4.5): the different angles, as seen from the source to a position on the crystal, fulfill Bragg's law only for specific x-ray wavelengths, while other wavelengths emitted along the same angle will only be absorbed. The perpendicular circular curvature focuses the reflected radiation along the non-dispersive plane, analogous to a focusing mirror in the optical range. Thus, it increases the solid angle for detection. With such a fixed setup (no scanning required), isotropic emitted photons of different energies are partially reflected for each Bragg angle condition and gathered along a line on the 2D detector where different energies correspond to different positions on this line (dispersive spectrom-

¹⁹This angle actually fulfills the Brewster condition for horizontally polarized X-rays, which then do not scatter along the polarization axis.

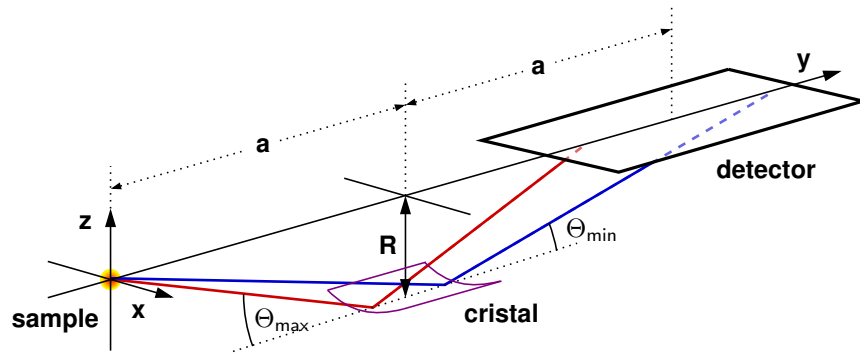


Figure 4.5: The geometry of the energy-dispersive von Hamos spectrometer. The X-ray beam defines the z axis in the beam direction, the y axis is vertical in VLab. The dispersion takes place on cylindrically curved crystals according to the Bragg condition. The curvature, with the radius R , thereby focuses the reflected radiation orthogonally to the dispersion direction. From geometric considerations and the distance $2a$ between sample and detector, a pixel-energy relationship can be established. (After [26])

eter type). Instead of curved crystals, several flat segments can also be used in a cylindrical arrangement.

5 Execution of the Experiment

Preliminary remark: This virtual experiment is designed for a duration of five days, in addition to the preparation time needed to create an experimental protocol. A “shortened version” requiring three experimental days is given by omitting all the grayed-out tasks in this chapter. However, this eliminates some of the main tasks, which are equally essential to understand and set up the optical path, a usual task at “real” beamlines. One of the main didactic goals of this virtual experiment is to precisely familiarize the user with the many complex components at a beamline. We thus recommend executing the long version, but at least attempt to understand and possibly briefly work on the grayed-out tasks.

Now we start doing this experiment. For this, we follow the usual approach when performing a real beamline experiment, step by step. Ultimately, the goal is to record (non-resonant) time-resolved Fe $K\beta_{1,3}$ X-ray emission spectra (XES, Sec. 4.3) after optical laser excitation of iron trisbipyridine, $[\text{Fe}(\text{bipy})_3]^{2+}$, in aqueous solution (Chap. 2). The extracted data should be analyzed, and the results interpreted in a meaningful photophysical context.

To do this, it is first necessary to measure and adjust the energy and spectrum of the incident X-ray beam for the actual experiment. Next, the X-ray beam is guided through the beamline, and you should obtain a beam on sample with useful properties, e.g., an optimum beam size. Plan approximate 1-1.5 days for these steps (Sec. 5.1).

Next, estimations and calculations of the optimal experimental parameters for the liquid jet, the substance dissolved in it, and its interaction with the X-ray probe beam is performed. Further estimations regarding the exciting optical laser beam (Chap. 3) are required. The X-ray emission spectroscopy setup, when correctly aligned, allows you to extract virtual experimental data delivering a suitable time-resolved signal. This requires about 1 day (Sec. 5.2).

Finally, after these preparative steps, the actual time-resolved X-ray experiment can be performed, which requires you to vary different parameters in a meaningful fashion. Subsequently, the measured data is extracted, evaluated, analyzed and interpreted. Plan approximate 2 days for this (Sec. 5.3 and 5.4).

When working on your tasks, please document your individual steps carefully, using suitable screenshots and writing down key parameters. Save the relevant data and visualize them with suitable software, e.g., Origin, Matlab or Python/Jupyter Notebooks. Origin and Matlab are provided free of charge to students of the University of Hamburg. If you want to use a

Python/Jupyter Notebook, we recommend the free Anaconda development environment (see [27] for installation instructions).

In any case, you should be able to display your extracted data graphically, to normalize data and to perform other operations including fits to mathematical functions. Displaying 2D- and 3D-graphs with false color intensities should be possible with your chosen plot software.

Now start the downloaded VLab software for your own X-ray lab beamtime and familiarize yourself with the environment. Have fun experimenting!

5.1 Alignment and Characterization of the X-ray Beam

1. **Select a suitable nominal X-ray energy E_0 for the XFEL radiation source in VLab by setting it in the undulator (Sec. 6.2.1).**

Keep in mind that you will perform an experiment using non-resonant X-ray emission spectroscopy (non-resonant XES) of the Fe $K\beta_{1,3}$ line. Consider two factors when making your choice: a) The required energy to access the absorption edge of atomic iron, which you can look up, for example, in the Database at the Center for X-Ray Optics (CXRO[23]). b) The EXAFS spectrum of the molecule (Fig. A.1).

Justify your choice with respect to a) and b). Also plot the transmission spectrum through iron from below the edge to above your chosen energy using the CXRO data.

2. **Confirm that you have a X-ray beam at all.**

The best device for this purpose is the beam imaging unit 2 (BIU2, Sec 6.4.1). To avoid saturation of the BIU camera detector, familiarize yourself with the operation of the beam attenuators (SAA, Sec. 6.4.5). Before doing so, all power slits (PS, Sec. 6.4.4) may be fully opened. Try different attenuator foils until the BIU2 camera is at maximum just below saturation.

Determine the size and position of the X-ray beam on the BIU2.

To do this, plot the horizontal and vertical profile width of the beam as a graph and determine the respective full width at half maximum (FWHM, App. B). Afterward, download about 20 single images (20 different X-ray pulses) and determine the horizontal and vertical position (x, y) of the maximum for each pulse. Then calculate the overall mean position and the standard deviation ($\pm 1SD$), where the latter is a measure of the spatial jitter of every X-ray pulse as seen on BIU2. Discuss your results in terms of relevance for the actual experiment (e.g., is the experiment feasible or not under these fluctuating conditions?).

3. The X-ray beam exhibits a broader energy distribution of ΔE_{SASE} around E_0 , the SASE spectrum. It is also called the “pink” X-ray beam, because it is neither truly monochromatic nor as broadband. By inserting the four-bounce crystal-monochromator (Sec. 6.3.2) into the beam path, the spectral bandwidth can be significantly reduced to ΔE_{Mono} .

Go to the monochromator and observe i) the pink and ii) the monochromatic X-ray beam on the BIU2. Document the change in intensity of

the beam in your lab book, using screenshots and 3D-plotted image data.

4. **Record the spectrum of the pink X-ray beam with the dispersive single-shot spectrum analyzer (SpA1, Sec. 6.4.3).**

Plot a) three single-shot spectra of an X-ray pulse and b) an averaged spectrum over many pulses. Indicate the spectral bandwidth (FWHM in eV) of the pink X-ray beam.

Note: Move the SpA1 crystal into the X-ray beam and adjust the detector arm to the appropriate Bragg angle. An adjustment guide is provided in the SpA1 subroutine for this purpose.

Compare the VLab spectrum to that of European XFEL (Fig. 6.15).

5. Repeat the previous task for the monochromatic X-ray beam. Discuss the differences in the spectra itself and the spectral bandwidth. How does this compare to your previous observation of the intensity decrease from the pink to the monochromatic X-ray beam in ex. 2
6. **Discuss whether a pink or monochromatic beam would be better suited for a non-resonant X-ray emission spectroscopy (XES) experiment of the Fe $K\beta_{1,3}$ line.**

Note: Keep in mind your results from the previous tasks and review the principle of non-resonant X-ray emission (Fig. 4.2).

7. The XFEL generates a pulse train of many successive X-ray pulses which fluctuate in their position in the beam tube and in their intensity (Sec. 6.2.2). **Characterize the changes in a) spatial position and b) intensity for different X-ray pulses. Use the signals from the four diodes of the intensity and position monitor (IPM, Sec. 6.4.2) for this purpose.**

Consider how you can obtain information from this signal about the changes in spatial position for the different X-ray pulses. Plot these in the lab book. The distance from the scattering foil to the intersection of the plane formed by the 4 diodes with the X-ray beam axis is 40 mm. The distance from the center point of the diode to its opposite diode is also 40 mm.

Note: Consider the horizontal and vertical position changes separately.

8. **Determine the transmission of a diamond attenuator foil in the beam attenuator unit (SAA, Sec. 6.4.5).**

To do this, measure the change in intensity of the X-ray beam on the BIU2 after inserting an additional 0.8 mm thick diamond foil (C) in the SAA. Repeat this for two additional undulator energies E_0 .

Compare your results with the expected transmission for such a foil according to CXRO.

9. **Focus the X-ray beam with the appropriate set of refractive beryllium X-ray lenses (CRL, Sec. 6.4.7) and determine the X-ray beam size at the sample location using the beam imaging unit “X-ray Eye”.**

Note: Each of the 10 CRL lens holders is individually equipped with up to several lenses (Tab. 6.1). In general, the lens stacks closer to the sample (the left side) contain more Einzellenses and thus focus more strongly than those

located upstream (the right side). The excitation laser should still be switched off at this time. To do this, set the pulse energy of the laser to zero. In VLab, a switched-off optical laser is also displayed by “red spheres”.

5.2 Preparatory Estimates for the Experiment

10. The objective of the experiment is to study the transition metal complex $[\text{Fe}(\text{bipy})_3]^{2+}$ in aqueous solution. For this purpose, the salt $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ is dissolved in deionized water to the desired concentration.

Calculate the molar mass of $[\text{Fe}(\text{bipy})_3]^{2+}$ and that of a full salt molecule with its two chloride counterions.

11. The actual sample is a powder consisting of small microcrystals. In addition to the salt described above, it also contains incorporated water. This water of crystallization remains in the crystal lattice after the crystallization process. Assume that three water molecules are bound in the crystal per $[\text{Fe}(\text{bipy})_3]\text{Cl}_2$ formula unit. A minimum volume of 50 ml is required to operate the liquid jet in the experiment.

Calculate the concentration if you dissolve 10 g of the sample powder in demineralized water to a total volume of 50 ml. Compare this value with the concentration of pure water.

12. The objective of this task is to gain an understanding of the contributions of each component (element) of the sample solution to the total attenuation of X-rays in the liquid jet.

Proceed as follows:

- Using CXRO, determine the transmittance in the energy range of 7000 eV to 7200 eV through a 25 μm thick jet composed entirely of water.
- Calculate the concentration (in mol/L) of an atomic iron foil ($\rho_{\text{Fe}} = 7.874 \text{ g cm}^{-3}$). Compare this to the concentration of $[\text{Fe}(\text{bipy})_3]^{2+}$ in H_2O in the previous task. Now you can calculate the thickness of an atomic iron foil that contains as many iron atoms as are embedded in the 25 μm thick $[\text{Fe}(\text{bipy})_3]^{2+}$ solution.
- For the same energy range, determine the transmission through a fictive iron foil with the thickness from b) in the 7000 eV to 7200 eV range.
- Now compare the contribution of iron and water to the total attenuation. Do this especially at 7050 eV and 7150 eV. This includes a comparative plot in the energy range from a).
- What are the X-ray physical mechanisms for the attenuation by water in this energy range?

13. TODO

14. In the time between two X-ray probe pulses, the sample should be completely refreshed in the interaction volume. This can be ensured by setting a sufficiently

high flow rate of the liquid jet. Within one X-ray pulse train, the repetition rate is 1.1 MHz. For a first estimation with idealized parameters, assume that the X-ray laser has a top hat beam profile (App. B) with the dimension $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$. **Calculate the minimum required flow speed of the flux jet to provide a fresh sample for each X-ray pulse.**

15. Three laser wavelengths in the visible range are available for excitation: 355 nm, 400 nm and 515 nm.²⁰

Calculate the concentration c of the solution for these three wavelengths for a desired optical density $\text{OD} = 1$ and a liquid jet thickness of $25\text{ }\mu\text{m}$.

Note: The optical density is given by $\text{OD} = \epsilon \cdot c \cdot d$ (Sec. 3.1 and exercise 13), where d is the thickness of the liquid beam and $\epsilon(\lambda)$ is the wavelength-dependent extinction coefficient of $[\text{Fe}(\text{bipy})_3]^{2+}$ in water.

16. **Select a laser excitation wavelength and justify your choice. The following pulse energies are available: $5\text{ }\mu\text{J}$ (355 nm), $15\text{ }\mu\text{J}$ (515 nm) and $50\text{ }\mu\text{J}$ (400 nm).**

17. The optical laser beam profile is approximately Gaussian-shaped (appendix B). It has an original diameter of 10 mm (FWHM) and this collimated beam is then focused onto the liquid jet by a lens with a focal length $f = 250\text{ mm}$.

Calculate the diameter of the focus.

Note: Consider the diffraction limit (Rayleigh criterion).

18. By positioning the focus of the optical laser beam behind the liquid jet, a larger excitation beam diameter than the calculated focus in exercise 17 can be achieved on the sample.

How would you choose the beam sizes of the optical laser and the X-ray pulse relative to each other to perform a time-resolved pump-probe experiment? Make a qualitative preliminary consideration for this.

Note: This consideration may influence the required flow speed of the liquid jet (ex. 14).

19. The optical laser should excite the sample efficiently. To achieve this, the laser intensity should be in a range, in which the absorption of the sample molecules scales linearly with the laser intensity. If the intensity is greater, i.e., more than one laser photon impinges on the ground-state absorption cross-section of the sample molecule, nonlinear multiphoton absorption processes can contribute to the excitation. Multiphoton absorption can excite the molecule to a variety of even higher states. In the case of $[\text{Fe}(\text{bipy})_3]^{2+}$, this would result in an excitation to states other than the singlet MLCT (Sec. 3.2), which complicates the interpretation of the experiment.

Ideally, the number of absorbed optical laser photons should be less than or

²⁰These are the usual wavelengths that can be generated by (second-, third and fourth) harmonic generation of commercial laser crystals. By means of a so-called TOPAS, all other UV/VIS wavelengths can also be generated, but the intensities of the harmonic wavelengths are typically more than one order of magnitude higher.

approximately equal to the number of sample molecules in the laser-excited volume.

- a) Calculate the number of $[\text{Fe}(\text{bipy})_3]^{2+}$ molecules in the volume illuminated by the laser. Select a suitable beam size for this purpose, take into account exercise 18.
 - b) Determine the pulse energy at which the number of photons per pulse is equal to the number of molecules from a).
 - c) Verify your calculation with the calculator built into the vLab / Laser. Document this with a screenshot.
20. Two X-ray crystal spectrometers are available at the FXE instrument of European XFEL: one von Hamos and one Johann spectrometer (Sec. 6.5.2). Which spectrometer would you use for the present X-ray emission experiment to record the Fe $K\beta$ lines?
21. Select appropriate crystals for the emission spectrometer. Calculate the Bragg reflection angles for the Fe $K\beta_{1,3}$ emission line. Consider the efficiency of the reflections, the energetic resolution, and the energy range to be covered. An online Bragg reflection calculator with integrated crystal database is useful for this purpose and provided by TU Dortmund [28]. A list of available crystals at FXE can be found on the FXE website [29].

5.3 Performing the Experiment with Data Acquisition

22. **Center the liquid jet on the X-ray beam.**
To optimize the position of the liquid jet in the X-ray beam, maximize the intensity of elastic X-ray scattering. Use the Large Pixel Detector (LPD, Sec. 6.5.4) for this purpose. Note: the beam should be centered through the central hole beforehand.
Document your achieved maximum intensity well. When readjusting, for example after each restart of the software, you only have to return to this maximum intensity.
23. **Adjust the X-ray emission spectrometer.**
Set the correct Bragg angle range for the selected crystal (ex. 21) and orient the detector perpendicular to the incoming dispersive spectrum. Make sure that you are in the line focus of the horizontally focusing emission beam. Check and optimize the cylindrical focusing using the 2D display of the Jungfrau detector, and document your geometric setting by suitable screenshots.
Download the 1D data and plot the $K\beta_{1,3}$ spectrum of the ground state sample (laser off).
24. Use the excitation laser with appropriated pulse energy (ex. 19), to prepare for your first pump-probe measurements:
- a) Adjust the focus of the excitation laser (ex. 18) to obtain your desired beam diameter in the sample plane.

- b) Adjust the spatial overlap between laser and X-ray beam. To do this, adjust the excitation laser beam by moving the focusing lens in x and y direction.
 - c) Document the profile of the excitation laser and the X-ray beam on the sample with the X-ray microscope. To do this, alternately move in strong X-ray attenuators to turn off the x-ray beam, and set the laser energy to zero to measure only the x-ray beam. Afterward, you can also take an image of both pulses together (with appropriate laser and x-ray intensity levels).
25. Record your first time-resolved spectra:
- a) Set the time delay to 100 ps. With this time delay, the laser pulse strikes the sample surely before the X-ray pulse arrives, and still not too early with respect to the expected excited state lifetime. Even if we do not yet know the exact time zero between the laser and X-ray pulses, it has already been (pre)determined to within a few ps.
 - b) Record an XES spectrum (1D, not the difference) with the laser on and a spectrum with the laser off. Plot the laser-on and laser-off together. Normalize the spectra to the area beforehand.
 - c) Now also save a difference spectrum. Is a transient signal recognizable? If so, briefly describe the curve and identify any striking features.
26. Verify the optimum spatial overlap, which preset with the X-Ray Eye, based on the intensity of the pump-probe signal.
Define a representative figure of merit (FOM) for the intensity of the time-resolved transient. Plot this figure of merit as a function of laser position, once along the x and once along the y directions.
27. Plot this FOM also as a function of the excitation laser intensity.
28. Measure the FOM as a function of the time delay between pump and probe pulses. Plot these and explain the measured curve(s).
29. Which measurement range and step sizes of the time delay are suitable for detecting the individual steps (or the individual intermediate states) of the photophysical/photochemical reaction of $[\text{Fe}(\text{bipy})_3]^{2+}$?
Record difference spectra in a series of meaningful pump-probe delays and with suitable signal quality.

5.4 Analysis and Interpretation

The goal is to determine the excited state pathway(s) of $[\text{Fe}(\text{bipy})_3]^{2+}$: Which state is present at long pump-prob time delays? Are there any short-lived intermediate states? What is the time-dependent population of the respective states? To do this, compare your measured spectra with reference spectra from a series of iron complexes

(fig: 4.4, data file is provided separately). This series covers, for each reference sample in its ground state, the different spin multiplicities ($2S + 1$) between one and five (singlet to quintet).

30. **Plot the reference spectra.** Compare your extracted $[\text{Fe}(\text{bipy})_3]^{2+}$ spectrum in its ground state (ex. 23) to the reference spectra.
31. **Take all possible differences between singlet and higher spin states from the reference spectra and plot these.** From exercise 25, compare (i) the measured 100 ps spectrum with the reference spectra and (ii) the measured transient with the differences of the reference spectra. In which spin state is the $[\text{Fe}(\text{bipy})_3]^{2+}$ molecule after 100 ps?
32. **Plot your time-resolved difference spectra from exercise 29.** How does the difference spectrum change qualitatively and quantitatively over time?
33. **Generate a 3D false-color plot of the time-resolved spectra, i.e., plot the emission intensity color-coded as a function of X-ray energy (y -axis) and time delay between pump and probe pulses (x -axis).** Discuss this plot.
34. **Plot the so-called kinetic traces, i.e., plot the emission intensity as a function of the time delay.**
To do this, choose suitable discrete emission energies and represent them by different colors.
From this, determine the rise time of the population of the excited state of $[\text{Fe}(\text{bipy})_3]^{2+}$.
35. **Can you also identify a possible ultrafast intermediate state in the relaxation cascade?**

6 Experimental Set-up

6.1 The X-Ray Laser European XFEL

The experiments are performed at the international Free-Electron X-Ray Laser research facility, European XFEL (EuXFEL, Fig. 6.1). Extremely intense and ultra-short X-ray flashes can be used to study tiny structures at atomic resolution, to “film” ultrafast processes such as chemical reactions, or to investigate matter under extreme conditions. The facility is used by research groups from all over the world who want to answer questions from a wide range of scientific disciplines with the help of this unique X-ray source.

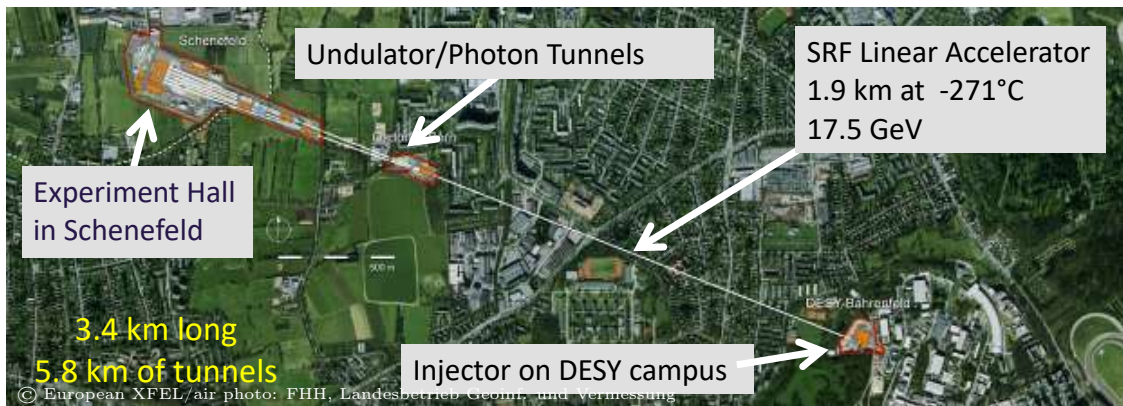


Figure 6.1: Aerial view of the western part of the city of Hamburg and the adjacent Schenefeld in Schleswig Holstein. The 3.4 km long underground facility runs from the DESY research center to the European XFEL campus, where the experimental stations, laboratories and administrative buildings are located.

Core element of FELs are the undulators (Sec. 6.2), devices that serve X-radiation from relativistic charged particles - here electrons. The EuXFEL electron source is located on the DESY site in Hamburg-Bahrenfeld. A 1.9 km long superconducting linear accelerator provides the electrons with energies of up to 17.5 GeV. The electron beam is then directed into a branched tunnel system (Fig. 6.2). contains three undulators, from which the synchrotron radiation then emerges: The undulators, labelled SASE 1 and SASE 2, are each 212 m long and generate hard X-ray radiation in the range of 5 keV to 25 keV. The shorter third undulator, SASE 3, is 125 m long and generates soft X-rays in the range of 0.25 keV to 3 keV. At the end of the 3.4 km long facility is the football field sized underground experiment hall, containing the

experimental platforms, installed in so-called *hutches*. From control rooms beside each of the instruments, the light is further adapted to the needs of the experiments.

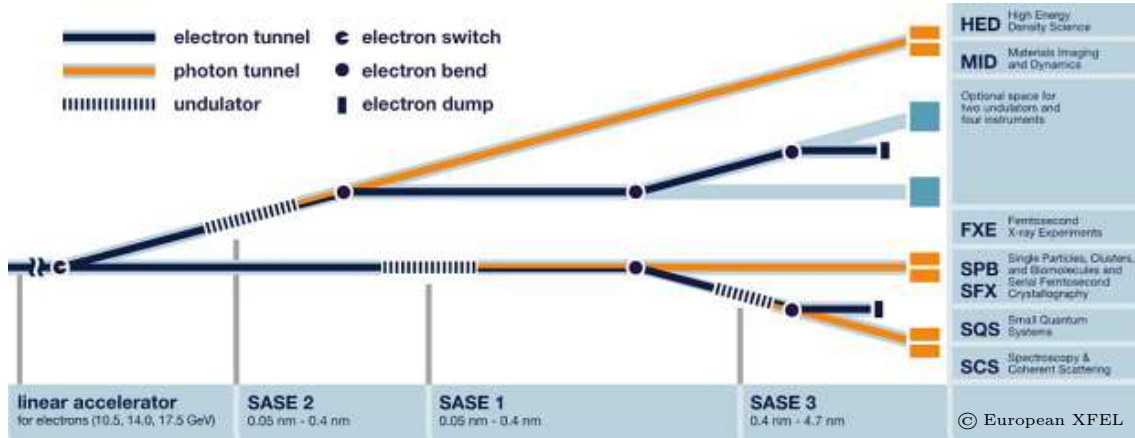


Figure 6.2: The EuXFEL tunnel system for the transport of electrons (blue) through the undulators (dashed) to the *electron dumps*. The photon tunnels (orange) transport the emitted radiation over about one kilometer to the experiments. Each of three photon tunnels serves two experimental stations (the new SXP station at SASE3 is not included here). Possible research applications for the use of the remaining two tunnels are currently being discussed in workshops or are already being planned.

European XFEL has been in operation since 2017 and generates extremely short X-ray pulses, in the range of 1 fs to 100 fs, with an intensity orders of magnitude higher than conventional synchrotron radiation sources and with 100 % lateral coherence. The facility currently (2024) serves seven experimental set-ups, similar in design, but adapted to their respective applications. These scientific instruments enable sophisticated studies in a wide range of fields, for example protein structure determination, reaction dynamics in gases, liquids and solids and processes in the interior of planets, and thus serve essential research areas in physics, chemistry, biology and materials science:

- HED (High Energy Density Science)
- MID (Materials Imaging and Dynamics)
- SCS (Spectroscopy and Coherent Scattering)
- SXP (Soft X-ray Port)
- SQS (Small Quantum Systems)
- SPB/SFX (Single Particles Clusters and Biomolecules and Serial Femtosecond Crystallography)
- FXE (Femtosecond X-ray Experiments)

The virtual experiment presented here realistically reproduces the FXE research platform. More details about the other experiment stations and the entire European XFEL facility can be found at: [30].

The FXE instrument at a glance

FXE's main research topics are studies of ultrafast chemical and biochemical reactions of solution-phase samples next to different solid-state application and nonlinear X-ray techniques. An ultrafast and powerful laser serves as pump source, allowing excitation wavelengths from the UV to the IR. Primary techniques for determining different observable are: X-ray diffraction (XRD), X-ray absorption spectroscopy (XAS, App. A) and X-ray emission spectroscopy (XES, Chap. 4).

At the end of the photon transport tunnel (Sec. 6.3) the X-ray beam is guided through a wall into the radiation-protected FXE experiment hut. Neighboring it directly are: the optical laser hut, the control hut and other huts (Fig. 6.3).

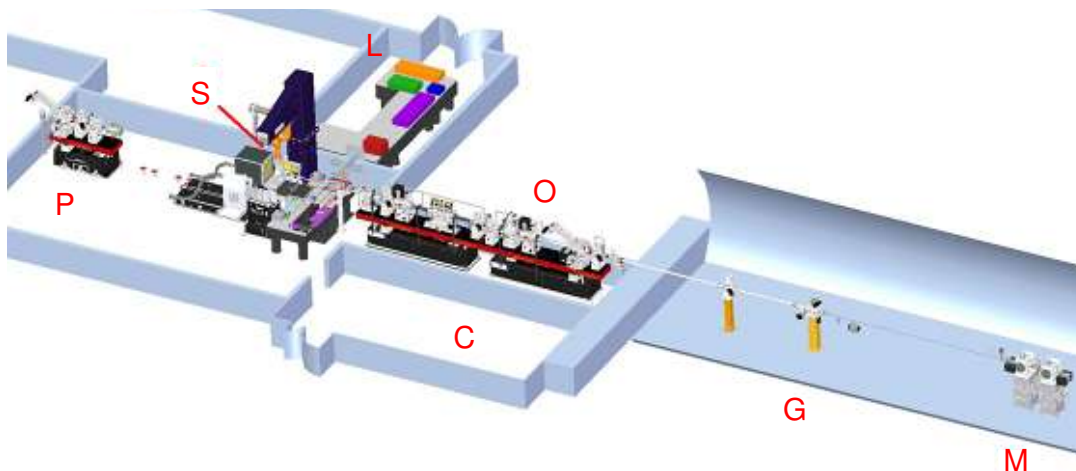


Figure 6.3: The vicinity of the experimental hut of the FXE instrument. The last components in the photon tunnel (on the right) contain the 4-crystal monochromator (M), and the beam splitter (G). Behind the radiation protection tunnel wall follows the optical bench (O), the sample environment (S), and at the end a diagnostic table (P). From a separate hut (L) the optical laser beam is guided to the experiment. In the control room (C) all components are remotely adjusted, and the experiment is performed. (From [31]).

The X-ray beam is manipulated, visualized and diagnosed on the roughly 9 m long *optical bench* (Fig. 6.4, Sec. 6.4). Main Issues are focussing the X-ray beam, ensure a spatial pump-probe overlapping and control the temporal offset between optical and X-ray laser. The actual experiment takes place in the *sample environment* (Sec. 6.5). The spectrometers and detectors are also located there. Behind this, a smaller *diagnostic table* (Sec. 6.6) and finally the *beam termination* are arranged.[31, 32]

6.2 Undulators as X-Ray Sources

6.2.1 SASE-Undulator

XFELs as well as synchrotron light sources use *undulators*, also called *insertion devices* (IDs), to generate (X-ray) radiation (Sec. 1.1). They consist of a linear alternating arrangement of dipole magnets, separated in the longitudinal direction

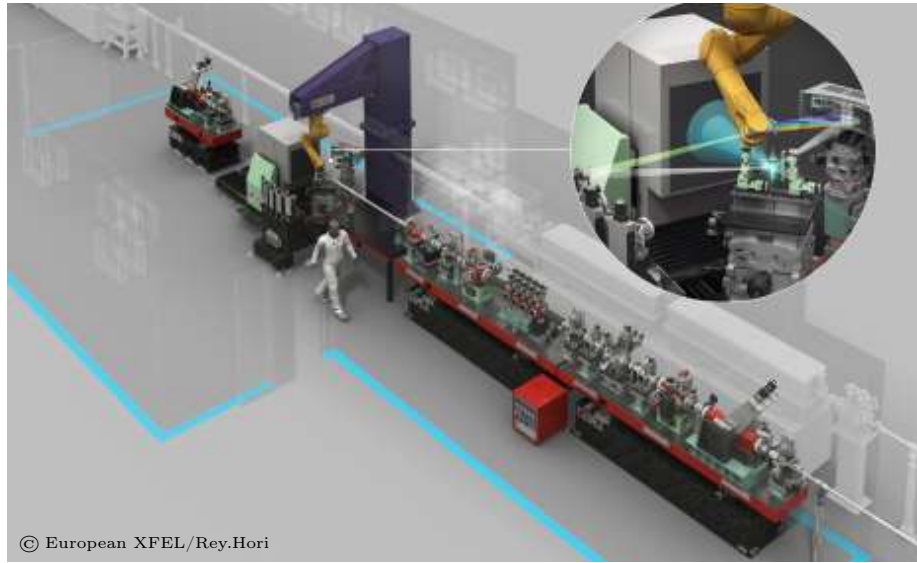


Figure 6.4: The FXE instrument: The X-ray beam comes from the bottom right of the tunnel. It is characterized on the optical bench and focused on the sample environment (below the blue tower) by means of Be lenses. At the end is the diagnostic table. The parallel beamline to another experiment (SPB/SFX) can be seen in the pale background.

by the period L . Their magnetic field is oriented perpendicular to the propagation direction of the relativistic electrons, thus forcing them onto a wiggly sinusoidally shaped path, which leads to emission of synchrotron radiation (SR, Fig. 6.5a).

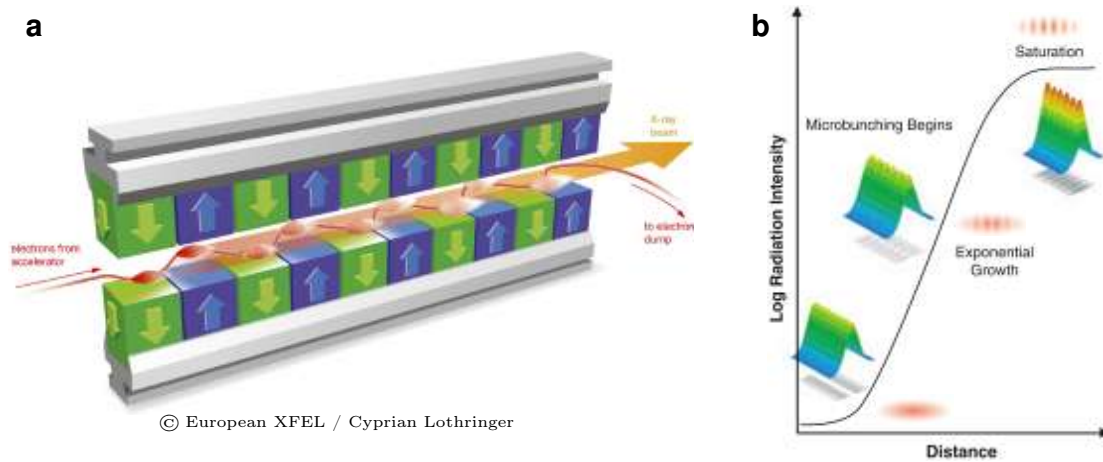


Figure 6.5: a) The principle of the undulator for generating electromagnetic radiation. Alternating magnets force relativistic electrons onto a wiggly slalom course, on which the radially accelerated particles emit radiation at each turn. This results in the emission of synchrotron radiation in forward direction. b) Amplification of SR radiation via SASE at a FEL source in a very long undulator using a very short duration (50 fs to 100 fs) electron bunch: Spontaneous radiation in perfect forward direction near the beginning is amplified in the following undulator sections by stimulated emission from the same electron bunch. The growing electric field strength leads to *microbunching* or accumulation of electrons within the nodes of the amplified radiation field (thus with attosecond separation), when amplification approaches saturation (maximum possible intensity). (Adapted from [33, 30])

Common undulators have pole periods L in the cm range, e.g., 4 cm for the case of the EuXFEL hard X-ray SASE undulators. Therefore, the emitted wavelength should also be in this range, or far away from X-radiation. However, the pole separation period is shorter in the reference frame of the relativistic electron (Lorentz contraction), reduced by the relativistic γ factor. The latter is defined by the electron's relativistic velocity v near the speed of light c , and also by the ratio of kinetic energy E_{kin} and the rest mass energy of an electron ($E_0 = m_e c^2 = 511 \text{ keV}$, with the electron mass m_e):

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - \beta^2}} = \frac{E_{\text{kin}}}{E_0} \quad (6-1)$$

Thus, γ can also be conveniently calculated with the accelerator energy E_{acc} via

$$\gamma = 1957 \cdot E_{\text{acc}} / \text{GeV} \quad (6-2)$$

which yields $\gamma = 3.42 \times 10^4$ at EuXFEL for its highest energy of $E_{\text{acc}} = 17.5 \text{ GeV}$ ²¹. The undulator pole period is reduced in the electron reference frame to:

$$L' = \frac{L}{\gamma} \quad (6-3)$$

The radiation emitted in forward direction experiences a relativistic blue shift of

$$\lambda = \frac{L'}{2\gamma} \quad (6-4)$$

and both processes (Lorentz contraction and blue shift) reduce the wavelength of emitted light from the cm-long magnetic pole period L into the sub-nanometer range thus delivering X-radiation:[22]

$$\lambda = \frac{L}{2\gamma^2} \quad (6-5)$$

The emitted radiation is largely enhanced in forward direction, and suppressed in backward direction (Fig: 6.6). In case of the EuXFEL, the X-ray beam exits the SASE2 undulator with a divergence of around $2 \mu\text{rad}$ (at 10 keV)[31]. SR is also horizontally linear polarized for vertically oriented magnetic undulator fields.

Fine-tuning of the emitted undulator wavelength is achieved by changing the time the electrons needs to pass through a full magnetic pole period. Since the relativistic velocity is practically constant, one can change this time by forcing the electron on a shorter or longer oscillatory path, and this can be conveniently adjusted by varying the magnetic field strength B . Stronger B fields force the electrons onto a larger-amplitude sinusoidal path, which thus takes more time and reduces accordingly the radiated frequency. The *undulator parameter* K takes this into account [22]

$$K = \frac{eBL}{2\pi m_e c} \quad (6-6)$$

with the elemental charge e . We vary the magnetic field strength in undulators by simply moving the opposing magnetic poles closer or further apart from each other,

²¹Specific operation conditions utilize lower energies.

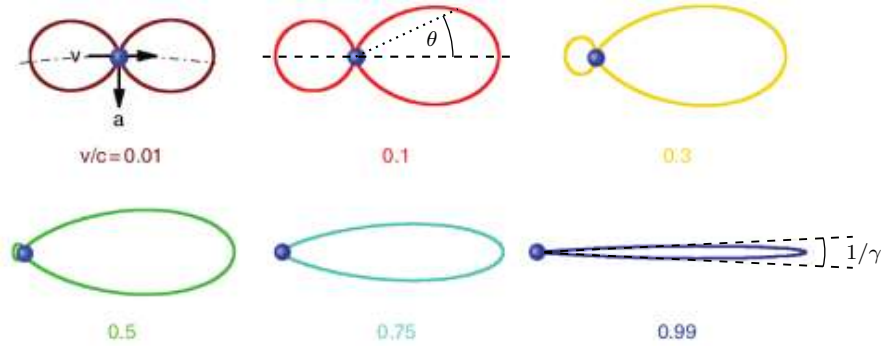


Figure 6.6: The angular distribution of electromagnetic radiation emitted by an accelerated charged particle traveling at different velocities. For small v/c the pattern resembles that of dipole radiation (similar to common radio antennas). As v/c approaches unity, the angular distribution becomes increasingly pencil-like sharp pointing in the forward direction with an increasingly reduced angular divergence angle of $\theta_{\max} = 1/\gamma$. (Adapted from [22])

thus varying the gap distance. In consequence, smaller gaps exhibit stronger magnetic fields in the electron path and this yields a slightly longer radiation wavelength. Overall, this leads to the *undulator equation*:

$$\lambda(\theta) \approx \frac{L}{2\gamma^2} \left(1 + \frac{K^2}{2} + \gamma^2 \theta^2 \right) \quad (6-7)$$

which fine tunes the radiation wavelength generated in undulator sources, accounting for magnetic tuning (K) and off-axis ($\gamma\theta$) radiation. The variable gap structure of the EuXFEL SASE1 undulator makes it possible to generate between 5 keV to 20 keV radiation without changing the electron beam energy (using K parameters of 1.65 to 3.9 [31]).

Undulators produce a significantly higher flux density and brilliance than pure deflection magnets (bend magnets), which dominated the first two synchrotron generations (Sec. 1.1). In synchrotrons, individual undulators are typically 1 m to 5 m long and exhibit comparatively long electron or positron bunches, on the order of 100 ps, which produce radiation pulses of the same pulse duration, or 1000 times longer than at femtosecond FEL sources.

At an XFEL, up to 35 undulators are connected together to form a unit of about 100 m to 200 m in length, which allow the SASE (Self-Amplified Spontaneous Emission) process to start (Fig. 6.5b). Here, the approximately 100 fs long electron bunches entering the first undulators generate X-radiation, exactly like in synchrotrons. One of these spontaneous emitted photons will point exactly along the entire undulator section, which then stimulates emission by copropagating electrons at each turn of their sinusoidal path. This leads to amplification of the electromagnetic field. The electric field vector slightly modifies the electron beam paths, such that they accumulate in the nodes of the copropagating electric field vector. This *microbunching* of electrons then marks saturation of this amplification process [34]. Each microbunch has been responsible for a specific wavelength emission, and with many microbunches in a single electron bunch, we have as many slightly different wavelengths centered around the undulator wavelength. This leads to the pink beam

(Sec. 6.3.2 and Fig. 6.15) with an energy bandwidth $\Delta E/E$ of the order of 10^{-3} , observed by the Spectrum Analyzer (SpA, Sec. 6.4.3 and exercise 4).

As a result of the SASE process, the X-ray intensity is unprecedentedly high: For example, a 1 nC electron charge cloud can produce 10^{12} to 10^{13} photons at 10 keV photon energy. In fact, in the hard X-ray range (> 5 keV), 2 mJ to 6 mJ pulse energies are meanwhile routine. This is several tens of thousand times more than conventional SR sources can deliver with the same electron charge cloud in a single SR pulse, although SR undulators are only less than 100 times shorter than the total FEL undulator length.[32]

6.2.2 Pulse Trains and Single Pulses

The superconducting accelerator at the EuXFEL is unique in its kind: It allows efficient acceleration of electrons of 20 MV m^{-1} to 40 MV m^{-1} up to a selectable final energy between 8.5 GeV and 17.5 GeV. In contrast to more conventional electron accelerators, it can accelerate up to 27 000 electron pulses per second to full energy, each for electron bunches containing charges of more than 1 nC. So-called warm accelerators are limited to about 100 Hz, which makes the EuXFEL about 100 times more intense. However, this is only achieved in so-called *pulse trains*, at a 10 Hz repetition rate.

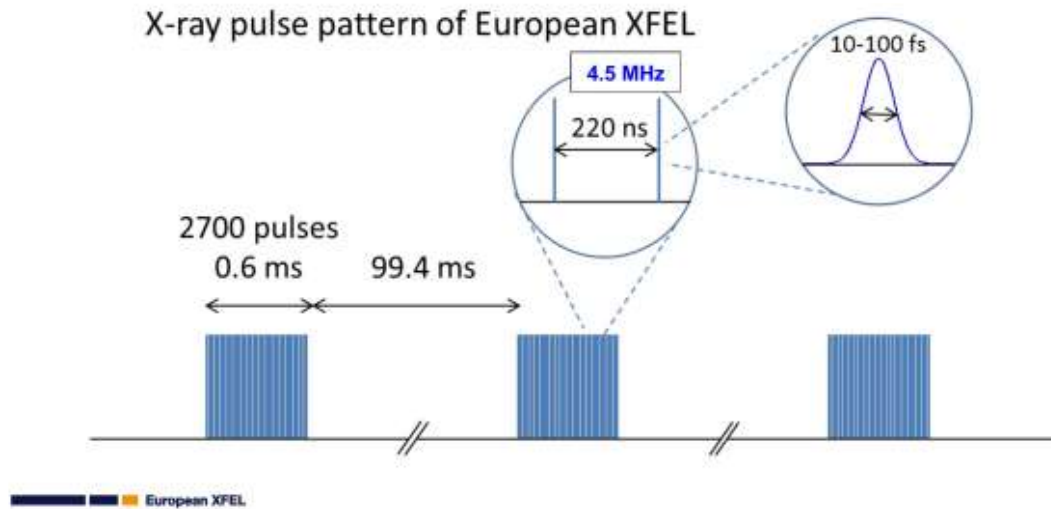


Figure 6.7: Typical EuXFEL pulse train pattern. Pulse trains are delivered at a 10 Hz repetition rate (or every 100 ms). Each pulse train contains up to 2700 single pulses at the maximum intra-pulse train repetition rate of 4.5 MHz, corresponding to a minimum pulse interval of 220 ns. The pulse duration of each pulse can be variably set in the 10 fs to 100 fs range.

A pulse train is a dense collection of individual electron pulses separated by time intervals of integer multiples of 220 ns (or an intra-pulse train repetition rate of up to 4.5 MHz). The maximum length of a pulse train is 600 μs at most, and each individual pulse train can thus contain a maximum of 2700 electron packets (Fig. 6.7). However, the filling of these possible packets is freely selectable within the 0.6 ms

time window and is set by the machine group according to user requirements. As of today (2021), up to 600 pulses at an intra-pulse train repetition rate of 1.1 MHz is the primarily adjusted pulse train sequence. To supply preferably all three photon tunnels (Fig. 6.2) with X-ray light, the electron bunches have to be divided up. This can also be done within a pulse train.

A single electron pulse has a pulse duration (also adjustable) of currently about 50 fs, and this corresponds approximately to the pulse duration of an X-ray pulse leaving the undulator. With a similar time resolution (determined by the cross-correlation between optical and X-ray pulse width) one can now plan and perform pump-probe experiments. For “ultra-fast” experiments, a time synchronization of both beams is crucially important (see chapter Arrival Time Monitor, Sec. 6.4.6).

The full SASE bandwidth of about 0.1 % can be used in the experiments, or monochromatized with the 4-crystal Si(111) monochromator near the end of the tunneling system (Sec. 6.3.2).

The X-ray beam shows an unwanted spatial pulse-to-pulse displacement, called the *jitter*. The cause of this fluctuation is still under investigation, but the M1-M3 horizontally deflecting mirrors (Sec. 6.3.1) are candidates for this. The horizontal jitter is significant, and amounts to about one beam diameter, while the vertical jitter is about a factor of ten smaller. Focusing the beam should reduce the jitter to an uncritical level. An upstream optical aperture (Sec. 6.4.4) can also help.[31, 32]

6.3 X-Ray Beam Transport System

The X-ray beam exits the SASE1 undulator with an energy-dependent divergence of about 2 μ rad (at 10 keV) from a source size of about < 100 μ m. The radiation fluence (see Appendix B) at the undulator exit is far too high for any X-ray optics. Thus, the first optical elements are located only at about 200 m from the radiation source, after a corresponding broadening of the beam to about 0.4 mm (Fig. 6.8).

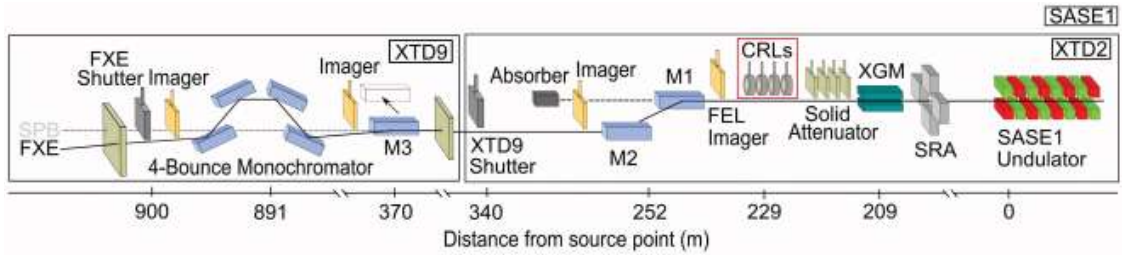


Figure 6.8: The about 900 m long X-ray transport system from the undulator source (right) to the FXE Instruments (left): SASE1 = Undulator for the generation of hard X-rays (Sec. 6.2.1), SRA = synchrotron radiation aperture, XGM = X-ray gas monitor, CRLs = compound refractive lenses, M = mirror devices (From [31]).

First, a *high-power synchrotron radiation aperture* (SRA) shields the rest of the tunnel from all synchrotron radiation emerging outside the X-ray beam divergence cone. An *X-ray gas monitor* (XGM) provides an absolute measurement of the beam

position and intensity, and a pneumatically operated *solid attenuator* (Sec. 6.4.5) is useful for adjusting the different components in the beam. A *refractive X-ray lens unit* (CRL, Sec. 6.4.7) collimates the divergent X-ray beam to a size of 1 mm to 2 mm (FWHM, depending on the X-ray energy) at the end of the tunnel, while retaining almost the entire intensity. Behind the lenses, the beam is parallel-shifted horizontally by two planar mirrors (M1 and M2, Sec. 6.3.1), so that highest-energy (> 100 keV) synchrotron radiation is absorbed by the concrete absorber behind M1. A third mirror (M3) guides the beam to the FXE experiment, or is moved out to serve the SPB/SFX experiment.

Along the over 900 m long path there are some scintillator-camera units, which determine the beam profile and position (imagers, they work like the BIUs, Sec. 6.4.1), i.e. they serve for the adjustment of the X-ray beam. Immediately before the FXE hut, still inside of the tunnel, there is the monochromator (Sec. 6.3.2) and a beam splitter (Sec. 6.3.3). [31, 32]

6.3.1 M1, M2 and M3 Mirrors

For the horizontal parallel offset (via M1 and M2) and the horizontal deflection (via M3) into the FXE branch 3 plane mirrors, each 1 m long, are used. They have an extremely low surface roughness of less than 5 nm rms, which is necessary for propagation of the wavefront without significant distortion. M2 is also equipped with a bending device that allows the beam to be easily focused horizontally for optimal illumination of, for example, M3 (rather than being diffracted beyond the edges). [31, 32]

6.3.2 The 4-Crystal Monochromator

Most synchrotron experiments, in particular, diffraction experiments, but also XAS and other applications require a well-defined photon energy. Furthermore, an energy change or even a scan may be obliged within a given experiment. After the SASE process, however, the beam is not monochromatic enough, but also not really polychromatic. It is therefore called a “pink” beam. Monochromatization is ordinarily achieved by diffraction, using periodic gratings for longer wavelength radiation, or crystals for wavelength shorter than a few nanometers.

Diffraction occurs from constructive interference, when radiation is scattered by planes of atoms in a crystalline material (or a grating). Precondition is a path length between reflections from adjacent crystal plane equal to an integer multiple n (the diffraction order) of the wavelength λ of the radiation. Bragg’s law describes the dependence from the interplanar spacing d of the (hkl) planes, and the angle θ of incidence above the plane surface (glancing angle): [22]

$$n \cdot \lambda = 2d_{hkl} \cdot \sin \theta \quad (6-8)$$

The diffractive (first) crystal in a *crystal monochromator* is planar (in contrast to focusing monochromators) and absorbs all non-constructively interfering wavelengths (Fig. 6.9a). Therefore, it usually must be cooled (cryogenically). A second crystal,

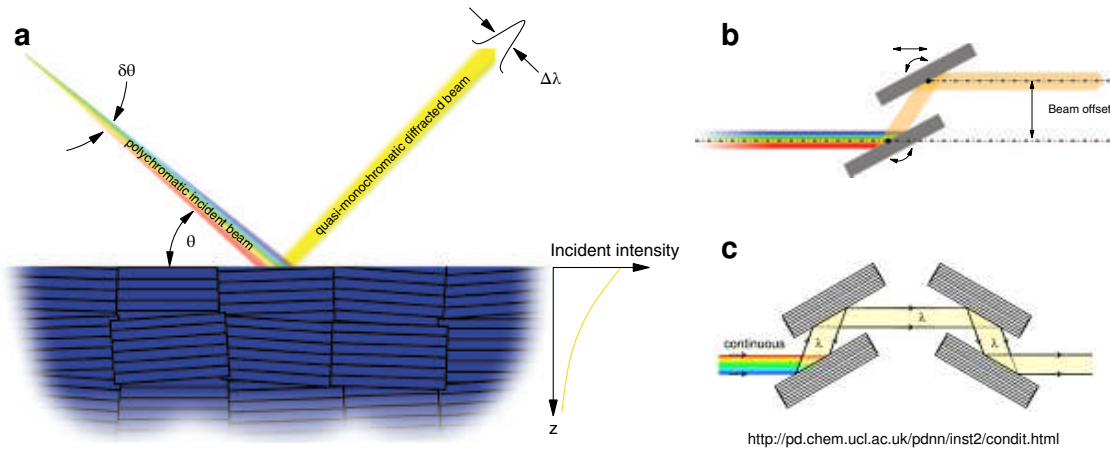


Figure 6.9: (a) Diffraction of an X-ray beam at a crystal surface and its intensity depending on the penetration depth. (b) Geometry of a double crystal monochromator. The first crystal monochromatizes the incoming polychromatic light, the second redirects the outgoing light parallel to it (From [22]). (c) Geometry of a 4-crystal monochromator. The beam is bent back to its original path after the offset. Note the opposite beam direction; in this document, the usual direction is right to left (upstream to downstream).

with correct adjustment with respect to both the angle and the distance, redirects the outgoing monochromatic X-ray beam parallel to the incoming one (Fig. 6.9b).

The geometry of a *four-bounce crystal monochromator* (4BCM) guides the monochromatic beam back to the original light path, and thus maintains the same position of the light paths of both the monochromatic and the pink light beams (when the monocrystals are retracted, Figs. 6.9c and 6.10). This arrangement is convenient for pump-probe experiments with an external laser beam, to ensure that the two beams always overlap congruently. It is often used at synchrotron radiation sources.

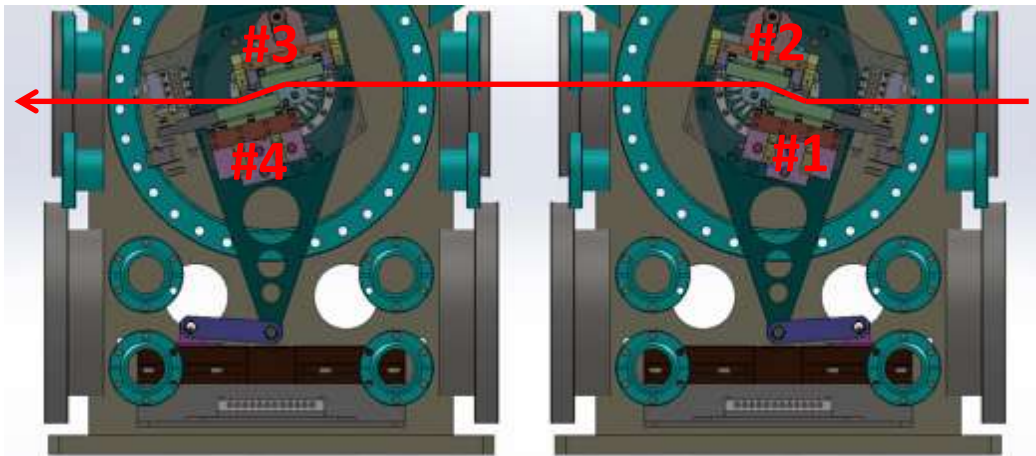


Figure 6.10: The 4-crystal monochromator for the FXE experiment. In the first chamber the beam is vertically parallel shifted by two Bragg reflections, and in the second chamber it is guided back to the original path by another two Bragg reflections. This allows experiments to be performed at the sample location without beam offset, with both, the monochromatic beam and the full SASE spectrum (From [31]).

For hard X-rays, silicon, germanium or diamond are commonly used as reflection material. The choice of material depends on factors such as its quality, its resistance to thermal stress and radiation damage, and its intrinsic minimum bandwidth. Here, the diffraction takes place on the (111) net planes of high purity silicon. Divergences of the incident beam, imperfections and residual absorptions of the crystal, as well as a finite penetration depth cause an increase of the bandwidth $\Delta\lambda$ of the reflected beam (Fig. 6.9a). The Darwin width of the FXE monochromator is $\Delta E/E < 1.5 \times 10^{-4}$ for photon energies above 5 keV.[22, 31, 32]

6.3.3 Beam Splitter (Diffraction Grating)

Finally, a retractable *beam splitter* is installed behind the monochromator near the end of the SASE1 tunnel section (G in Fig. 6.3 and Fig. 6.11). It splits the X-ray beam mainly into three parts: the 0th order (the main beam) and both ± 1 order diffractions. The latter can serve different diagnostic components, such as a spectrum analyzer (SpA1, Sec 6.4.3) or a time arrival detector (TDA, Sec 6.4.6), without altering the incident beam itself significantly.

Exchangeable (not in VLab) 10 μm thick grooved diamond gratings causes the diffraction. They have a pitch of 200 nm or 150 nm, and the efficiency of the diffraction orders is roughly 1%. So the main beam is attenuated by only a few percent, and shows a negligible change in beam profile (Fig. 6.11b,c). Different energies are diffracted with different angles. This leads to a horizontal angle broadening of the diffracted 1st order beam - due to the only approximately 0.2% energy distribution of the SASE pulse, although negligible here.[31, 32, 35]

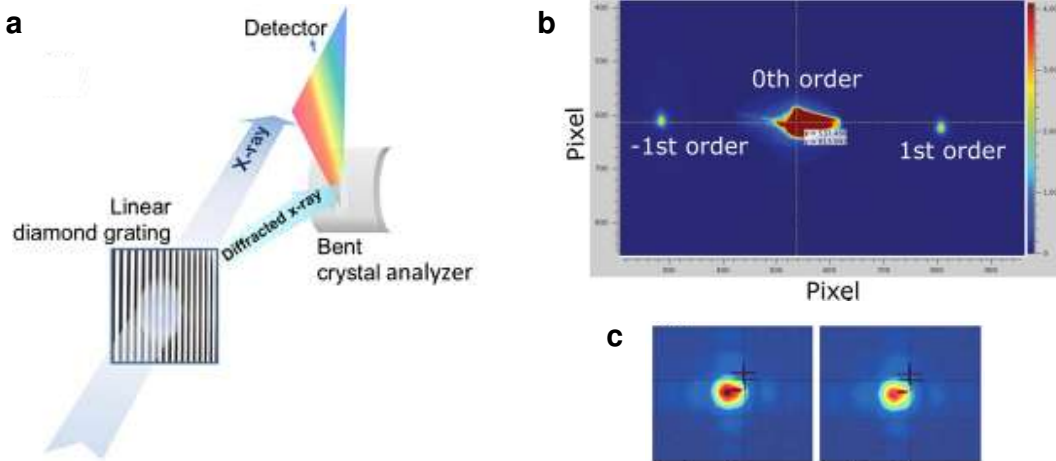


Figure 6.11: (a) Schematic drawing of the beam splitter device. One of 1st-order diffraction (the other one is not shown) serves a spectrum analyzer (From [35]). (b) Beam profile at BIU1 at a photon energy of 9.3 keV. The distance between the 0th order and ± 1 order diffraction is about 270 pixels, which corresponds to $\Delta y \approx 6.7$ mm (From [31]). (c) The main beam with (left) and without (right) upstream diamond grating indicates no significant difference due to the beam splitter (recorded at LCLS FEL in Stanford, USA with a comparable setup, From [35]).

6.4 Components of the Optical Bench in the Experimentation Hut

Various components for diagnosis, visualization and manipulation of the X-ray beam are installed on the optical bench (Fig. 6.12). Most of them are already in use in numerous synchrotron radiation and existing FEL facilities worldwide. They have been adapted and further developed for the EuXFEL. The purpose, design and operation of the main components are described in brief below.

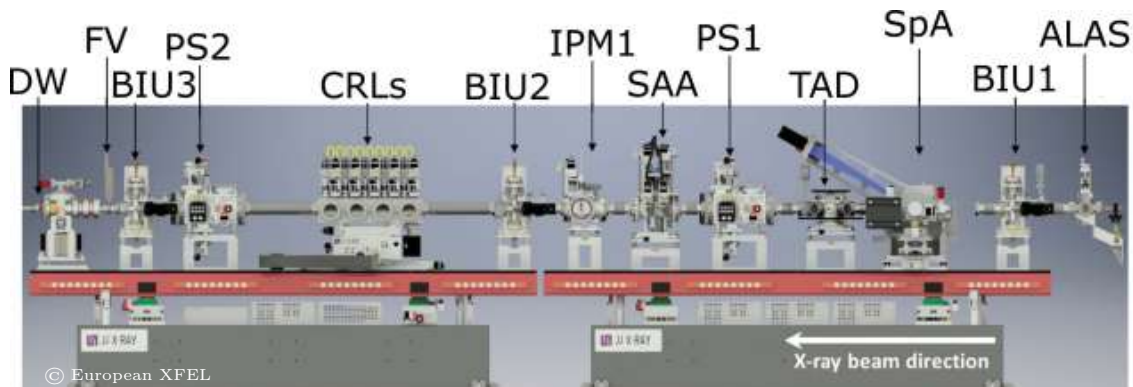


Figure 6.12: The optical bench in the FXE hut (the X-rays come from the right through the tunnel wall). Currently (2019), implemented components: BIU1,2,3 = beam imaging units; SpA = spectrum analyzer; TAD = time arrival detector; PS1,2 = power slits; SAA = solid attenuator assembly; IPM1 = intensity position monitor; CRLs = compound refractive lenses. Other instrumentation: ALAS = alignment laser; FV = fast valve; DW = diamond window (From [31]).

6.4.1 Beam Imaging Cameras

A *beam imaging unit* (BIU, Fig. 6.13a,b) images the lateral or 2D profile of the X-ray beam perpendicular to its propagation direction (Fig. 6.11b,c). That allows a quantitative analysis of the beam spot size and its intensity distribution, but only synchronized to the 10 Hz pulse train frequency (due to the slow readout time of slightly better than 100 ms). So, a BIU cannot measure the profile of a single pulse, but maps the average of the profile of a whole pulse train.

Three identically built BIUs are installed on the optical bench, and another is located on the diagnostics table behind the sample environment. BIU1 indicates whether the beam from the tunnel has entered the experimental hut, BIU2 and BIU3 are used to align the lens array (CRL, Sec. 6.4.7). However, they can also be pulled down to diagnose the beam profile constricted by the high power slits PS1 and PS2.

Two different scintillator materials (in VLab only one, a N-doped diamond foil) are installed on a movable, water-cooled copper rod (Fig. 6.13b). For beam diagnostics, the rod is positioned at an angle of 45° with respect to the X-ray beam propagation direction. A CCD camera, mounted perpendicular to the screens, is synchronized to

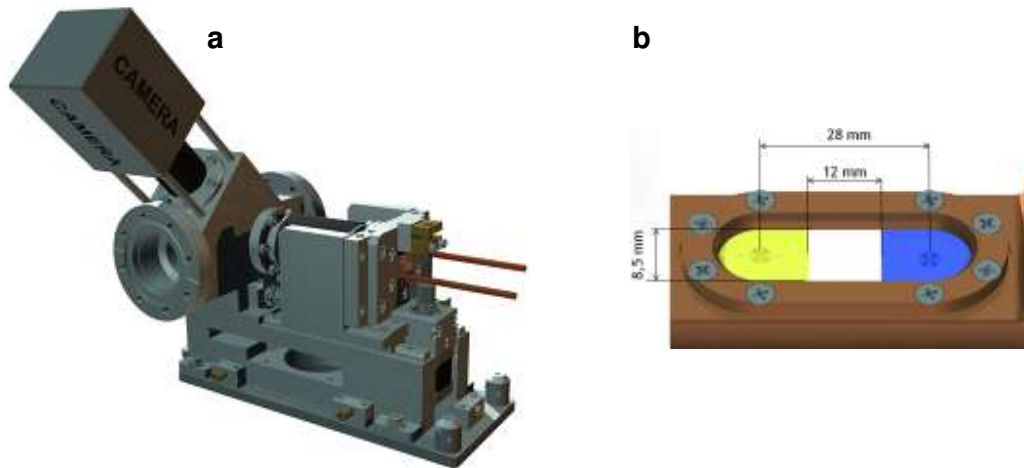


Figure 6.13: (a) Beam Imaging Unit. In the foreground is the stepper motor that guides the carrier into the beamline. The X-ray beam passes through the tube in the center of the image, above which is the CCD camera mounted at a 45° angle to the beam. (b) The copper rod carries the Ce:YAG (yellow) and the pCVD (blue) scintillator foils. The white field in the center is transparent, and the crosshairs are included for orientation (From [36]).

the pulse trains, i.e. it delivers images with a repetition frequency of 10 Hz. In this optical arrangement (with imaging lenses in the optical path), the optical resolution is roughly $25\text{ }\mu\text{m}$ per pixel.

A scintillation screen emits photons in the visible wavelength range after irradiation with hard X-rays. Currently installed at BIU are: a Ce-doped YAG crystal (yttrium aluminum garnet, $\text{Y}_3\text{Al}_5\text{O}_{12}$) and a polycrystalline diamond (pCVD), both $50\text{ }\mu\text{m}$ thick. We prefer the former for images at low X-ray intensity and of single pulse trains, the latter for high intensities or when high transparency is required. The high transmittance of the diamond scintillators allows the continuous recording of the pulse profile *in-situ*, so that beam changes can be detected immediately. Further crystals of different thicknesses and materials are available, especially suitable for higher energy X-ray pulses as well as for high pulse frequencies up to 4.5 MHz.[31, 32, 36]

6.4.2 Intensity and Position Monitor

Two *intensity and position monitors* (IPM, Fig. 6.14) are included upstream and downstream of the sample position. They can determine the beam position with single-shot resolution within a pulse train to show the X-ray beam propagation through the FXE instrument, and are also designed for measuring the integrated intensity of the beam. However, unlike the BIUs, IPMs cannot map a beam profile.

The IPM is equipped with a retractable screen which scatters (dominated by Compton scattering) a small proportion of the incident X-ray beam. The screen is made of a $15\text{ }\mu\text{m}$ -thick nanocrystalline diamond foil (grain size $< 15\text{ nm}$). The backscattered as well as fluorescence photons are collected by four identical avalanche photodiodes

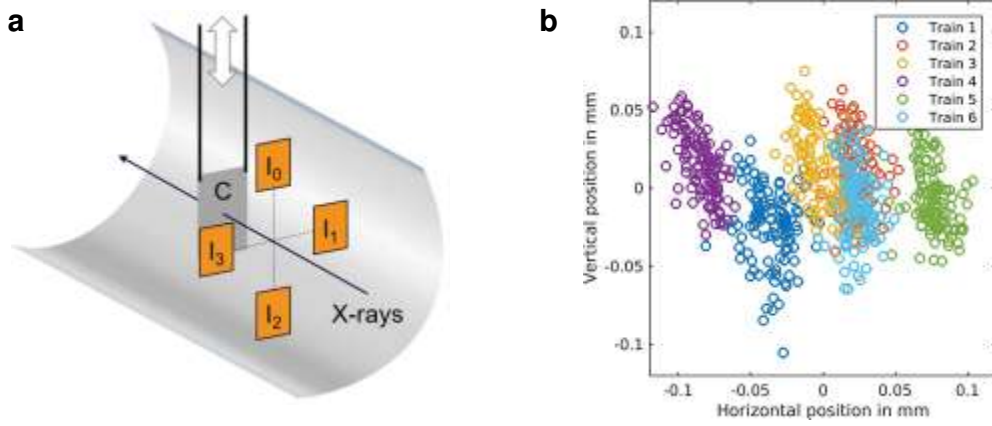


Figure 6.14: (a) Intensity and Position Monitor(IPM). The X-ray beam is partially scattered by a thin diamond foil (C). The backscattered (and fluorescence) photons are registered by four APDs ($I_0 - I_3$), arranged in a quadrant. The intensity differences between opposite diodes can determine the horizontal and vertical beam position. The total signal is proportional to the pulse energy. (b) Single-pulse-resolved beam positioning for 10 Hz consecutive trains arriving at a 1.1 MHz intra-train pulse repetition rate, with an incident beam size of 700 μm vertical and 230 μm horizontal (FWHM) and a X-ray energy of 9.3 keV. Intra-train jitter occurs predominantly in the vertical plane, while the inter-train jitter is dominated by a horizontal contribution. (From [31], simplified in the VLab)

(APDs, Sec. 6.5.3) with a temporal resolution below the X-ray pulse separation time of 220 ns, enabling the read-out of single pulse energies. The square arrangement of the diodes allow calculation of the horizontal as well as vertical beam position from the normalized intensity difference of opposite APDs, while the sum is proportional to the pulse energy (Fig. 6.14). The entire diode assembly can be moved closer to or further away from the backscattering foil, depending on the actual incident flux and photon energy.

The IPM spatial information as well as the determined single pulse energy can be calibrated by comparison with the beam position and intensity on an upstream and/or downstream BIU. Since the BIUs are synchronized to the pulse trains, this requires an 10 Hz operation with only a single pulse within the pulse trains, to assign the slower BIU images to the same pulse measured by the IPM.[31, 32]

6.4.3 Spectrum Analyzers

Every single EuXFEL X-ray pulse has an individual spectral fine structure due to the stochastic nature of the SASE process (Sec. 6.2.1). The femtosecond long electron pulses have a low longitudinal coherence within the undulator and are broken down into many smaller packets (microbunching). Each of this micro bunches radiates a slightly different amplified emission energy. Consequently, each individual X-ray pulse shows a characteristic spectrum with many (about a hundred, Fig: 6.15) spectral peaks within the energy bandwidth of $\Delta E/E \approx 10^{-3}$, while maintaining the weighted central wavelength. These fluctuations in spectral composition can affect the interpretation of the measurements.

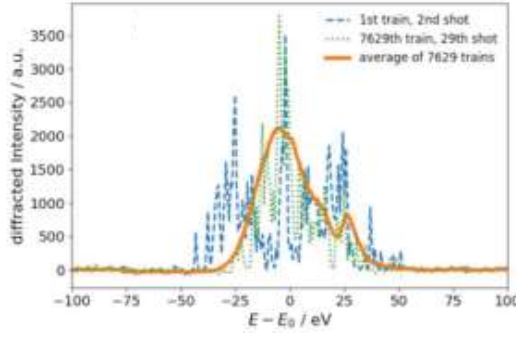


Figure 6.15: Two arbitrarily selected EuXFEL/SASE1 single pulse spectra (green and blue curve) and the average spectrum after 12 min collection time (orange) at the incident X-ray photon energy of $E_0 = 9.3$ keV (From [31]).

For some XFEL experiment, in particular for absorption spectroscopy and for high-resolution diffraction experiments, it is advantageous to know the spectral pulse-to-pulse fluctuations exactly, sometime it's even essential. For non-resonant X-ray emission spectroscopy (XES, Sec. 4.3) this spectral information is not crucial: XES electrons are excited into the continuum by the XFEL pulse, but the spectra are generated by inner-shell transitions. Thus, slight energy differences of the continuum electrons do not play a role in measuring the XES spectrum. However, knowledge of the single-shot spectral fine structure can be useful to pre-calibrate the Johann spectrometer.

The diagnostics for these are provided by a *Dispersive Single-Shot Spectrometer*, also called the *Spectrum Analyzer* (SpA, Fig. 6.16a), using the first-order diffraction from the upstream beam splitter (Sec. 6.3.3, Fig. 6.11a). The spectrometer, based on a curved Bragg diffraction crystal, offers an adjustable resolution up to about $\Delta E/E \approx 10^{-5}$ (or about 100 meV at 10 keV) and is capable of recording a shot-to-shot incident SASE spectrum (but limited by the detector read-out speed to 0.8 MHz).

The SpA utilized a thin ($< 100 \mu\text{m}$) cylindrically curved single crystal of Si(111), Si(110), Ge(110) or diamond/CVD (chemical vapor deposition) foil, with a reflection dependent bent radius between 50 mm to 300 mm. When the X-ray beam impinges onto the convex side of the crystal, different (vertical) portions of the beam make different incident angles, each satisfying the Bragg condition (eq. 6-8) for a (slightly) different wavelength. Hence, for different energies the reflection point walks on the curved crystal surface, as do the reflected rays on the detector plane (Fig. 6.16b). The development of these extremely thin and highly precisely curved single crystals yields a transmission of more than 90 % for all energies in the 5 keV to 20 keV range. As a result, it is also possible (not in VLab) to place the SpA1 directly in the main beam without significant intensity losses. Both methods (diamond in the direct beam, Si in the 1st order diffraction beam from the beam splitter) have already been tried out at FXE, each with advantages and disadvantages.

The dispersion Δx is recorded by a Gotthard detector (1D streak detector, Sec. 6.5.3), mounted on a movable arm (2θ -arm). This allows a detector orientation normal to

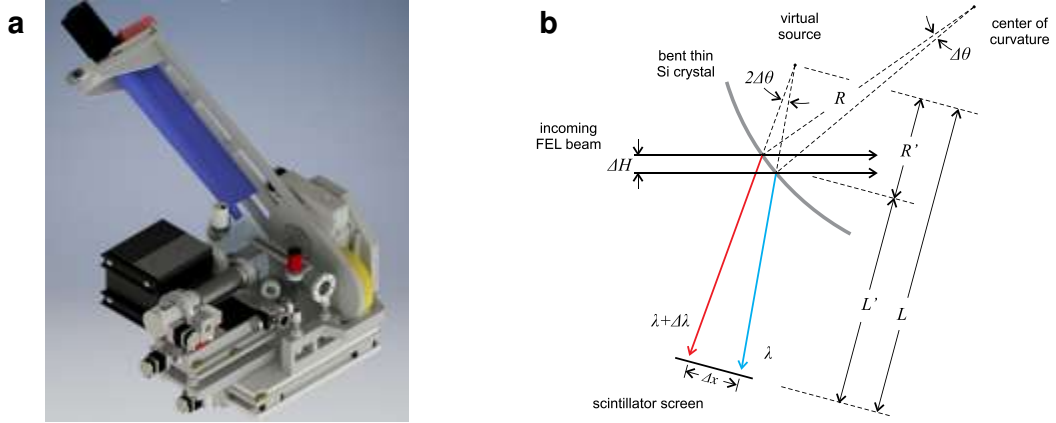


Figure 6.16: (a) The motor section in front of the SpA moves a carrying copper rod into the (1th-order) X-ray beam, allowing the mounted curved diffraction crystal(s) five degrees of freedom (three translations and two rotations). The main beam passes through the small tube in the middle. A rotatable He-filled enclosure (blue) allowed a $2\theta_{\text{Bragg}}$ motion of the detector at the end of the arm (From [31]). (b) The dispersion geometry of the SpA. Different parts of the beam see different Bragg angles on the cylindrically bent crystal membrane leading to a dispersive diffraction of the FEL beam to form the spectrum, which is recorded on a detector positioned normal to the diffracted beam (From [37]).

the diffracted beam for all dispersion angles θ . Considering the geometry of the spectrometer, the dispersion relation $\Delta\theta(E)$ can be calculated and, in conclusion, $\Delta x(E)$, with x being the pixel or location coordinate of the detector. A slightly non-linear function $\Delta E(E)$ is obtained which depends on the spacing between the strips of the Gotthard detector (50 μm), the distance from the membrane to the detector plane ($L' = 1\text{ m}$), and the radius R of curvature of the diffraction crystal. The calibration can additionally be determined precisely with the help of the primary monochromator. The beam size ΔH and the crystal curvature R define the theoretical recordable energy window.

A second SpA on the diagnostic bench behind the sample allows X-ray absorption measurements. These can be normalized and represented correspondingly using

$$n\sigma d = \ln \frac{I_0}{I_1} = \ln \frac{\text{SpA}_1}{\text{SpA}_2} \quad (6-9)$$

with the sample concentration n , the energy-dependent X-ray absorption cross-section σ , and the layer thickness d . Due to the irregular intensity distribution of the SASE energy spectrum (see Fig. 6.15), one does not obtain an end-to-end absorption spectrum for a single pulse, but only fragments within the spectral width. By accumulation of many such pulses, however, this unevenness is equalized, so that a continuous absorption spectrum is obtained. For example, a single burst at 10 Hz repetition rate has up to 2700 pulses, and this is an excellent way to measure XANES. Thus, the difference between the two SPA spectra can show which energies are absorbed by the sample.[31, 32, 35, 37]

6.4.4 Power Slits

With pulse energies in the millijoule range, the X-ray beam can cause some damage to all materials introduced in it. There is no material that would not vaporize in the focused FEL beam. For example, 30 cm thick pure copper was perforated in less than 30 s, and stainless steel is also riddled with ease. Only some ceramic materials, such as B_4C , could offer some resistance. However, even the unfocused beam is not harmless for all materials.

The *power slits* (PS) provide an optical aperture to attenuate, shape or collimate the X-ray beam. Furthermore, they can block and absorb unwanted parts of the beam, e.g. parasitic scattering from upstream optics and spontaneous radiation. Commercially available X-ray slits were redesigned for the EuXFEL, consist of 8 cm long and equally wide B_4C blocks tilted at a small angle (8°) to the beam. Each of the four slits (two in horizontal direction and two vertical) is individually controlled and allows the user to set a fixed aperture. E.g. the spatial jitter (Sec. 6.2.2) can be reduced or eliminated, but at the expense of transmitted intensity.[31, 32]

6.4.5 Solid attenuator assembly

The FXE instrument is designed to take the full intensity (minus some unavoidable losses) of the X-ray beam delivered by the undulators onto the sample. However, in some cases it is necessary to attenuate the beam to protect the sample, beamline components or detectors from damage. This is done by the *Solid Attenuator Assembly* (SAA, Fig. 6.17), which can attenuate the intensity of the beam as a whole by up to several orders of magnitude. In contrast to the power slits, the beam profile is not changed.

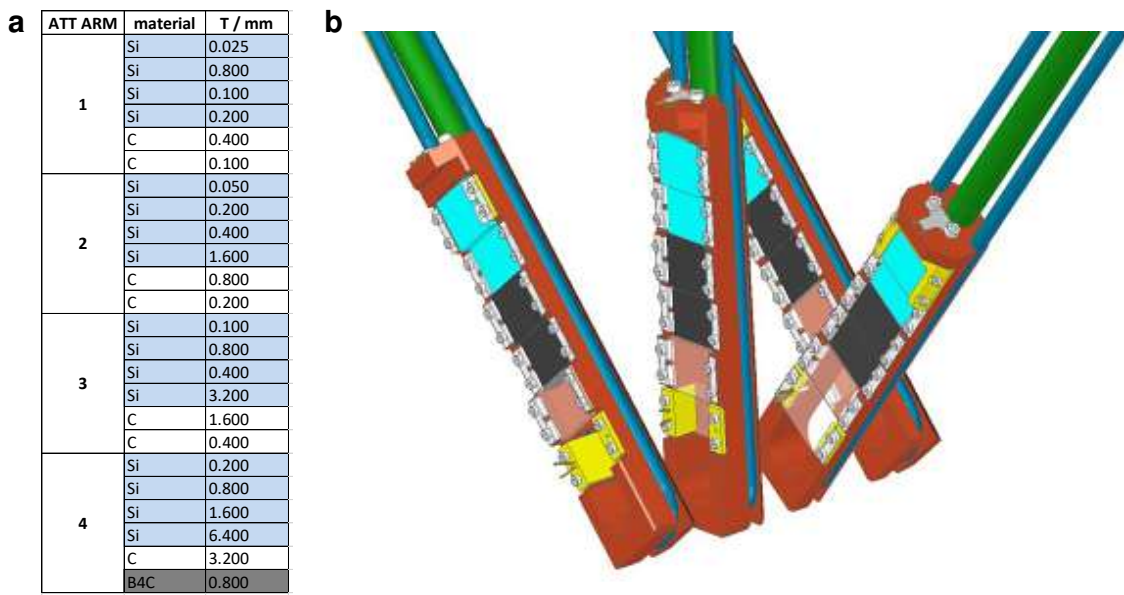


Figure 6.17: (a) Assembly of the 4 arms of the attenuator unit SAA with selected materials and their thickness.[32] (b) Arrangement of the 4 attenuator arms in SAA, here the X-rays run from left to right. (From [32])

Foils made of diamond (C^*), silicon (Si) and boron carbide (B_4C) of different thicknesses are available. By combining the 4 available arms, nearly any desired attenuation can be set at any possible X-ray energy. It must be noted that thicker Si ($> 200 \mu m$) is never used in the full beam, i.e. the beam must first be attenuated with the first (and perhaps the second) arm with diamond until the intensity is about 1 % or less. The current configuration of each of the 4 arms is given in Fig. 6.17. Behind this unit (and the PS1) the IPM1 and BIU2 are arranged as beam diagnostic components to control the beam properties.[31, 32]

6.4.6 Time Arrival Detector

In pump-probe experiments, knowledge of the relative time shift between the two light pulses is of crucial importance. In the laser laboratory, this is achieved with sufficient precision using a beam splitter and a precisely adjustable linear stage with a reversing mirror for one of the two beams (the only question then is at which shift do both light pulses actually arrive simultaneously). This is more difficult to ensure with two independent light sources, and with an FEL and an optical laser the light pulses, generated many hundreds of meters apart, is a challenging task. This is mainly due to the thermal stability of all optical elements in both light sources, i.e. as far as possible, all mirrors used should not expand at all, and that requires high thermal stability. Over the years, this stability between an optical laser and an FEL machine has been guaranteed to be around 250 fs to 500 fs, and in some cases, stabilities of the relative arrival time between optical laser and FEL pulses of less than 100 fs have been demonstrated. This stability has to be guaranteed on the nanosecond to microsecond scale of the single arriving pulses of a pulse train as well as on the minute and hour scale, which proves to be difficult to impossible.

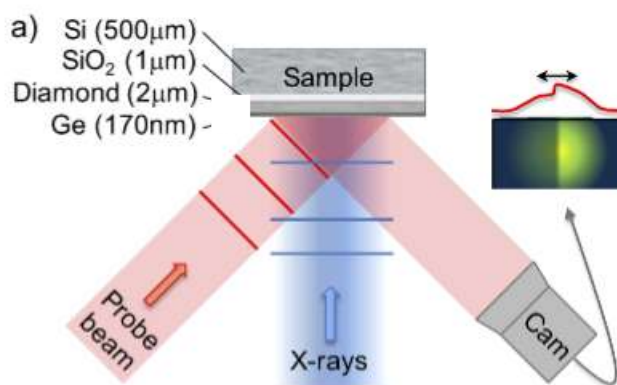


Figure 6.18: Example of a setup for spatial decoding of the relative arrival time of an X-ray pulse. (From [?]).

An alternative to ultrastabilization of the relative arrival time is a measurement of the relative arrival time for each individual X-ray pulse, and this method is also used worldwide. The basic setup is to determine the relative arrival time between FEL and laser pulse as close as possible to the sample location, and two methods

have been established for this purpose: one based on the spatial encoding scheme and another based on spectral decoding within a so-called chirped pulse.

Spectral decoding of the relative arrival time

Here, the X-ray and laser light pulses are overlapped on a specimen (transparent foil) with an (adjustable) angle (Fig. 6.18). At the position on the sample, where part of the optical pulse overlaps perfectly in time with part of the X-ray pulse, time zero is found (afterwards). The X-ray pulse transmits through the sample and generates many charge carriers in the conduction band. These in turn reflect optical light better, and one will measure an increased reflectivity for all regions of the optical pulse that pass through the sample after the X-ray pulse. This can be imaged optically, and one can find an edge at time zero via the lateral distribution of the reflected intensity. If the relative arrival time of the next pair of pulses shifts, then the location of this edge in the reflected optical image shifts accordingly. The beam cross-sections are then a measure for the time range covered, and the angle set is a measure for the time resolution.

Spectral decoding of the relative arrival time

Here, one exploits the relative arrival time of the different colors in a so-called chirped optical pulse. The chirped pulse, which is about 1 ps long, overlaps almost colinearly with the X-ray pulse on the foil. After the arrival of the X-ray pulse, all subsequent colors of the optical pulse are reflected more significantly, and this also leads to an intensity edge in the optical spectrum (Fig. 6.19). Different relative arrival times make this edge appear at different wavelengths, correspondingly. After calibration, a measure of the relative arrival time between each optical and X-ray pulse pair is also obtained.

Some remarks on these arrival time measurements

Basically one uses a length measurement for the time measurement. Here, light travels a distance of 30 μm in about 100 fs. In order to determine the time to an accuracy of about 1 fs, the paths must be known or stabilized with an accuracy of 0.3 μm . This is then necessary for ALL beam paths: for the laser beam path to the sample, for the path for the arrival time measurement, but also for the X-ray pulse. For the spatial decoding, a corresponding spatial resolution is necessary, slightly relaxed by the trigonometric conditions of the angled X-ray and laser pulses. For the spectral decoding, an exact knowledge of the chirp along the laser pulse is necessary. Both measurement methods are implemented in the program. In reality, each pump-probe event (i.e. each single shot measurement) is provided with a time correction, which is derived from these arrival time measurements. This results in an enormously extensive evaluation, the time correction of which must be carried out first before the corresponding time-aligned measurements (sorted into so-called time bins) can be averaged. The typical pulse-to-pulse time jitter is about 200 fs to 300 fs (FWHM), for some stabilized FELs (PAL-FEL, EuXFEL, SwissFEL) it can be close to 100 fs (FWHM). To be able to precisely break down ultrafast processes within

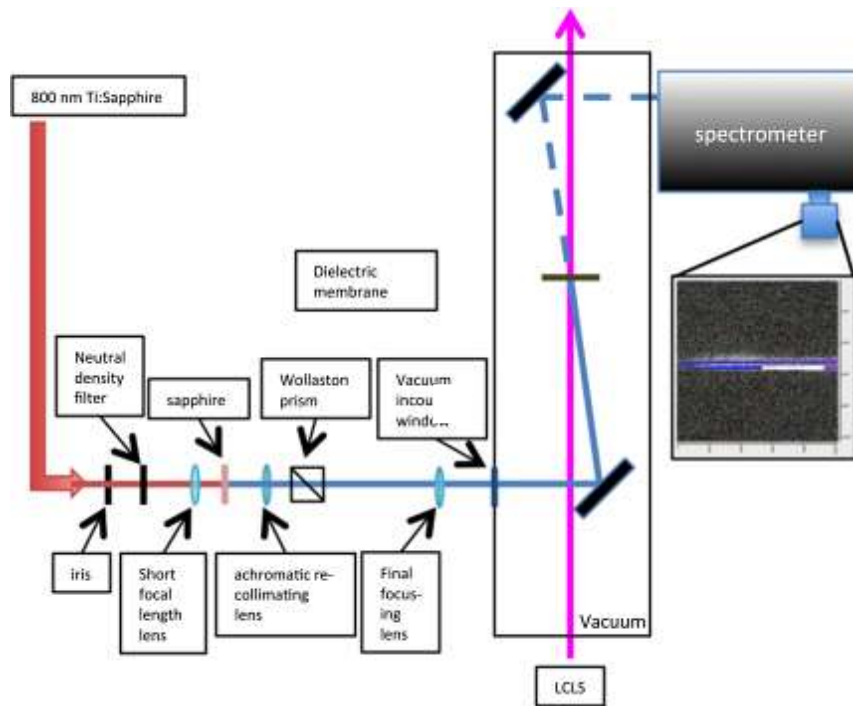


Figure 6.19: Spectral decoding of the arrival time of the FEL pulse. (From [?]).

the first 100 fs, we need these time measurements. For slower processes (> 1 ps) this is no longer necessary.

In this VR program, we use the first diffraction order beam generated by the diamond grating (in the tunnel). The associated longer travel distance (compared to the main beam used for the actual measurement) is quite small, but, most of all, geometrically precisely defined. This means that the time measurement is still precise. With the measurement of an ultrafast event around time zero, it is a constant problem to mark this time “zero” exactly. To do so knowledge of the physical process is indispensable. The measuring methods of the relative arrival time are still merely relative measuring methods. The arrival time in/situ at the sample is not measured. And even then, a precise knowledge of the physical process is required. For example, the macroscopic heating of the sample is usually delayed relative to the arrival of the pulse, because it must first expand in the entire sample. At the precise time zero, a very small signal is measured, which rises to a maximum value within 1 ps. This time zero is therefore somewhat to the left of the pulse. This time zero is thus somewhat left of the actual signal rise. On the other hand, a change in electronic absorption becomes visible very promptly (e.g. after laser ionization of a 5p electron in xenon, the 1s or 2s-5p transition becomes possible immediately). Then the time zero will be seen rather as the mean rise of this signal.

6.4.7 Compound Refractive X-ray Lenses

Compound refractive lenses (CRLs) are, next to curved mirrors, commonly used at synchrotron and FEL beamlines to focus X-ray beams to sizes in the micrometer – and occasionally sub-micrometer – range, as is required for many experiments. For

example, a small spot size results in large X-ray fluence and thus increased diffraction signal in protein crystallography. Furthermore, a small spot size is required for an optimized angular resolution on the detector during diffraction experiments to prevent blurring of the diffraction spots. In pump-probe experiments, the X-ray beam should be focused smaller than the optical laser beam (usually with a Gaussian intensity distribution), leading to a reasonably constant concentration of excited sample molecules in the probe-spot volume.

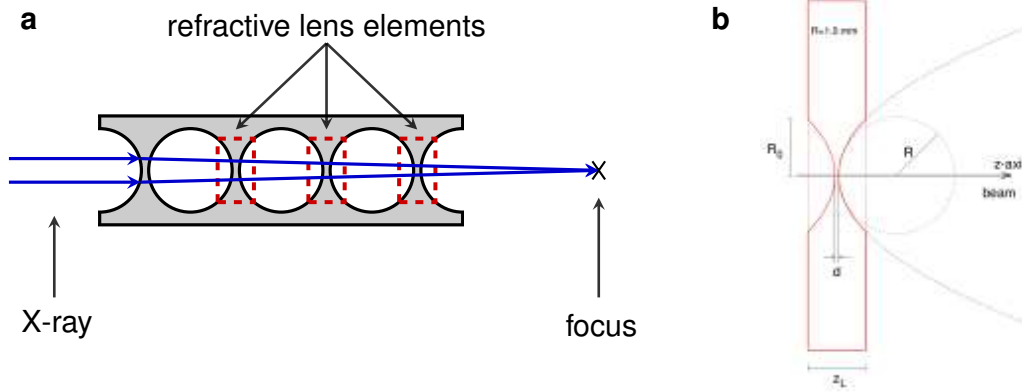


Figure 6.20: (a) X-ray focusing by a CRL. Various cylindrical holes in an aluminum block acts as a chain of refractive concave X-ray lenses (red dashed boxes), and focuses a hard (5 keV to 20 keV) X-ray beam in one dimension (Adapted from [38]). (b) Cut through a single CRL lens for EuXFEL: a $Z_L = 1$ mm thick heated block of Be is pressed by two punches to a concave parabolic lens, shown for a radius of curvature $R = 1$ mm. In contrast to a), the curvature is 2-dimensional, as is the focus. The thickness d at the center is only about $30 \mu\text{m}$ (from [39]).

To understand the working principle of the CRLs, we look at the dimensionless, wavelength-dependent index of refraction $n(\lambda) = n_R(\lambda) + i \cdot n_I(\lambda)$, an optical property of a material. The real part n_R describes the refraction and the imaginary part n_I the attenuation. The refractive index measures the *phase velocity* of light and can drop to values less than unity under certain circumstances, as it is the case in the X-ray regime. In this region of so-called anomalous dispersion, we express $n(\lambda)$ as [22],

$$n(\lambda) = 1 - \delta(\lambda) + i \cdot \beta(\lambda) \quad (6-10)$$

with the wave attenuating imaginary part of the absorption index β and the refractive index decrement δ . δ assumes very small positive values for all materials, e.g. for Be it varies between 1.6×10^{-8} at 12 keV and 1.5×10^{-10} at 25 keV [39]. In consequence, the X-ray refraction indices are lower than but very close to unity. So the refractive power is extremely low and focusing is performed with concave-shaped lenses. Optical lenses, in contrast, focus with convex lenses.

A rather simple implementation of CRLs is a linear arrangement of holes - each with only a radius of several hundred μm - perpendicular to the X-ray beam in an aluminum block (Fig. 6.20a). The thin concave walls between the holes act like a series of connected lenses. The refraction of X-rays through such a single lens is still small, but a compound of tens or hundreds of lenses can readily focus X-rays in

one or two dimensions. Nowadays, parabolic single lenses made of Al, Be or Li are commonly used (Fig. 6.20b). The low atomic number of these materials results in low X-ray absorption and avoids absorption edges with extremely varying refractive indices.

The focal distance F of a single refractive X-ray lens depends on the lens radius R , with $F = R/2\delta$. Due to the small values of δ it is necessary to stack several lenses in series to focus at a finite distance. The focal distance of N lens elements is given by [39]:

$$F = \frac{R}{2N\delta} \quad (6-11)$$

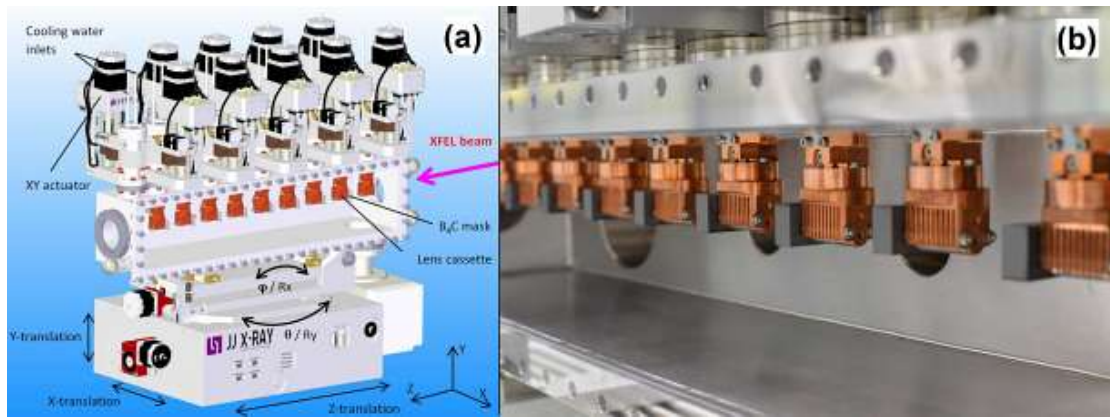


Figure 6.21: (a) The CRL unit of the MID instrument, which is almost identical to that one of FXE/VLab. 10 individual retractable actuator arms are mounted on a linear stage that can be moved in three degrees of freedom for fine adjustment of the focus. A water cooling system ensures temperature stability. (b) Each CRL cassette can hold up to 10 individual lenses. The focus can be set for each energy in the 5 keV to 20 keV range using a suitable lens combination. Apertures made of B_4C in front of each cassette (gray rectangular blocks, as the X-ray beam runs from left to right) protect the copper holders from intense X-rays during insertion and extraction (From [40]).

The FXE/VLab CRL (Fig. 6.21) device for tunable beam focussing is located 5 m upstream from the sample. It provides 10 retractable carrier cassettes, each of which can be loaded with a specific configuration of up to 10 individual Be lenses with different radii of curvature. By using various lens combinations (Tab. 6.1), the beam size can be collimated depending on the X-ray energy (correction of chromatic aberration). Optimization calculations show how many lenses with which different radii should be selected for each arm. The implementation of CRLs needs little space and keeps the optical axis in place (generates no beam offset).

Table 6.1: List of lenses mounted at FXE (and VLab) CRL. N is the number of lenses in each cassette and R denoting the central curvature of the lens. Arm 10 is situated most downstream (leftmost in VLab) and Arm 1 is the most upstream.[32]

Arm	10	9	8	7	6	5	4	3	2	1
N	10	10	8	7	5	9	4	10	6	1
R/mm	0.5	0.5	0.5	0.5	0.5	1.0	0.5	1.5	1.5	0.5

In addition, the entire CRL unit can be moved by up to 50 cm in the direction of the beam for fine focus adjustment. The effect can be shown by caustic scans around the sample position and a focal spot sizes of about $10\text{ }\mu\text{m}$ (FWHM) can be obtained in both horizontal and vertical (Fig. 6.22).

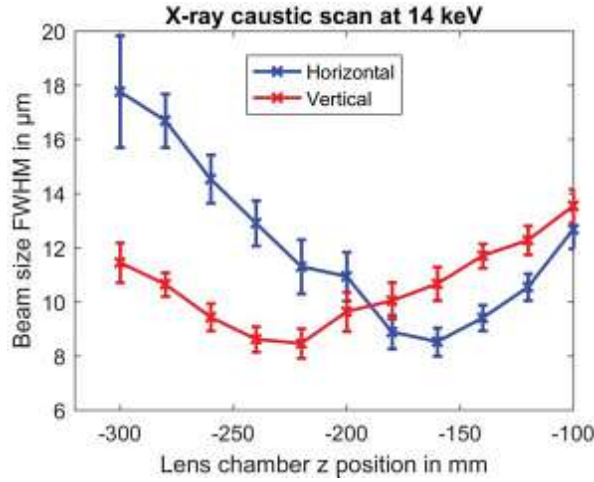


Figure 6.22: A caustic scan shows the horizontal (blue) and vertical (red) focal properties. The data were obtained at the sample position by moving the entire lens chamber in the beam direction. The error bars indicate a 1σ standard deviation of the single-pulse X-ray beam size. An astigmatism is seen in this measurement and a minimum focus of less than $10\text{ }\mu\text{m}$. (From [31])

An additional CRL unit in the tunnel (Sec. 6.3), 229 m behind the Undulator collimates the divergent FEL radiation from the undulator. [22, 31, 32, 38, 39]

6.5 FXE Sample Environment

The actual experiment takes place in the *sample environment*. On the sample table, the pump laser and the probing X-ray beam hit the sample. On the side are the secondary spectrometers and detectors, and behind them the Large Pixel Detector. A few meters behind the sample there is a downstream beam diagnostic unit, which provides further information on X-ray spectrum, intensity and position (Fig. 6.3 and 6.23).

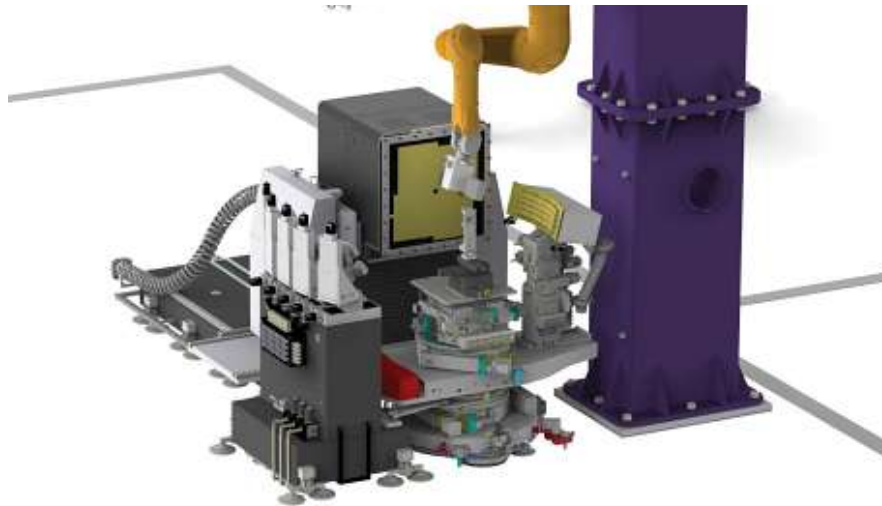


Figure 6.23: Overview of the sample environment. In the center is a goniometer table on which the liquid jet injector is mounted, to the left of it is the Johann spectrometer, and to the right the Von Hamos spectrometer. The robotic arm (yellow) attached to the tower (blue) holds the APD (Johann), Gotthard (von Hamos) and Virgo (both) detectors needed for the spectrometers. In beam direction behind the sample is the LPD for forward scattering. (From [31])

6.5.1 Goniometer Table for the Sample

The sample stage of the FXE experiment must be flexible and adaptable to the different properties of the samples and the different spectroscopic methods with the highest precision. Crystalline and other solid samples can be precision-positioned on a *goniometer stage* with respect to the beam angles and coordinates. For liquid samples, a *liquid beam injector* is mounted on the stage. The sample is either in atmospheric air, in vacuum, or in a chamber under inert gas atmosphere (He, possibly also Ar, N₂, or others). Cooling and heating devices to adjust the sample temperature are also available.

Diffraction studies on crystalline samples

With a φ -circle and two cradles (ω and χ), the sample can be adjusted for diffraction experiments on solid samples (not subject of this experiment) in grazing incidence, reflection or transmission geometry. In addition, there is an ω and χ -scanner (each within a 50 x 50 mm area) for 10 Hz positioning of solid samples to minimize falsifying effects of radiation damage. The X-ray beam is always defined as z -axis, positive in beam direction (y vertical). For experiments in grazing incidence, close to total reflection, surfaces can be compensated with micrometer accuracy. Full imaging in all diffraction directions can be sampled using a detector mounted on the robotic arm.

Liquid jet experiments

A continuous supply of a fresh sample for each pump-and-probe event by liquid jet injectors (Fig. 6.24b) considerably reduces/prevents otherwise disturbing effects

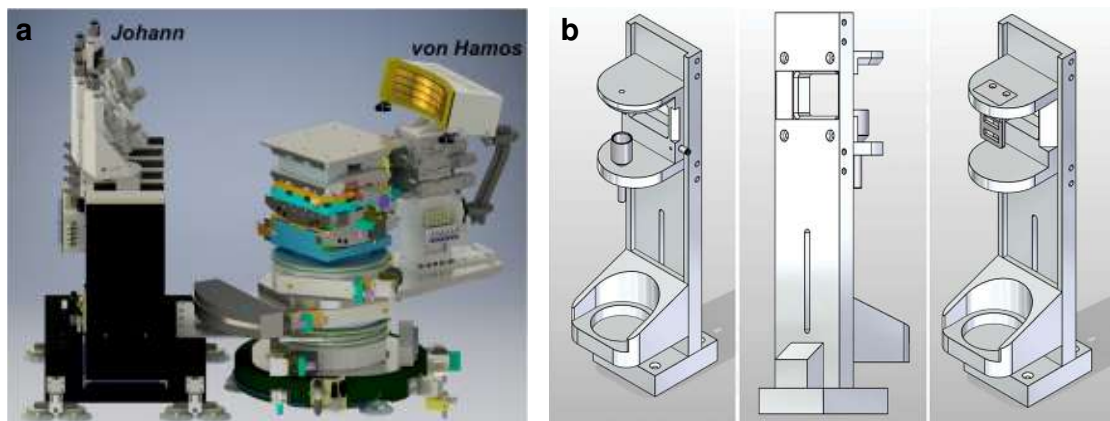


Figure 6.24: (a) The goniometer stage in the center, and laterally the two rotatable secondary spectrometers. A stacked set of different rotation and translation stages allows the alignment and also the controlled shifting of the samples. (b) In addition to the nozzle, the liquid beam injector mounting fixture can also place devices for beam characterization or diagnostics, such as: Ce:YAG or diamond film screens (the rectangular holders on the far right), with positional accuracy ($\pm 10 \mu\text{m}$). (From [31]).

of radiation damage. The supply is expected to function reliably even at high repetition rates. In this process, the liquid is fed through a nozzle via a reservoir in a closed circuit. The temperature of the solution should be adjustable to within 0.2 K. Rectangular sapphire nozzles (8 mm wide) with jet thicknesses of 0.1, 0.2, 0.3 and 0.5 mm are currently available. This creates a liquid jet with a flat (non-curved) surface, which is advantageous for optical excitation. The available HPLC and other pumps allow flow velocities of up to about 30 m/s with flow rates of up to 500 mL/min with these nozzles. The choice of appropriate pump and nozzle combinations are also suitable for many liquids, including aggressive ones. Very common are water, acetonitrile and certain alcohols. In addition, a device for a high-speed jet at 10 - 70 m/s with adjustable thicknesses of 1 - 100 μm is also available, but currently limited to water as a solvent.

6.5.2 Secondary Spectrometers

The energy-dispersive detection of emission spectra is performed by diffraction on crystals according to the Bragg conditions (eq. 4-2), for energies above about 2 keV (or $\lambda < 6 \text{ \AA}$). The two spectrometers used in the FXE sample environment operate with slightly different principles: the Von Hamos spectrometer is a dispersive spectrometer type, the Johann spectrometer is a monochromatic one. The Von Hamos is better suited for recording an entire spectrum, the Johann is better for single small wavelength ranges of interest. Due to the quasi-simultaneous recording of the entire emission spectrum, the Von Hamos does not require normalization to the accumulated incident intensity for now. The Johann spectrometer measures very efficiently the intensity of a single wavelength (about 100 x more efficiently than the Von Hamos). It requires normalization to the accumulated incident intensity at each measurement point to measure the *whole spectrum*. This necessity for normalization

(or the lack of that necessity) is the normal case with every kind of X-rays, finally synchrotron radiation.²²

The Von Hamos spectrometer

The Von Hamos spectrometer (chap: 4.5) used on the FXE experiment consists of up to 16 cylindrically curved crystals (Fig. 6.25). These are each attached to three adjustable mounts and so can be individually tilted horizontally and vertically. In addition, the distance to the sample as a whole can be adjusted. For different emission energy ranges, optimally matched crystal materials and reflection planes are available. In addition, the spectra of different chemical elements in the sample can be recorded simultaneously by skillful combination of the crystals (e.g. 8 crystals aligned to the Fe-K α emission lines and further 8 to the Cu-K α lines). The selection of the crystals is made taking into account the available imaging area on the detectors. Depending on the application, a MHz-compatible strip detector (GOTTHARD, Chap. 6.5.3) or a 10 Hz compatible pixel detector (JUNGFRAU, chapt. 6.5.3) are suitable.

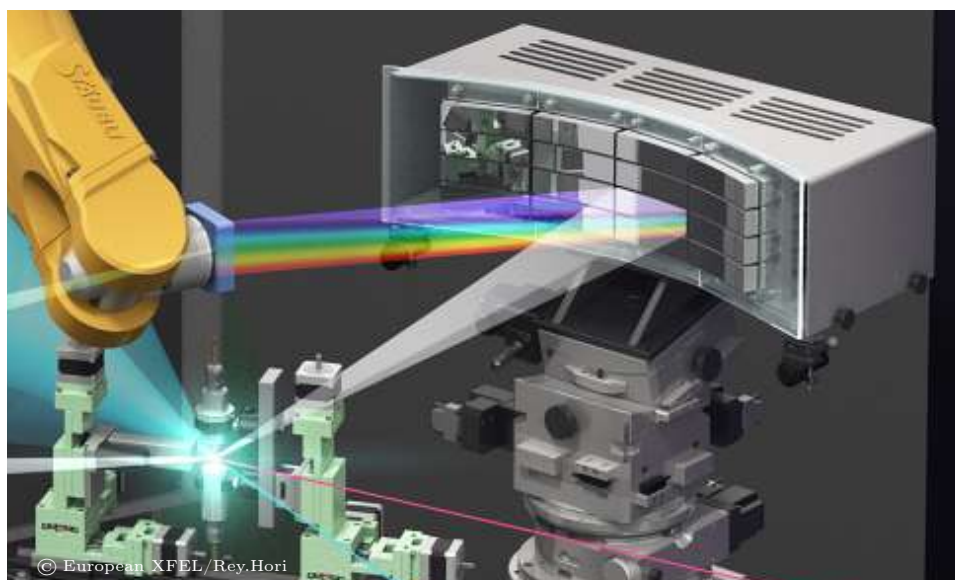


Figure 6.25: The Von Hamos spectrometer in action: as an example the vertical dispersion of the different emission wavelengths is shown on one of the 16 crystals. A detector attached to the robot arm (yellow) records the dispersed emission spectra. (However, the light beam between the crystal and the detector widens much more than shown here).

Johann Spectrometer

In *Johann geometry*, a spherically curved crystal focusses the energetically dispersed emission spectrum onto a detector or photodiode (Figs. 6.26b and 6.27).

²²Focused FEL radiation also enables novel phenomena: e.g. in the very high intensity ranges, nonlinear processes become possible, where e.g. stimulated X-ray emission is observed, within the extremely short spontaneous lifetime of about 1 fs.

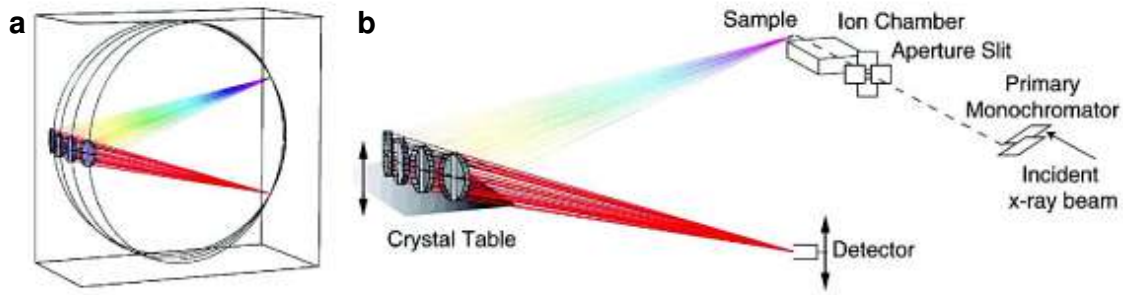


Figure 6.26: (a) A Rowland circle. (b) Construction and operation of a Johann spectrometer. Polychromatic emission from the sample (shown in rainbow color) strikes the spherically curved crystals at a fixed Bragg angle, and only monochromatic light is reflected onto the detector. In order to record the other emission wavelengths and to obtain a spectrum, the angle of incidence on the crystals must be moved stepwise. (From [44])

If the source of the X-ray emission as well as the analyzer crystal are located on the so-called *Rowland circle* (Fig. 6.26a), a theoretical circle with half the radius of the crystal and tangential to it, then the reflected radiation will be focused on another point of this circle. The Bragg condition is fulfilled only for one wavelength at a time. On the other hand, its solid angle range is about 2 orders of magnitude larger than that of a Von Hamos (or cylindrically curved) crystal of about the same size, and leads to a higher photon collection rate. To collect a full spectrum, the crystal and possibly the detector must be moved along the Rowland circle. This is done in a *scan* by a fine rotation of the spherically curved crystal as the imaging location shifts accordingly on the detector. [45].

The Johann Scanning Spectrometer (JSS, Fig. 6.27 and cover picture) at the European XFEL is equipped with five slightly curved analysis crystals made of silicon or germanium. The analysis radius (Rowland circle) is usually 1 meter, but crystals with radii of up to 2 meters are also available if needed. Each crystal is independently adjustable with four degrees of freedom. The entire spectrometer is installed on a movable granite block.

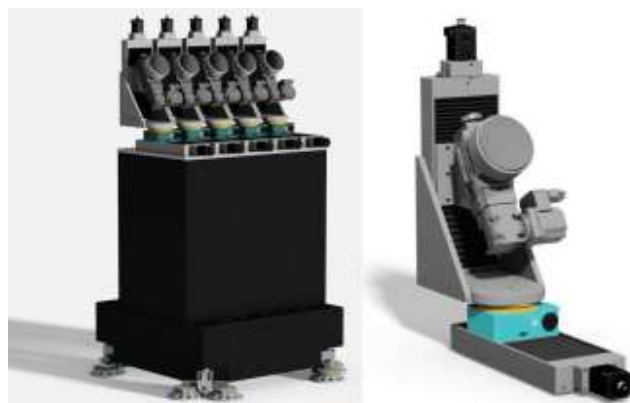


Figure 6.27: The monochromatic Johann spectrometer. Emission source, reflection crystals and focus are on a Rowland circle. Up to 5 mirrors can be used simultaneously to enhance the solid angle of detection.

There are 3 types of detectors (chap. 6.5.3): i) an avalanche photodiode is a simple detector without spatial resolution (0D detector). For this, the robot moves step-wise along the Rowland circle during the rotational scan. ii) With the help of a streak or line detector (1D detector), which, skillfully placed, can record the entire spectrum during the rotational scan by a secant corresponding to the Rowland circle traversed without any necessary movement. iii) A pixel detector (JUNGFRAU) fulfills the previous condition with 2D spatial resolution.

6.5.3 Detectors for X-Rays

Different types of detectors are used to detect X-ray light. The simplest are photodiodes, but they give very limited information about the emitted intensity. For processes, where one would like to have spatially resolved information, strip detectors or 2D pixel detectors are used. Most detectors are made of silicon, in which X-rays generate conduction band electrons and their charge quantity is registered. The sensitivity of these Si-based detectors is therefore also energy-dependent, and at higher X-ray energies around 20 keV, sensors of 0.5 to 1 mm thickness are needed to measure a sufficient signal.

The avalanche photodiode

Avalanche photodiodes (APDs) are detectors without spatial resolution but with an internal gain dependent on the operating voltage. In principle, they can be considered analogous to photoelectron multiplier detectors, which are the laboratory standard in optical spectroscopy. They have proven to be extremely useful in X-ray spectroscopy for the entire energy range of 1.5 - 20 keV.

APDs exploit an *avalanche effect* for signal amplification (Fig. 6.28): the diodes are operated in reverse bias, i. e. the entire applied voltage is present at the pn junction without any significant current (only a few nA). If conduction band electrons (more precisely: electron-hole pairs) are generated by impact ionization with an X-ray photon, they are then accelerated in the pn junction and in turn lift further electrons into the conduction band. This process is repeated several times until the electrons move over into the n-region without any further significant voltage drop. This leads to a short steep (about 0.5 ns long) current rise. The drop is determined by the RC combination (approx. 5 - 20 ns). This current pulse can be transported via a (50 Ω) BNC cable to an oscilloscope, which records a corresponding voltage pulse. Intensities of the incident X-ray photons are measured with a very good S/N ratio, which also enables sensitive detection of single X-ray photons (similar to a photoelectron multiplier for visible light).

The APDs used here are large area APDs (LAAPD) with 10 x 10 mm². They operate at a reverse voltage of about 200 V, and with a breakdown voltage of about 380 V. The signals are read out with a fast 12-bit analog-digital converter. The temporally short pulses of 20 - 50 ns are shorter than the 220 ns minimum pulse separation time. Thus, the APDs are able to resolve single X-ray pulses even at the maximum possible 4.5 MHz repetition rate (within one pulse train). The PIN diodes

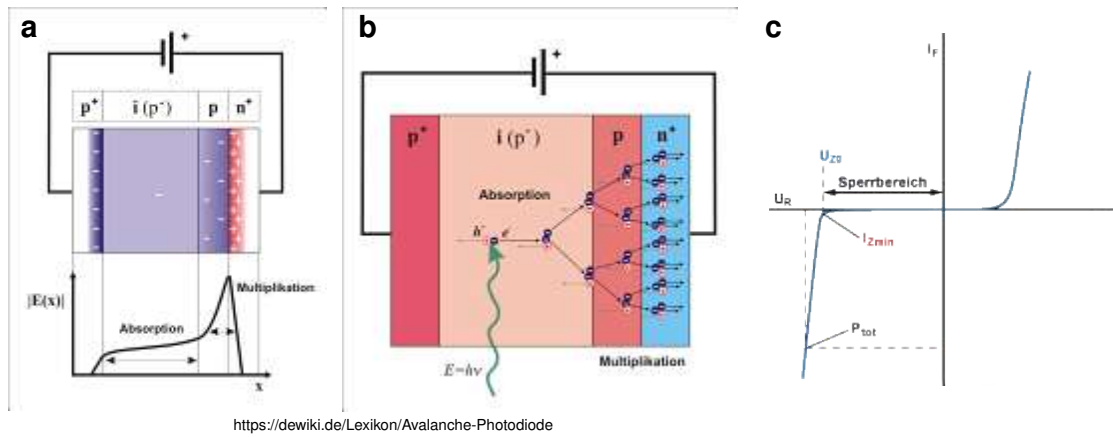


Figure 6.28: (a) Basic structure of the avalanche photodiode (IPM). (b) The highest voltage is applied at the pn junction, in this area additional charge carriers are generated by impact ionization. (c) Voltage drop within an APD.

used in other FELs have a similar structure, but usually have microsecond-long signals per pulse, and are therefore less suitable for the European XFEL.

GOTTHARD

We use two types of detectors developed at the Paul Scherrer Institute: GOTTHARD and JUNGFRÄU (see next chapter).²³ GOTTHARD (not yet commercially available) is the further development of the predecessor model MYTHEN. It is a strip detector, with 8 mm long strips and 1024 x 50 μm (GOTTHARD 1.0) or 2048 x 25 μm (GOTTHARD 2.0) cells each. This detector can thus record 1D information, e.g. the emission spectrum imaged by the Hamos spectrometer. The internal electronics allows recording single pulses with a repetition rate of up to 0.9 MHz (in the 1.0 version). The currently developed version 2.0 will not only be able to record narrower pixel widths of 25 μm and twice as many of them, but also single pulses at the maximum repetition rate of 4.5 MHz.

JUNGFRÄU

This detector, also developed at PSI, is not yet commercially available. It consists of square 75 x 75 μm² pixels. For this experiment there is a 0.5M and 1M version. The internal electronics only allow the accumulation and storage of a whole pulse train. A possibility under development includes the option to record up to 16 single pulses at MHz repetition rates. Pulse train averaging is possible when the detector is not saturated in the course of the pulse train, and this must be determined beforehand or set in such a way.

²³At the Paul Scherrer Institute near Zurich in Switzerland, various detectors have been developed for over 20 years, originally for high-energy experiments at CERN. It turned out that these detectors are also suitable for the detection of X-rays at synchrotron radiation sources. Meanwhile, some of these detectors have been commercialized (via the spin-off company DECTRIS). The different detector types are named after popular Swiss mountains: e.g. PILATUS, EIGER, MYTHEN, and just GOTTHARD and JUNGFRÄU.

6.5.4 Large Pixel Detector for Forward Scattering

The *Large Pixel Detector* (LPD, Fig. 6.29a) was specifically developed by the Rutherford Appleton Laboratory (RAL) in the UK as the main analytical instrument for scattering and diffraction experiments. It was designed for the special requirements, high photon energy, brilliance and repetition rate, of the European XFEL and the development to the present prototype took about 10 years, a typical development time for new detectors.

The LPD is a 1 mega pixel detector that can resolve single pulses within a pulse train (i.e. at up to 4.5 MHz). The edge length of the square pixels is $500\text{ }\mu\text{m}$ (hence the name “large pixels”) and the detector can be positioned at distances from 10 cm to 8 m behind the sample in the beam direction. In forward scattering geometry, a central hole in the detector allows the destructive directly transmitted beam to pass through the detector without damaging sensors, and thus can be directed onto the beam diagnostic units in He atmosphere.

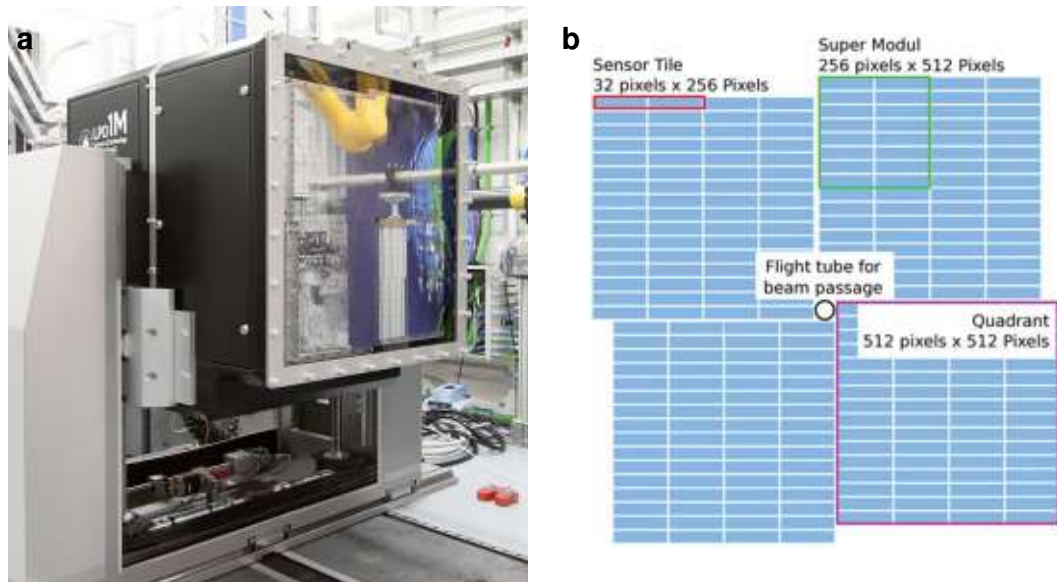


Figure 6.29: (a) The LPD, movable in the direction of the beam (and perpendicular to it), with the central aperture in the middle through which the X-ray beam passes onto the diagnostic table. (b) The sensor surface is divided into four quadrants, which control the size of the central aperture by an iris-like clockwise or counterclockwise arrangement. Each quadrant consists of 4 super-modules, each with 16 tiles of 128×32 pixels. (From [31])

The detector-sensitive $50 \times 50\text{ cm}^2$ square area consists of 1024×1024 pixels of $500\text{ }\mu\text{m}$ size, arranged in 256 interchangeable rectangular tiles (128×32 pixels). The area is divided into four quadrants, which can be moved independently in the detector plane to form an adjustable central hole to allow the direct beam to pass through (Fig. 6.29b). Each quadrant in turn consists of four largely independently operating so-called *super-modules*, each with 16 tiles. This modular system ensures a high degree of flexibility with regard to service and maintenance despite constant operational readiness.

The 500 μm thick Si sensors enable a quantum efficiency of more than 80 % in the photon energy range of 1 - 13 keV and roughly 40 % detection efficiency at 20 keV (based on Lambert-Beer). The LPD has - due to the large pixel volume - a high dynamic range of up to 10⁵ photons at 12 keV and at the same time offers a single photon sensitivity (from about 12-15 keV) in the highest pre-amplifier setting. The detector electronics support 3 parallel gain levels (1x, 10x, 100x), and with frame rates of 4.5 MHz, up to 512 full frames can be stored in the detector memory, which are transferred to the memory medium every 100 ms. It thus operates in the burst mode of the European XFEL pulse sequence pattern. A fast frame “veto” function has been added to reject poor quality images based on on-line diagnostics. This increases the overall quality of the image sets stored on the detector in a burst.

6.6 The diagnostic table near the end of the hutch

This bench contains the components BIU4, SpA2, IPM2, as already discussed. In addition, an absolute intensity monitor based on the ionization of rare gases (XGM) is added at the entrance to this bench. The XGM allows measuring the absolute calibrated X-ray intensity. The post-diagnostic bench allows to measure the properties of the transmitted x-ray beam. E.g., using SpA1 and SpA2 before and behind the sample, one can measure (femtosecond time-resolved) XANES spectra of the sample with the pink x-ray beam. Due to the spiky spectral features one will need to average many pulses to achieve a XANES signal with sufficient signal quality. You can observe this by inserting both SpA devices simultaneously.

The final beam stop

The beam is normally focused on the sample, which is 8 m away from the end of the hutch. A sequential set of B4C and a Cu beam stopper is then sufficient to fully attenuate the beam. In case of operation errors, e.g. when focusing with the CRL lenses to the end of the hutch, a so-called „burn-through“ monitor is interfaced right in front of the beam stop materials. Focused x-radiation can drill a hole into this sealed monitor and generate visible light. Once light is detected it promptly switches off the electron photoinjector and thus turns off the x-ray beam. Extensive testing in advance is required to ensure that this - in principle life-threatening - scenario never occurs, and this requires thorough certification by the radiation protection group.

A X-Ray Absorption Spectroscopy

An x-ray absorption spectrum of an absorbing atom within a molecule or in condensed matter exhibits a specific fine structure around its absorption edge, called *X-ray absorption fine structure* (XAFS). This contains information about the geometric and electronic structure in the local surroundings of the absorbing atom. One usually distinguishes between two energetic regions, which are differently analyzed and interpreted (Fig. A.1): the region right around the absorption edge itself is called the *X-ray absorption near edge structure* region (XANES), or sometimes also *near edge x-ray absorption fine structure*, NEXAFS. It contains the pre edge region and the first 20-50 eV above the edge energy. For higher energies this merges into the EXAFS region (see next chapter).

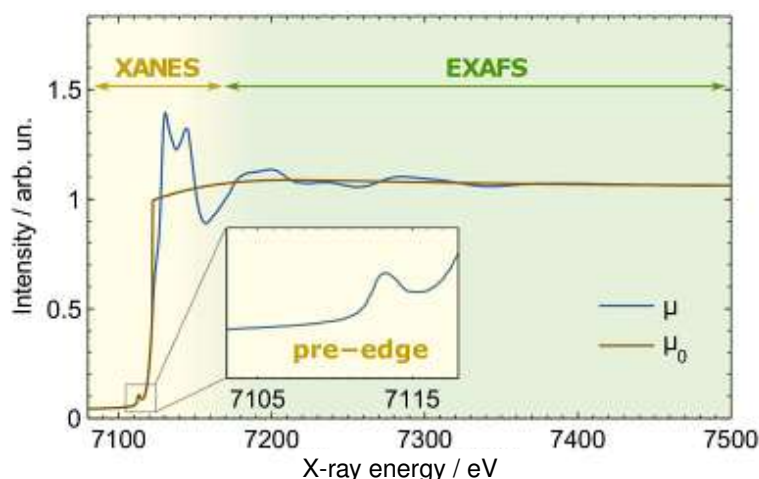


Figure A.1: X-Ray absorption spectrum of $[\text{Fe}(\text{terpy})_2]^{2+}$ (terpy: terpyridin) at the Fe K edge displaying the XANES and EXAFS regions including the background absorption contributions μ_0 . The inset zooms into the weak pre edge features. (Taken from: [24, 25])

Within a molecule or any chemical compound, in general in any condensed matter environment, the x-ray ejected electrons interact with their neighboring atoms, which changes the density of accessible states for these ejected electrons. The density of states can decrease and increase, and this influences according to Fermi's Golden Rule the absorption probability. The Golden Rule states that the total probability for interaction of the exciting x-ray photon with the electron is governed by two main factors: that for the actual light-matter interaction itself and for the excited electron finding an accessible quantum state (in the continuum).

A.1 X-ray Absorption Near Edge Structure (XANES)

If the energy of the photon $E_{\text{X-ray}}$ is equal to the difference between the energy $E_{n,j}$ of a near-edge nuclear electron (with principal quantum number n and total angular momentum j) and that of a not fully occupied valence orbital $E_{n',j'}$,

$$E_{\text{X-ray}} = |E_{n,j}| - |E_{n',j'}| \quad (\text{A-1})$$

the electron can be lifted into this vacancy (Fig. 4.2). These transitions just below the ionization energy are observed in the XAS spectrum as discrete absorption structures in the pre-edge region. Therefore, this region contains information about unoccupied molecular orbitals, here: predominantly with metal orbital character. In the case of a $3d^6$ transition metal complex, the K-edge shows a weak $1s \rightarrow 3d$ transition. This transition is forbidden for O_h symmetries, according to the dipole selection rules $\Delta l = \pm 1$. However, a vibronic transition can cause a $3d$ - $4s$ mixing, which leads to a weak absorption structure in the spectrum in the region of the pre-edge. Its intensity contains information about the symmetry of the molecule, the oxidation state of the metal or its spin states. In contrast to K-edge spectroscopy of $3d$ transition metal complexes, for the L-edges of $4d$ or $5d$ transition metal complexes the pre-edge transitions are partly dipole allowed and cause a more intense absorption [[10, 24]].

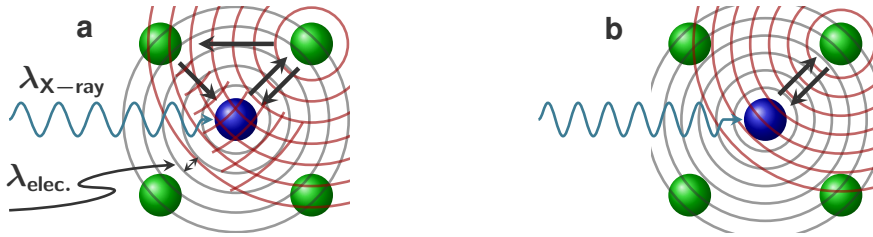


Figure A.2: Emitted (gray) and backscattered photoelectron spherical wave (red). (a) Multiple scattering process (XANES) and (b) single scattering process (EXAFS). (Nach [24, 25])

When the x-ray photon energy is larger than the binding energy of the electron, the excess energy is transformed into kinetic energy via $E_{\text{kin}} = E_{\text{X-ray}} - |E_{n,j}|$ of the emitted *photoelectron*. Its wavelength λ or its wavenumber k is calculated using the De Broglie relation with Planck's constant $\hbar$:

$$k = \frac{2\pi}{\lambda} = \frac{\sqrt{2m_e E_{\text{kin}}}}{\hbar} \approx 0.512 \sqrt{\frac{E_{\text{X-ray}} - |E_{n,j}|}{\text{eV}}} \cdot \text{\AA} \quad (\text{A-2})$$

The photoelectron is ejected as a spherical matter wave and can be scattered off neighboring atoms (more precisely: scattered off their electron density). The backscattering process itself influences the absorption spectrum shape. In the region around 10-50 eV above the absorption edge the ejected photoelectron can efficiently scatter off several neighboring atoms (Fig. A.2a) due to the energy-dependent scattering cross section, which is larger for low kinetic energies. This marks the so called

XANES region. The interference between outgoing and backscattered wave at the original location of the absorbing electron modulates the absorption shape as a function of x-ray energy, and as such delivers information about the local structure around the absorbing atom.

²⁴. These reveal then e.g. bond lengths and angles. The De Broglie wavelengths of low energetic photoelectrons (up to about 25 eV) are then larger than typical atomic bond lengths and this leads to quite complicated Absorption spectra in this energy range, which are often difficult to simulate and to interpret. A workaround is their comparison to striking features in the XANES spectra of reference compounds, allowing the so called „fingerprinting“.

Added information can be extracted from the precise energy of the absorption edge itself, which can deliver valuable information about the oxidation state of the central absorbing atom. When the oxidation state increases (by removal of valence electrons) the shielding of the 1s electron from its nuclear core decreases slightly and this leads to an increase of $E_{n,j}$.

A.2 Extended X-ray Absorption Fine Structure (EXAFS)

When the x-ray energy exceeds the electron binding energy significantly (about 25 - 50 eV above the absorption edge) then the electron leaves the atom as a matter wave with a wavelength that is smaller and - at even higher energies - much smaller than the atomic bond lengths. In this case single scattering events dominate (Fig. A.2b), which can be more conveniently modeled and interpreted than multiple scattering events.

The absorbing atom emits a spherical wave, which is partially backscattered from the neighboring atom, or more precisely: from its potential (Fig. A.2). This partially reflected wave interferes with the outgoing wave, or it interferes with itself. In the ideal case (in contrast to the XANES region) it does not interfere further with other neighboring atoms, this is fulfilled mainly due to the fact that the backscattering amplitude decreases with kinetic energy of the photoelectron. When the outgoing and reflected wave are in phase with each other at the position, where the core electron was originally bound, then electron density has returned to the central absorbing atom. Hand-waving we can say that the electron did not depart from the atom, or in other words: the x-ray absorption cross section is slightly lower than it would be for the isolated atom. At slightly higher kinetic energies this constructive interference gradually turns into a destructive interference, and this leads to a slightly higher absorption cross section. This sequence between constructive and destructive interference repeats itself for increasing energies and leads to a wiggly modulation of the absorption cross section above the edge (Fig. A.1).

The analysis of these modulations in the so called EXAFS spectrum can thus be

²⁴E.g., the XAFS region of gaseous Kr exhibits nearly no fine structure due to the lack of neighboring atoms (it shows a similar spectrum as μ_0 in Fig. A.1). If bound - e.g., via physisorption - onto a graphite surface, one observes a fine structure of an oscillating modulation of the absorption coefficient ([2])

used to determine the bond lengths between the absorbing atom and the back scattering atom. This quantitatively done using the so called EXAFS equation: This introduces the property $\chi(k)$ which describes the pure oscillatory part of the linear absorption coefficient $\alpha(k)$ above the absorption edge and as a function of the photoelectron wave vector k .²⁵ Hereby k is calculated via the excess kinetic energy of the photoelectron via eq. A-2. The quantitative value of $\chi(k)$ is scaled to units of the absorption edge jump (or the difference between the atomic cross section above ($= \mu(E)$) and below the edge considered). The analysis of these modulations in the EXAFS region allows us to determine the distance to its neighboring atoms. The analysis of the modulations in the EXAFS region thus allows to measure the interatomic distances between absorbing atom and its nearest neighbors. The quantitative description is given by the so called EXAFS equation: For this purpose we introduce a property $\chi(k)$, which describes the purely oscillatory part of the linear absorption cross section $\alpha(k)$ relativ to the (atomic) background BG. The wave vector k is given by eq. A-2. $\chi(k)$ is normalized to the absorption edge jump itself, and therefore quantifies the weak oscillations relativ to this absorption edge jump. A measured full EXAFS spectrum in energy space is then transferred into k-space using $\mu_{BG}(k)$ for the atomic background signal (sometimes called $\mu_0(k)$) to generate $\chi(k)$, see Fig. A.1). This is then normalized to the absorption edge jump (set to unity) by subtracting the μ below and above the absorption spectrum of the free atom. This yields the correct normalization of $\chi(k)$ via

$$\chi(k) = \frac{\mu(k) - \mu_{BG}(k)}{\mu_{BG}(E_0)} \quad (\text{A-3})$$

The EXAFS equation describes then the oscillations in the term $\sin(2kR_j + \delta_j(k))$ by the nearest neighbor distances R_j :

$$\chi(k) = \sum_j \frac{N_j}{kR_j^2} f_j(k) \sin(2kR_j + \delta_j(k)) e^{-R_j/\rho(k)} e^{-\sigma_j^2 k^2} \quad (\text{A-4})$$

Hereby N_j describes the number of atoms in the coordination shell j (defined by exhibiting the same distance R_j), $f_j(k)$ is the energy (thus k) dependent backscattering amplitude of these atoms in that coordination shell, where the electron wave experiences a phase change δ_j for these identical atoms, $e^{-\sigma_j^2 k^2}$ is the term containing the Debye-Waller factor σ_j , and $\rho(k)$ describing the mean free path of the emitted photoelectron. This is usually given by the universal mean free path curve for photoelectrons in solid matter. Usually the parameters N_j , σ_j and R_j are free fit parameters, while the backscattering amplitudes and phases utilise tabulated or calculated values. Fourier transformation of the EXAFS equation (A-4) then yields the radial distribution function including peaks at the atomic distances for each coordination shell.

²⁵The linear absorption coefficient $\alpha(k)$ describes - according to the law of Lambert-Beer - the transmission losses inside a solid material (at constant density). The extinction coefficient $\epsilon(\lambda)$ is a related quantity, which relates to the case of a molecular concentration in a liquid. All these considerations, however, can be transferred to the general description in terms of an atomic absorption cross section σ (in barn or cm^2) (chapter 3.1).

B Photon Flux, Fluence, Brilliance, etc.

Very often, both the spatial distribution and the temporal pulse shape of the laser intensity, including that of an X-ray laser²⁶, is Gaussian and can be described as

$$I(r, t) = I_0 \exp \left[\frac{-2r^2}{w_0^2} \right] \exp \left[\frac{-t^2}{\Delta t^2} \right]. \quad (\text{B-1})$$

Here the variable r describes the distance to the center of the beam and t the time. Furthermore, w_0 is the beam waist, and in the above plot, w_0 corresponds to the distance from the center of the beam (or from $r = 0$) at which the intensity drops to $1/e^2$. The half pulse length Δt is here the time relative to the center of the pulse (by $t = 0$), after which the intensity has dropped to $1/e$. Both Δt and w_0 can be rewritten in FWHMs (full width at half maximum):

$$d_{fwhm} = w_0 \sqrt{2 \ln 2} \approx 1.18 w_0, \quad (\text{B-2})$$

$$t_{fwhm} = \Delta t \sqrt{4 \ln 2} \approx 1.67 \Delta t. \quad (\text{B-3})$$

The *fluence* (flux density) F is then obtained by temporal integration of the intensity over the complete laser pulse

$$F(r) = \int_{-\infty}^{\infty} I_0 \exp \left[\frac{-2r^2}{w_0^2} \right] \exp \left[\frac{-t^2}{\Delta t^2} \right] dt \quad (\text{B-4})$$

$$= I_0 \exp \left[\frac{-2r^2}{w_0^2} \right] \sqrt{\pi} \Delta t \quad (\text{B-5})$$

²⁶However, the temporal SASE pulse structure is as a start dominated by the erratic intensities of the individual *microbunches*, but these are within a Gaussian envelope, or, averaged over several pulses, yield a Gaussian curve.

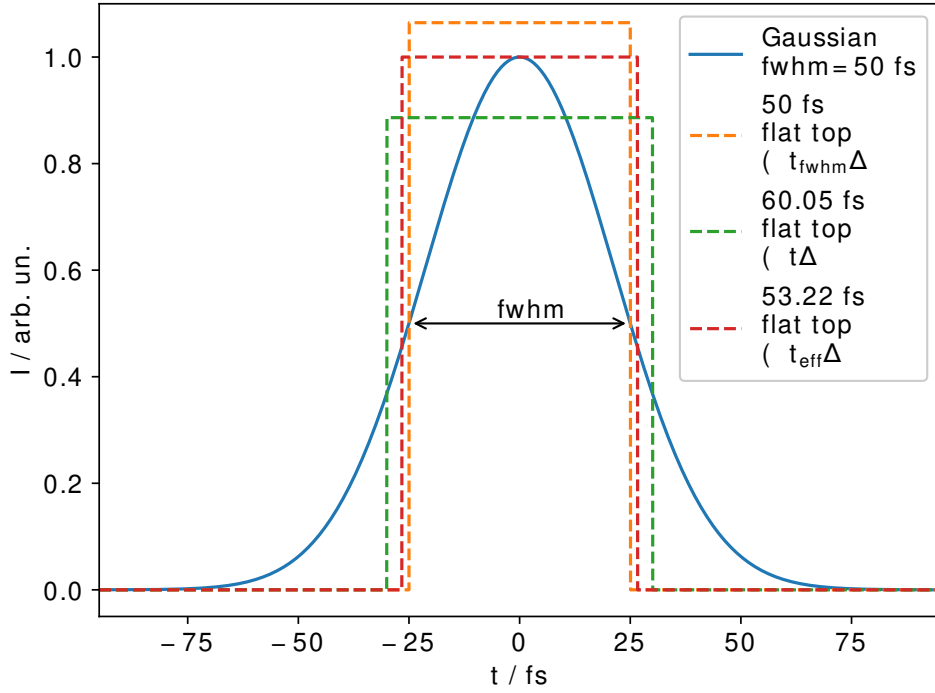


Figure B.1: Gaussian temporal intensity profile of a laser pulse with a pulse length of $2\Delta t = 60$ fs. For comparison, two flat top profile pulse lengths of 50 fs and 60 fs of the same area are plotted.

The fluence is a measure of the areal density of the incident photons and is measured in J/cm^2 . The pulse energy E_p is then obtained by further integration of the fluence over the area (e.g. the lateral extent on the sample under investigation)

$$E_p = \int_0^{2\pi} \int_0^\infty \int_{-\infty}^\infty I_0 \exp\left[\frac{-2r^2}{w_0^2}\right] \exp\left[\frac{-t^2}{\Delta t^2}\right] r \, dt \, dr \, d\rho \quad (\text{B-6})$$

$$= \int_0^{2\pi} \int_0^\infty F(r) \, r \, dr \, d\rho \quad (\text{B-7})$$

$$= I_0 \underbrace{0.5\pi w_0^2}_{\text{eff. Fläche}} \underbrace{\sqrt{\pi}\Delta t}_{\text{eff. Pulslänge}} \quad (\text{B-8})$$

$$= I_0 A_{\text{eff}} \Delta t_{\text{eff}} \quad (\text{B-9})$$

$$= I_0 A_{\text{eff}} \Delta t_{\text{eff}} \quad (\text{B-10})$$

The effective area A_{eff} can then be calculated either with the fwhm width d_{fwhm} or the $1/e^2$ beam waist w_0 can be described,

$$A_{\text{eff}} = 0.5\pi w_0^2 = \frac{\pi d_{\text{fwhm}}^2}{4 \ln 2} \approx 1.443 \frac{\pi d_{\text{fwhm}}^2}{4} \approx 1.13 d_{\text{fwhm}}^2 \quad (\text{B-11})$$

Thus, if the fwhm width d_{fwhm} or the $1/e^2$ beam waist w_0 , each without correction factors, is used approximately to calculate the area of the laser beam, this leads to a deviation of a factor 0.5 or 1.443. Analogously, the effective pulse length Δt_{eff} can

be described both with the fwhm pulse length Δt_{fwhm} and with the $1/e$ pulse length $2\Delta t$,

$$\Delta t_{\text{eff}} = \frac{\sqrt{\pi}}{2} 2\Delta t \approx 1.253 \times (2\Delta t) \quad (\text{B-12})$$

$$= \sqrt{\frac{\pi}{4 \ln 2}} t_{\text{fwhm}} \approx 1.064 t_{\text{fwhm}} \quad (\text{B-13})$$

Thus, approximating the effective pulse length Δt_{eff} by the fwhm pulse length Δt_{fwhm} or the $1/e$ pulse length $2\Delta t$ without correction factors leads to a deviation of a factor 1.064 and 1.253.

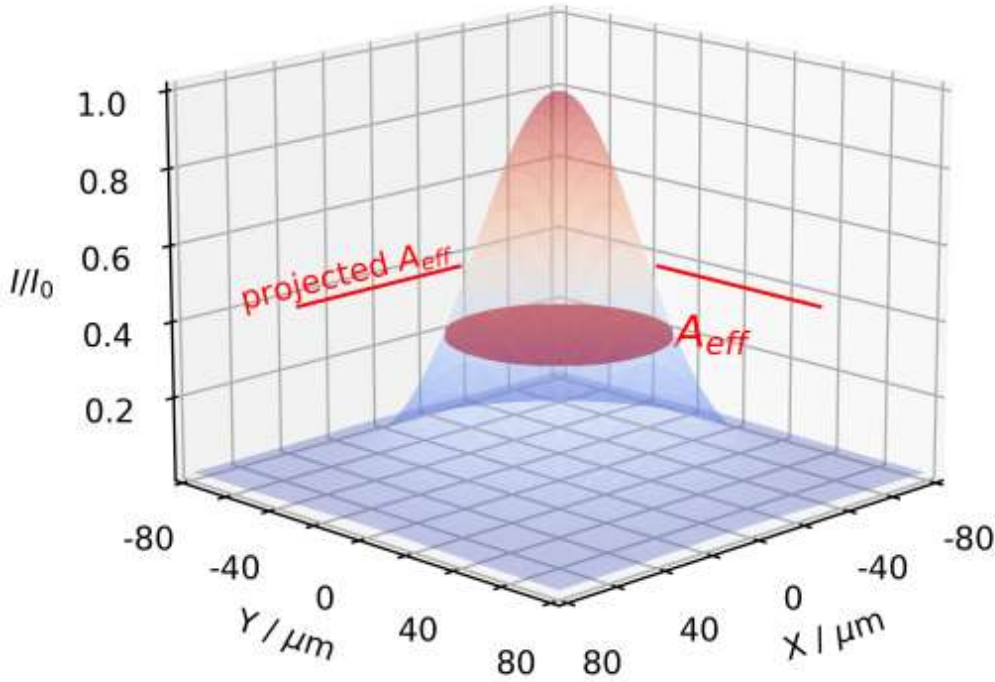


Figure B.2: Gaussian beam profile with $d_{\text{fwhm}} = 50\mu$. The red area denotes A_{eff} .

The *fluence* F of radiation, e.g. on a sample, is a measure of the areal density of incident photons, and is measured in $\text{joules}/\text{cm}^2$. It is given by

$$F = \frac{\text{Laser pulse energy}(E_p)}{\text{effective beam size}(A_{\text{eff}})} \quad (\text{B-14})$$

The beam size can be defined in different ways (Fig. B.3), but for practical purposes we take the half"=width at half maximum (FWHM) as a measure for the beam size.

To distinguish the difference between femtosecond and pico"= or nanosecond long pulses, this eq (A-1) is also divided by the pulse duration. This then contains useful information regarding possible nonlinear excitations (e.g. two"=photon absorption), and we can define a pulse *intensity* as a so"=called *flat top* profile via (see Fig. XY)

$$\text{Pulsintensit\"at}(I_p) = \frac{F}{\text{Pulsdauer}} = \frac{E_p}{A_{\text{eff}} \Delta t_{\text{eff}}} \quad (\text{B-15})$$

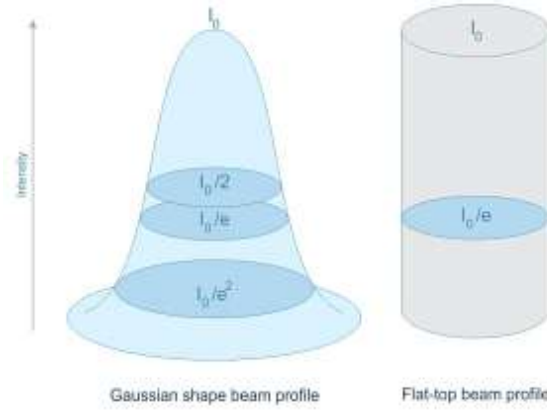


Figure B.3: For the definition of the beam size. In optical laser research, the widths at I_0/e and I_0/e^2 are usually used for the calculation, in X-ray research, areas of the half-width (FWHM), i.e. at $I_0/2$, are common. The 2D Gaussian profile is also often approximated with a so-called flat-top profile, here the widths at $I_0/2$ as well as I_0/e are used, but the integration over the whole pulse profile shows that only the cylindrical profile delivers an identical pulse energy.

For repetitive light sources, such as amplified laser systems with 1-10 kHz repetition rates, or synchrotron light sources with MHz repetition rates, one can use Eq (A-2) in different ways. Once via the single pulse duration, and this then provides the so-called *peak* intensity, or over a longer time period lasting many cycles, and then one obtains an *average* intensity. At the European XFEL, there are two different values for the averaged intensity, on the one hand the intensity averaged over several 10 Hz cycles, but also the more intense median intensity averaged over the duration of the entire pulse train. While the long-time average yields values around 30 W, this intensity averaged over one pulse train is orders of magnitude higher (in the kW range).

Now, in order to estimate the intensity obtained on the sample according to Eq. (A-2) of any light source, we need information about the radiation source. For FEL radiation, the end of the last undulator marks the so-called radiation source. There the beam size (which we now call source size) is about 10 μm , and the natural divergence is a few μrad . These values then determine how well I can focus the outgoing radiation on the sample with ideal optics, according to

$$\text{Brilliance} = \frac{I_0}{A_Q \cdot \Delta\Omega \cdot \Delta E} = \frac{\text{Photonen/sec}}{\text{mrad}^2 \text{mm}^2 (0.1\% \text{B.W.})} \quad (\text{B-16})$$

I_0 is the intensity at the source in Ph/sec, A_Q is the source size, $\Delta\Omega$ represents the divergence of the outgoing radiation, and ΔE denotes the considered (or available) energy bandwidth. A typical energy bandwidth ($= \Delta E/E$) for FEL radiation is about 0.1 %, this is also the approximate value for monochromatized synchrotron radiation. Therefore, light sources are often compared with this normalization, because this can be readily converted into the flux (in Ph/sec), that is available for an experiment. A comparison of the different X-ray sources in these units is shown in Fig. B-4.

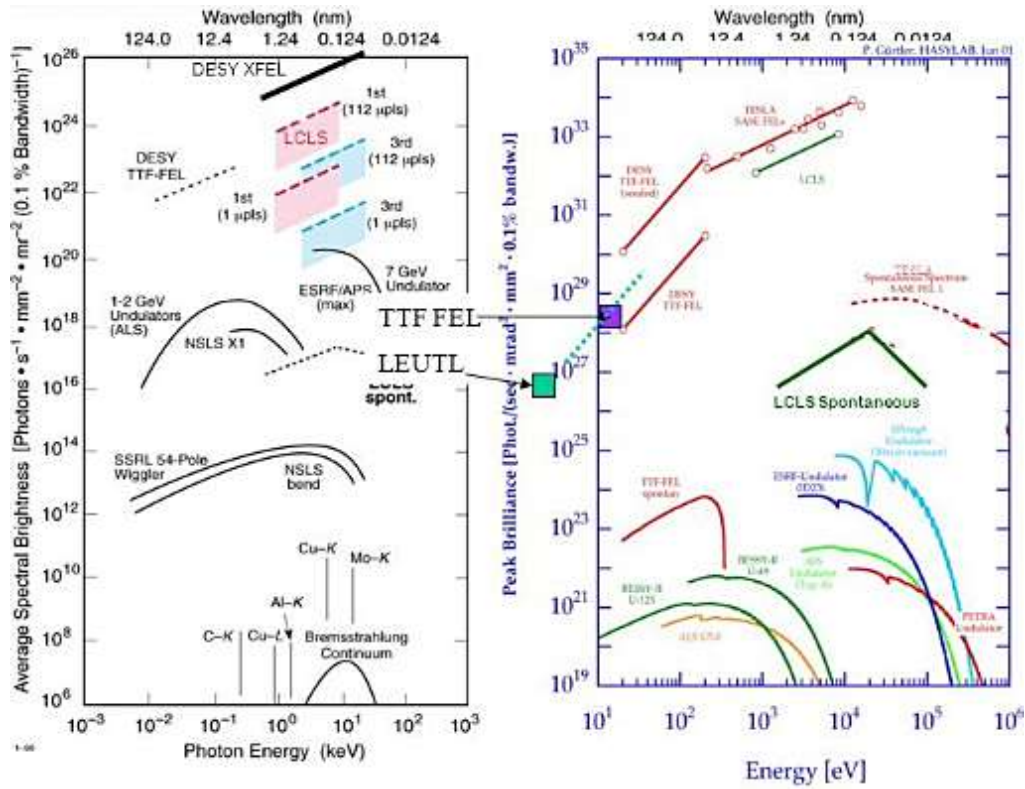


Figure B.4: Comparison of the brilliance of different light sources. For single pulses, the so-called *peak brightness* is calculated (right-hand picture), while averaging over the (maximum) repetition rate yields the so-called *average brightness* (left-hand). E.g. European XFEL and LCLS have similar *peak brightness* values, but the *average brightness* at European XFEL is correspondingly higher due to its > 100-fold pulse rate (27'000 sec⁻¹ versus 120 sec⁻¹) (in this figure EuXFEL is called "TESLA SASE FEL" or "DESY XFEL"). Fig. (still) without reference.

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Declaration

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Hamburg, December 08, 2020

A.B, K.K., K.E., C.B.

Version 1.1

Contents updated following extended software update with added functionalities and shortened to focus better on this experiment

Hamburg, August 25, 2021

Several incremental updates occurred over the student runs in 2022 and 2023

Version 2.0

Contents completely overhauled after 2 years of student experiments. This version now fits a 3 day and a 5 day long experimental run, and takes the latest software updates into account.

Hamburg, January 25, 2024

K.E., C.B.